

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
```

```

        df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))

    df = df[cols]
    #df['campaign'] = "ALL"

    return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()

```

```

        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)

    return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : column

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : column

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv

- r_{train} : $\text{train_cv}/\text{test_cv}$

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

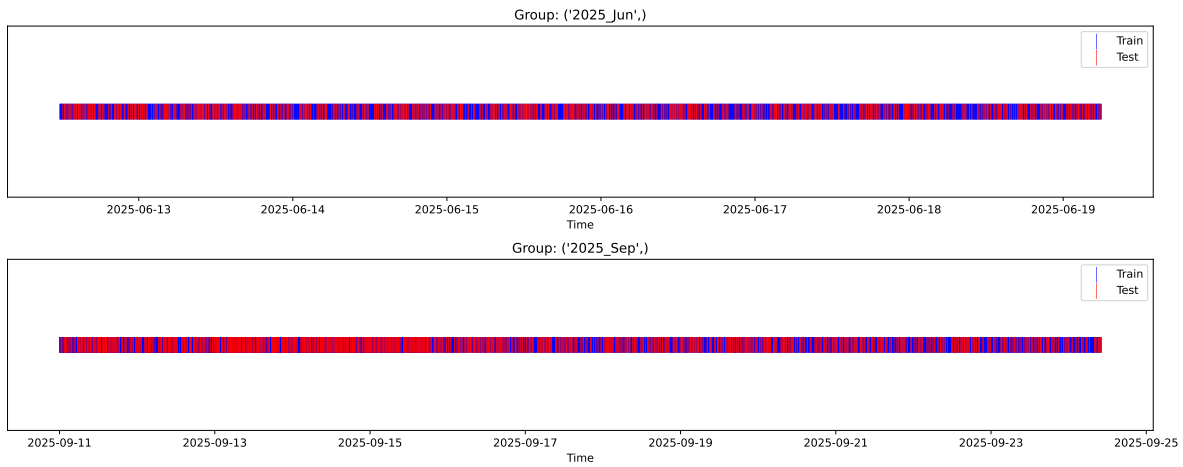
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

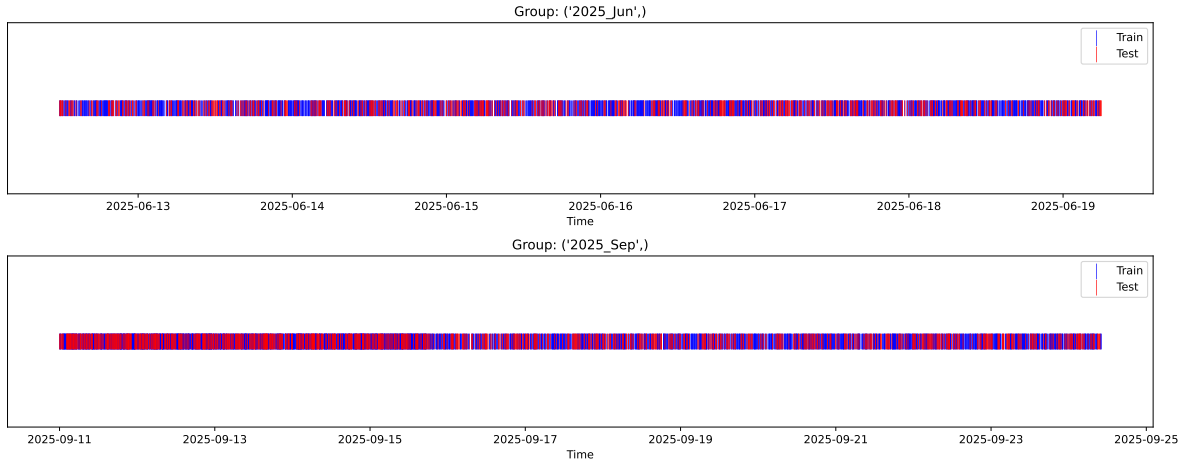
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

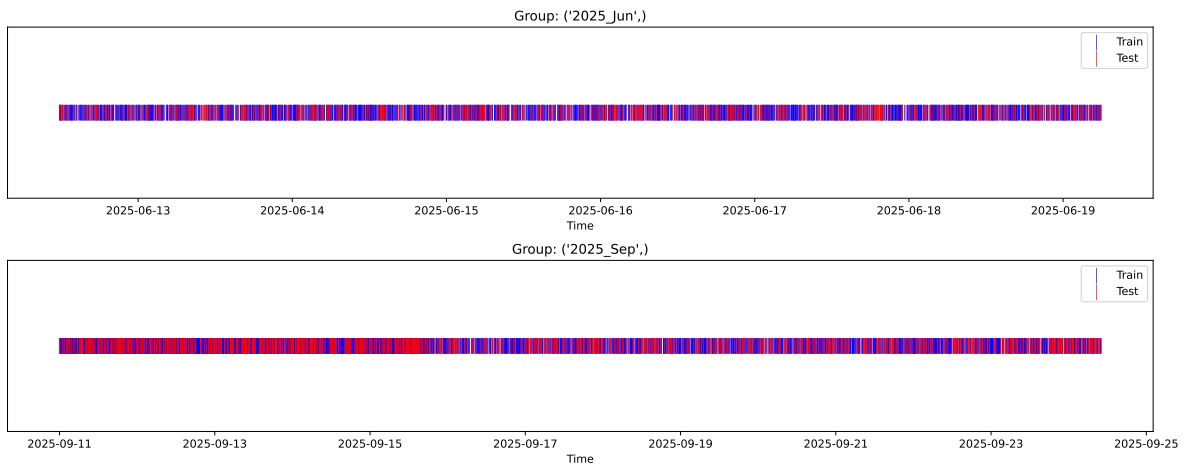
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



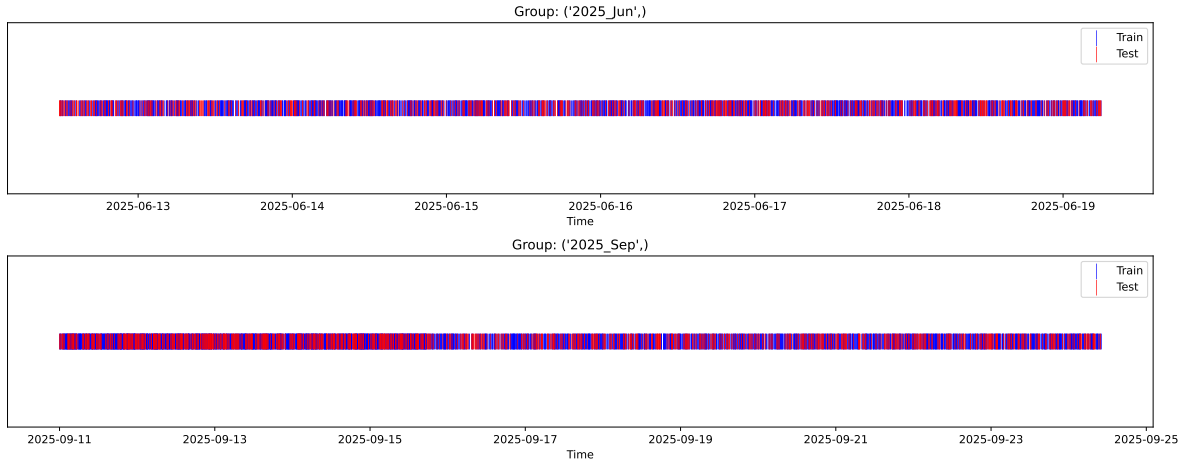
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



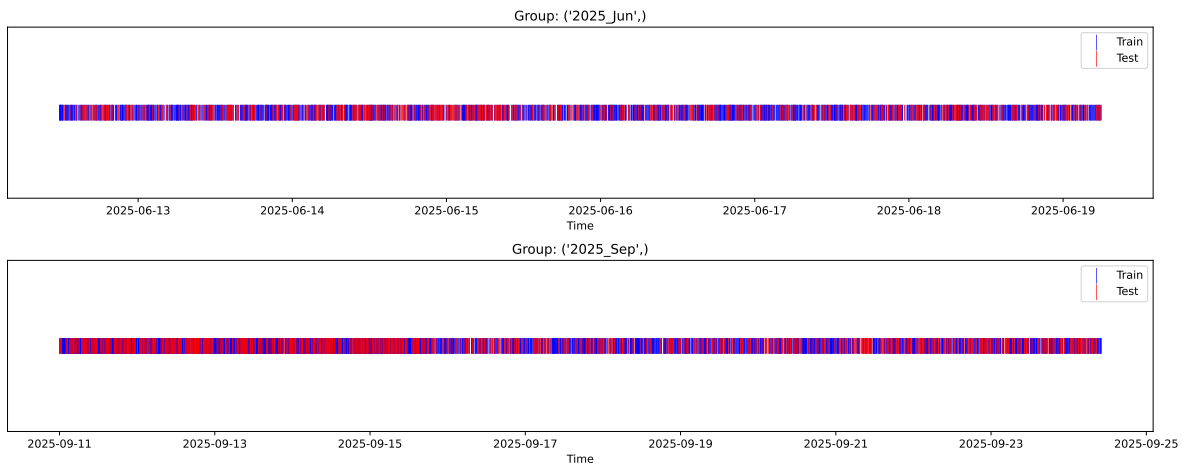
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



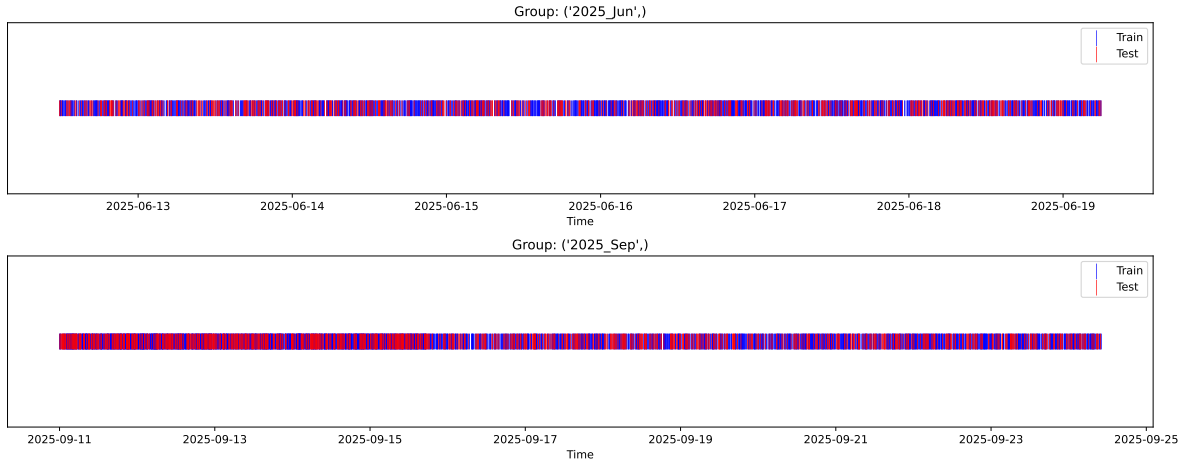
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 1

Len(group) : 5452, Len(training) : 3817, Len(test) : 1635

Distribution y_train vs y_test: y_train: mean=2.389, std=0.817, min=0.973, max=13.845
y_test: mean=2.385, std=0.771, min=1.088, max=7.277

Fold 0 — y_test: mean=2.37, std=0.77, y_train: mean=2.40, std=0.84

Fold 1 — y_test: mean=2.38, std=0.80, y_train: mean=2.39, std=0.82

Fold 2 — y_test: mean=2.39, std=0.83, y_train: mean=2.39, std=0.81

Fold 3 — y_test: mean=2.39, std=0.79, y_train: mean=2.39, std=0.83

Fold 4 — y_test: mean=2.40, std=0.86, y_train: mean=2.39, std=0.80

train scores CV: [0.43340769 0.4327579 0.42597048 0.43756683 0.4734002]

test scores CV: [0.44580332 0.44620218 0.46005545 0.43492237 0.36264806]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4406	0.4299	0.0346	0.4776	0.4782	0.0483	1.1123	1.0249

Coefficients:

$\text{coef_r_CO2} = -0.0821$

$\text{coef_r_NH3} = -0.1711$

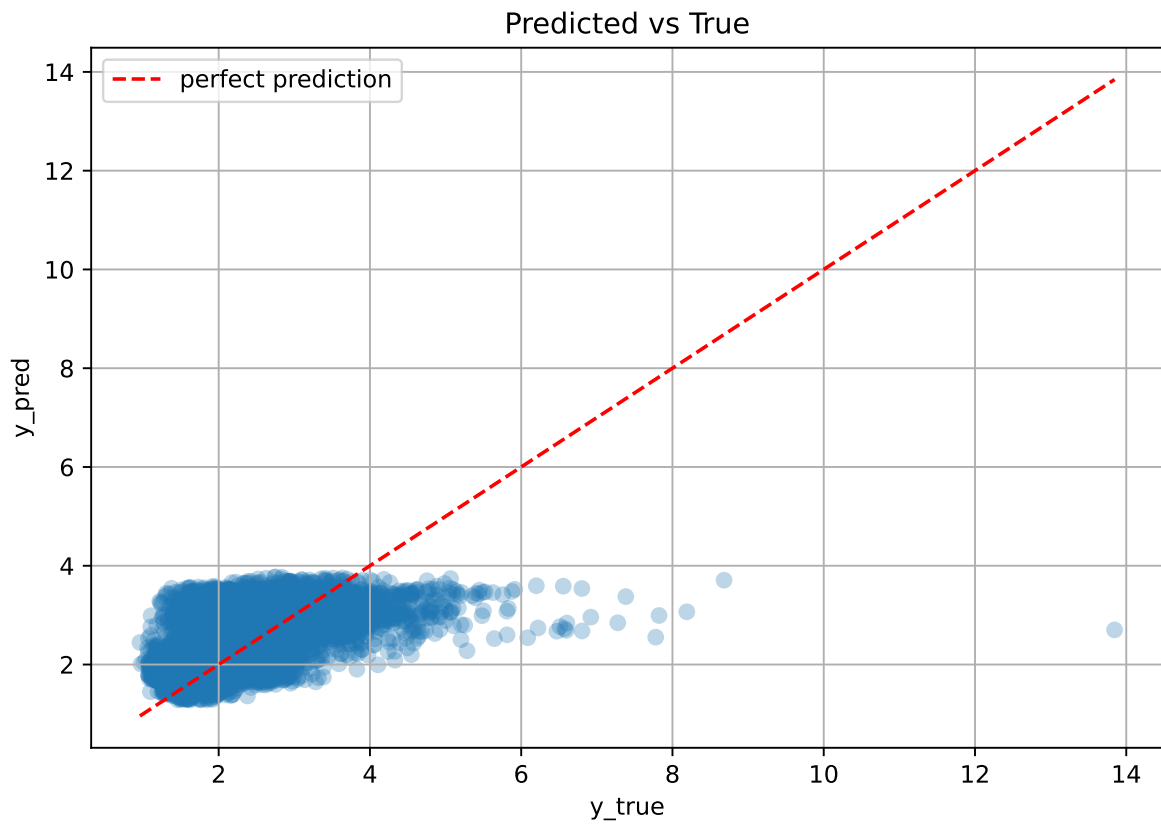
$\text{coef_r_THI} = 0.4407$

$\text{coef_r_temperature} = -0.4486$

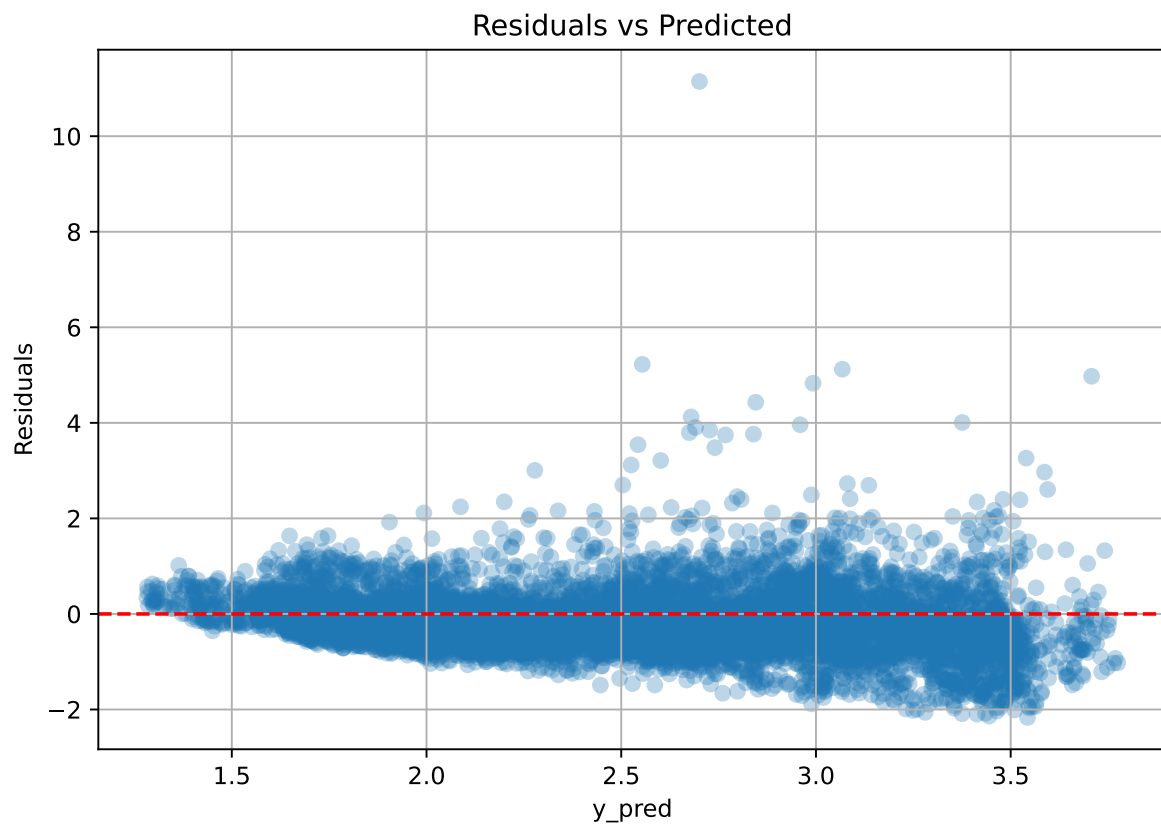
$\text{coef_r_umidity} = -0.5957$

$\text{intercept} = 2.3894$

2.3.2 Scatter



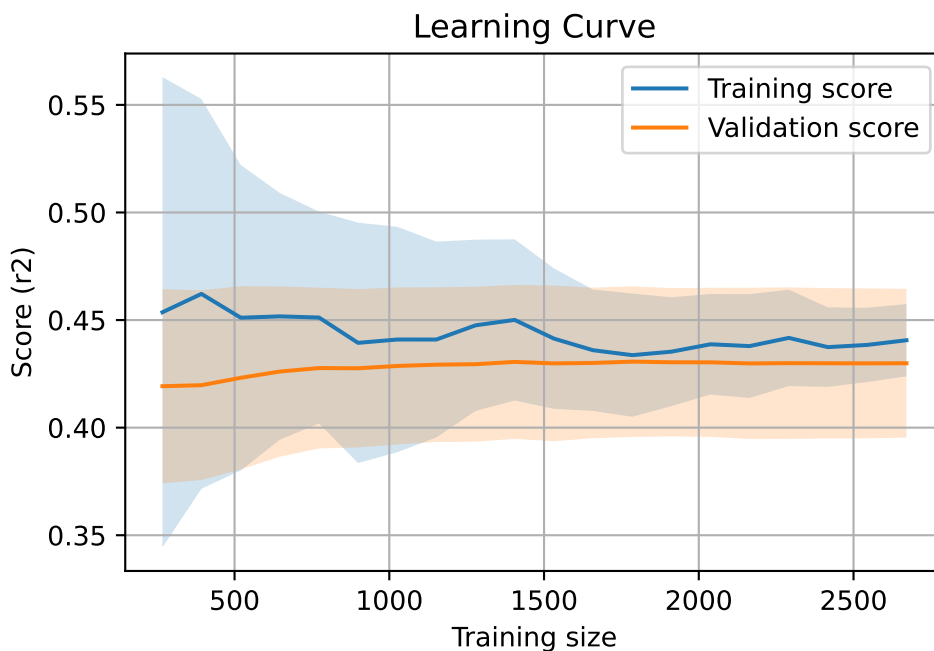
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 1

`Len(group)` : 5452, `Len(training)` : 3817,

`Len(training_real)` : 2672, `Len(test_of_training)` : 1145, `Len(test)` : 1635



2.3.5 Gridsearch — Group: 2

Len(group) : 2716, **Len(training)** : 1902, **Len(test)** : 814

Distribution y_train vs y_test: y_train: mean=2.260, std=0.774, min=0.959, max=7.822
y_test: mean=2.299, std=0.764, min=1.068, max=6.514

Fold 0 — y_test: mean=2.24, std=0.74, y_train: mean=2.27, std=0.79

Fold 1 — y_test: mean=2.27, std=0.74, y_train: mean=2.26, std=0.79

Fold 2 — y_test: mean=2.23, std=0.72, y_train: mean=2.27, std=0.80

Fold 3 — y_test: mean=2.26, std=0.77, y_train: mean=2.26, std=0.78

Fold 4 — y_test: mean=2.28, std=0.80, y_train: mean=2.25, std=0.76

train scores CV: [0.3588487 0.36320323 0.33531576 0.36111957 0.37208129]

test scores CV: [0.37354902 0.36389667 0.43899582 0.36954466 0.34431042]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3581	0.3781	0.0321	0.3225	0.3234	-0.0546	0.8555	0.9472

Coefficients:

$\text{coef_r_CO2} = -0.0293$

$\text{coef_r_NH3} = 0.0918$

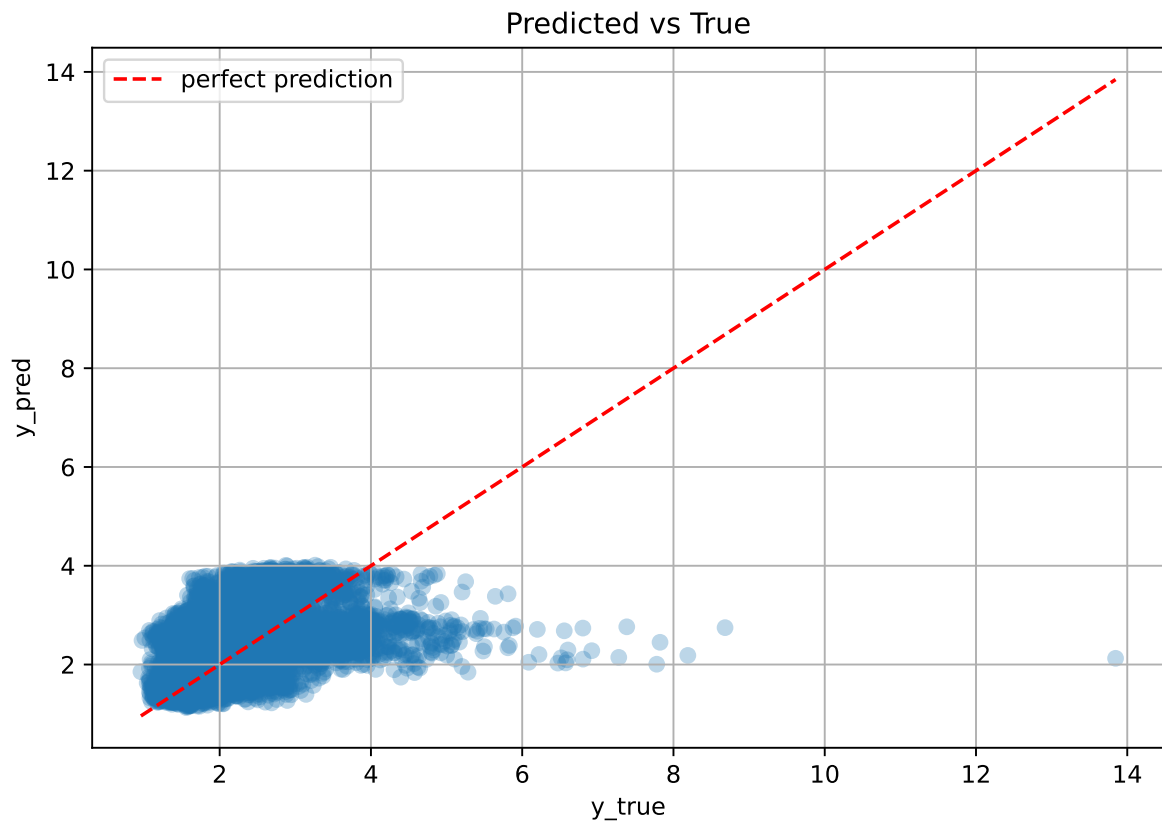
$\text{coef_r_THI} = 0.6273$

$\text{coef_r_temperature} = -0.8810$

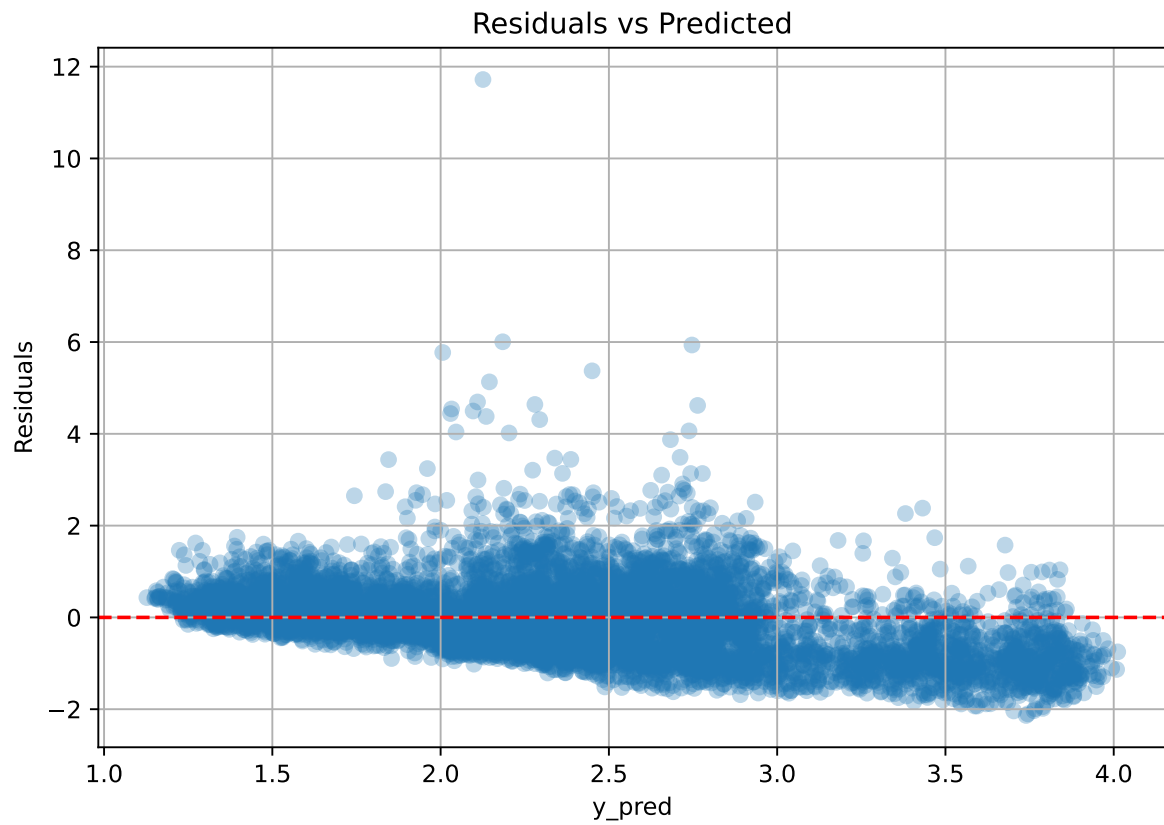
$\text{coef_r_umidity} = -0.7713$

$\text{intercept} = 2.2605$

2.3.6 Scatter



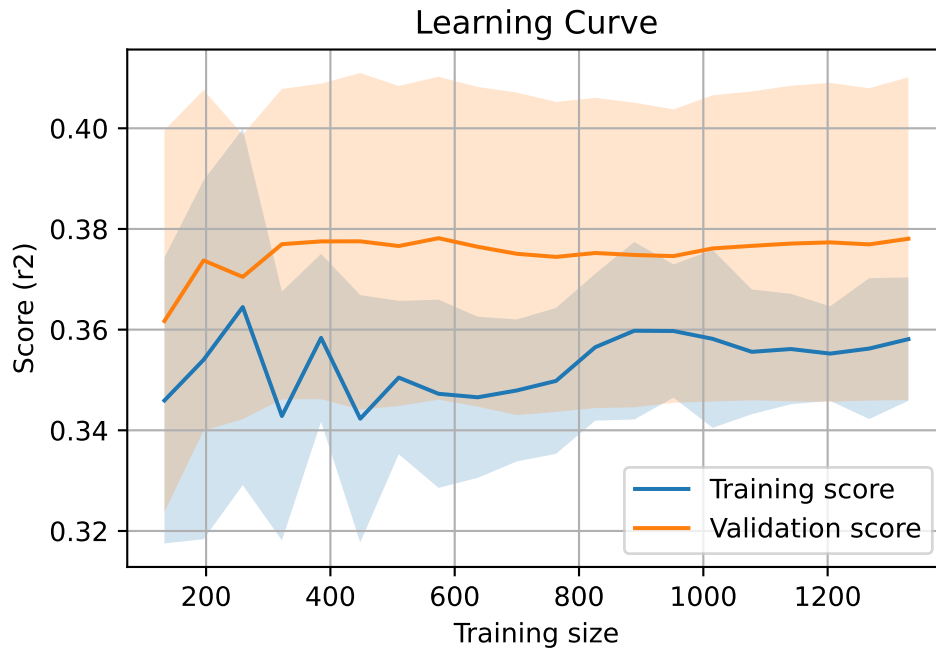
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 2

`Len(group)` : 2716, `Len(training)` : 1902,

`Len(training_real)` : 1332, `Len(test_of_training)` : 570, `Len(test)` : 814



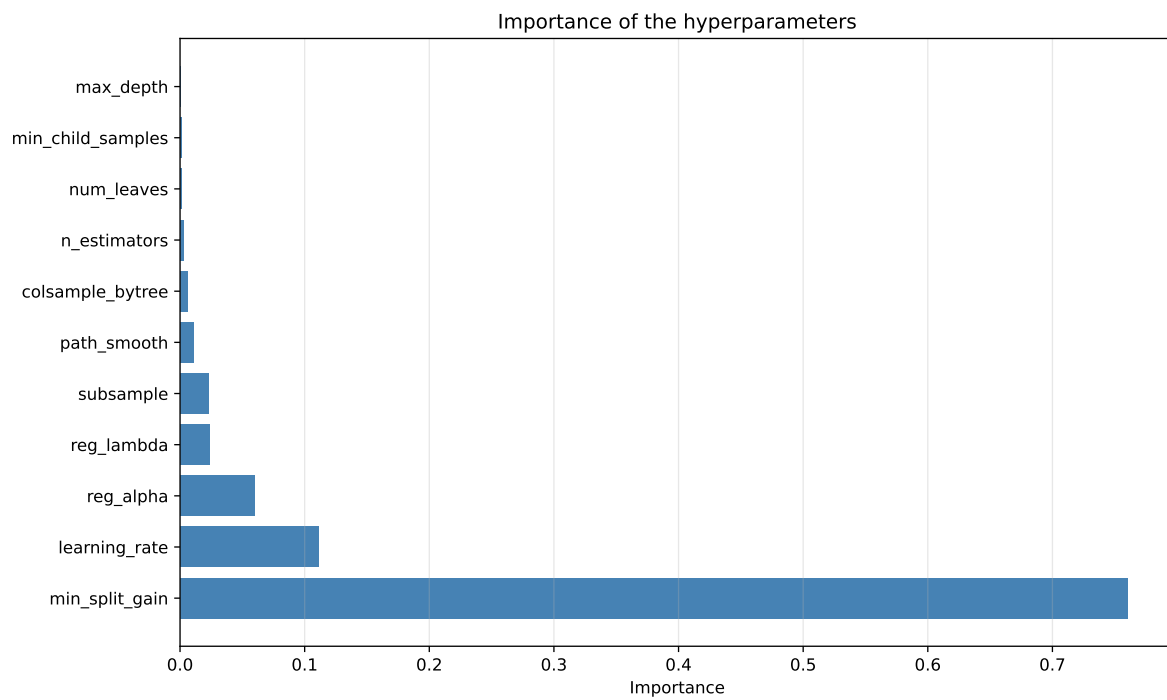
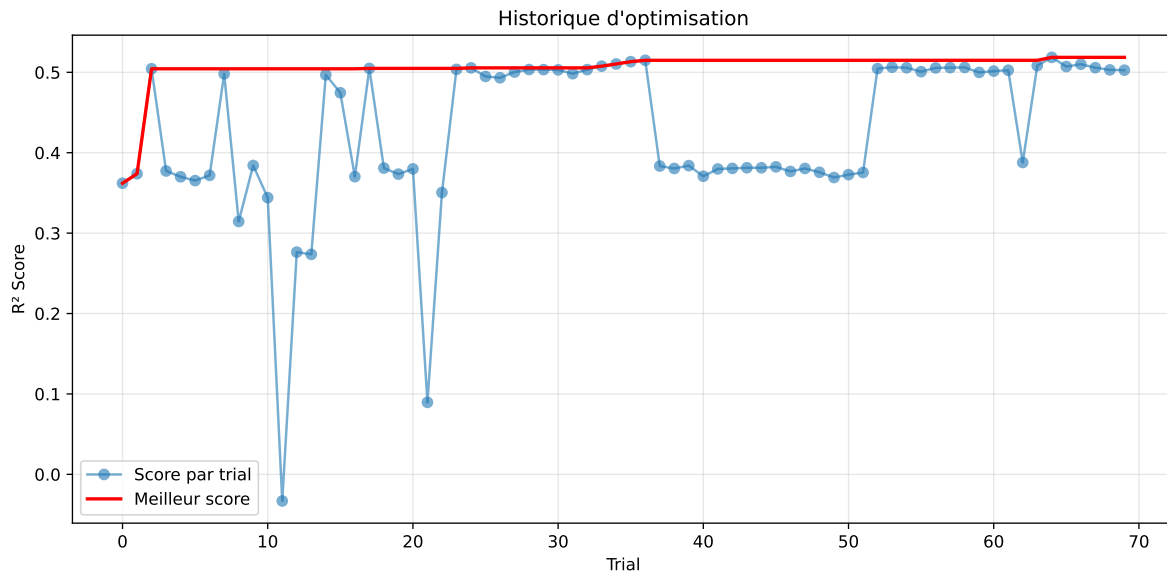
2.4 Model : LGBM

2.4.1 Gridsearch — Group: 1

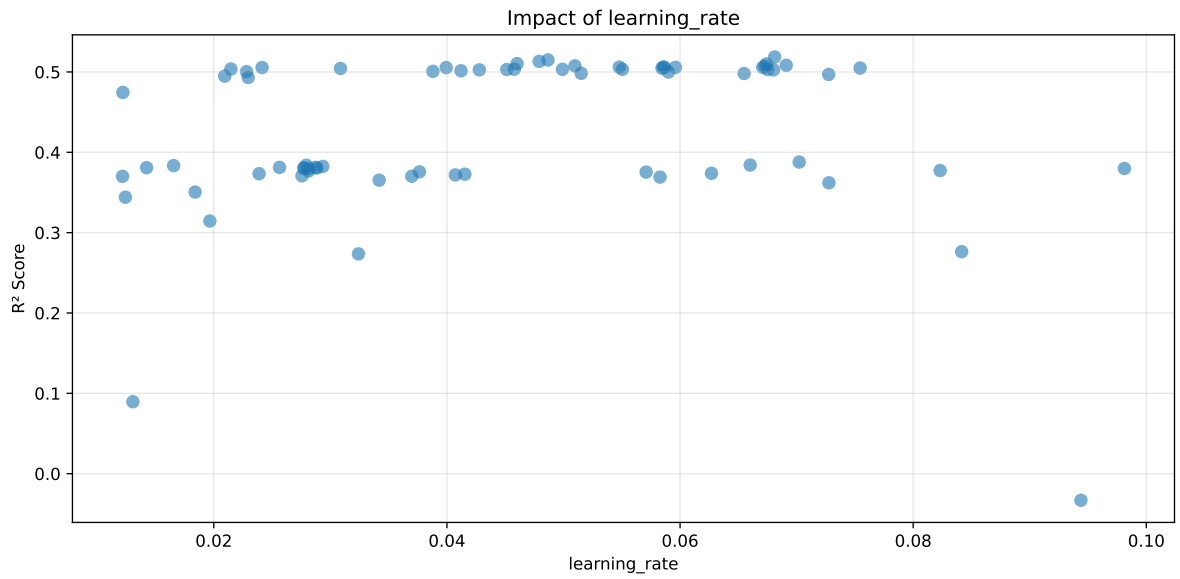
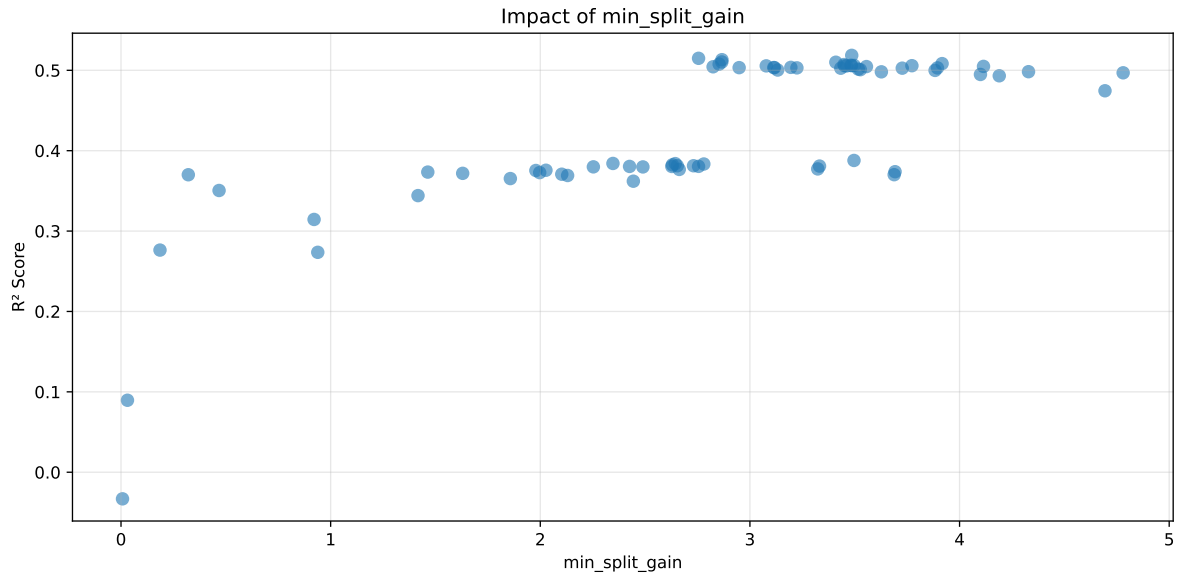
Len(group) : 5452, Len(training) : 3817, Len(test) : 1635

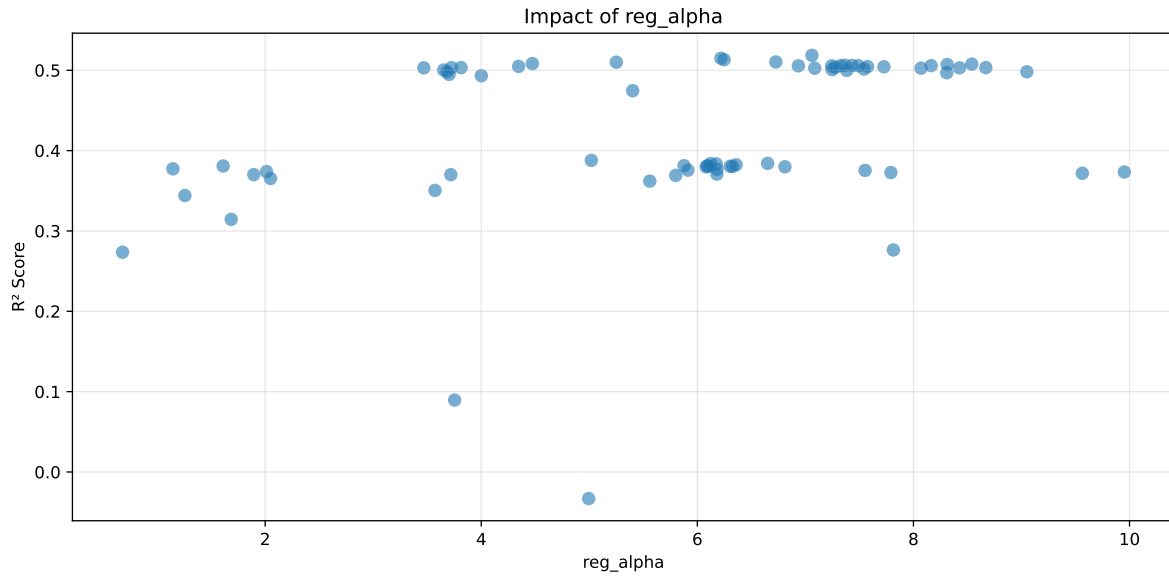
Distribution y_train vs y_test: y_train: mean=2.389, std=0.817, min=0.973, max=13.845
y_test: mean=2.385, std=0.771, min=1.088, max=7.277

===== TOP Importance FEATURES =====
feature importance r_NH3 102 r_umidity 94 r_CO2 63 r_temperature 34 r_THI 12



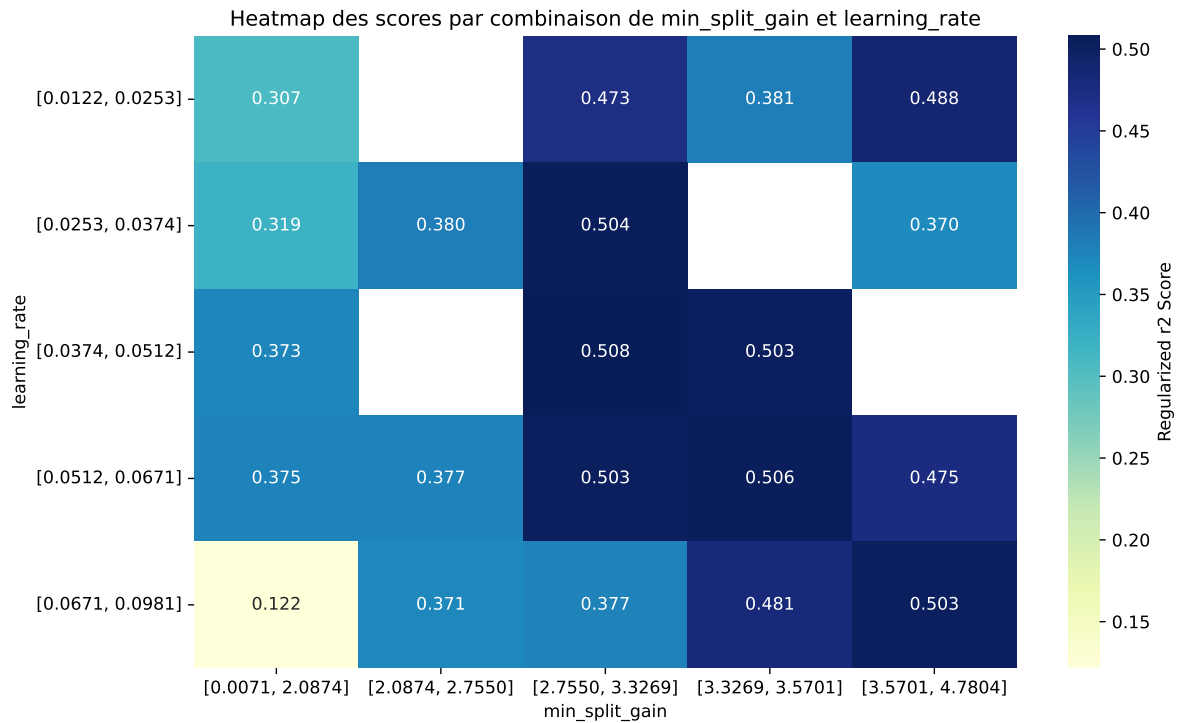
Paramètres avec >5% d'importance : ['min_split_gain', 'learning_rate', 'reg_alpha']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5444	0.5186	0.5684	0.0379	0.5186	0.5894	0.0708	1.1364	1.0497
2	0.5402	0.5149	0.5644	0.0365	0.5149	0.5866	0.0717	1.1392	1.0491
3	0.5387	0.5132	0.5636	0.0366	0.5132	0.5862	0.073	1.1422	1.0496
4	0.5353	0.5104	0.5601	0.0363	0.5104	0.5843	0.0739	1.1448	1.0488
5	0.5331	0.51	0.5584	0.0363	0.51	0.5781	0.068	1.1334	1.0451
6	0.5301	0.5083	0.5559	0.0363	0.5083	0.5776	0.0693	1.1363	1.0428
7	0.5318	0.5076	0.5578	0.0357	0.5076	0.5825	0.0749	1.1475	1.0475
8	0.5287	0.5071	0.5549	0.0374	0.5071	0.5774	0.0702	1.1384	1.0424
9	0.53	0.5063	0.5542	0.0354	0.5063	0.5788	0.0725	1.1433	1.0469
10	0.5298	0.506	0.555	0.0356	0.506	0.579	0.073	1.1443	1.0469
---	---	---	---	---	---	---	---	---	---
70	0.7298	0.6026	0.6477	0.0491	-0.0331	0.6638	0.0612	1.1015	1.211

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.06811581847425988, 'reg__num_leaves': 19, 'reg__subsample': 0.5661845195554986, 'reg__colsample_bytree': 0.8696145337239631, 'reg__min_child_samples': 46, 'reg__reg_alpha': 7.060389558806824, 'reg__reg_lambda': 5.108070836912308, 'reg__min_split_gain': 3.485625164808146, 'reg__path_smooth': 2.0486437033096188}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.048669817344678394, 'reg__num_leaves': 23, 'reg__subsample': 0.6019864840961671, 'reg__colsample_bytree': 0.845347446134046, 'reg__min_child_samples': 30, 'reg__reg_alpha': 6.217854743620681, 'reg__reg_lambda': 40.57807734507179, 'reg__min_split_gain': 2.755046370824527, 'reg__path_smooth': 2.1458577708962903}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.047907240476474745, 'reg__num_leaves': 23, 'reg__subsample': 0.5984046145654294, 'reg__colsample_bytree': 0.844858797087247, 'reg__min_child_samples': 40, 'reg__reg_alpha': 6.247614393459522, 'reg__reg_lambda': 40.28922999308201, 'reg__min_split_gain': 2.866541934808906, 'reg__path_smooth': 2.318516705268718}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.04601632018402271, 'reg__num_leaves': 15, 'reg__subsample': 0.5992115895933491, 'reg__colsample_bytree': 0.832613537566718, 'reg__min_child_samples': 32, 'reg__reg_alpha': 6.726794941004357, 'reg__reg_lambda': 41.57749408255629, 'reg__min_split_gain': 2.8659037502911655, 'reg__path_smooth': 2.2283234177192557}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.06741033801059419, 'reg__num_leaves': 33, 'reg__subsample': 0.5675129977014007,

'reg__colsample_bytree': 0.8060794906314217, 'reg__min_child_samples': 47, 'reg__reg_alpha': 5.2500226556207545, 'reg__reg_lambda': 32.84403131823668, 'reg__min_split_gain': 3.409836613065854, 'reg__path_smooth': 1.877942635999056}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.06910543020281655, 'reg__num_leaves': 26, 'reg__subsample': 0.6206835531954661, 'reg__colsample_bytree': 0.7931203074193528, 'reg__min_child_samples': 33, 'reg__reg_alpha': 4.472414403619734, 'reg__reg_lambda': 46.02639959390031, 'reg__min_split_gain': 3.9166939823337916, 'reg__path_smooth': 3.043836961882734}

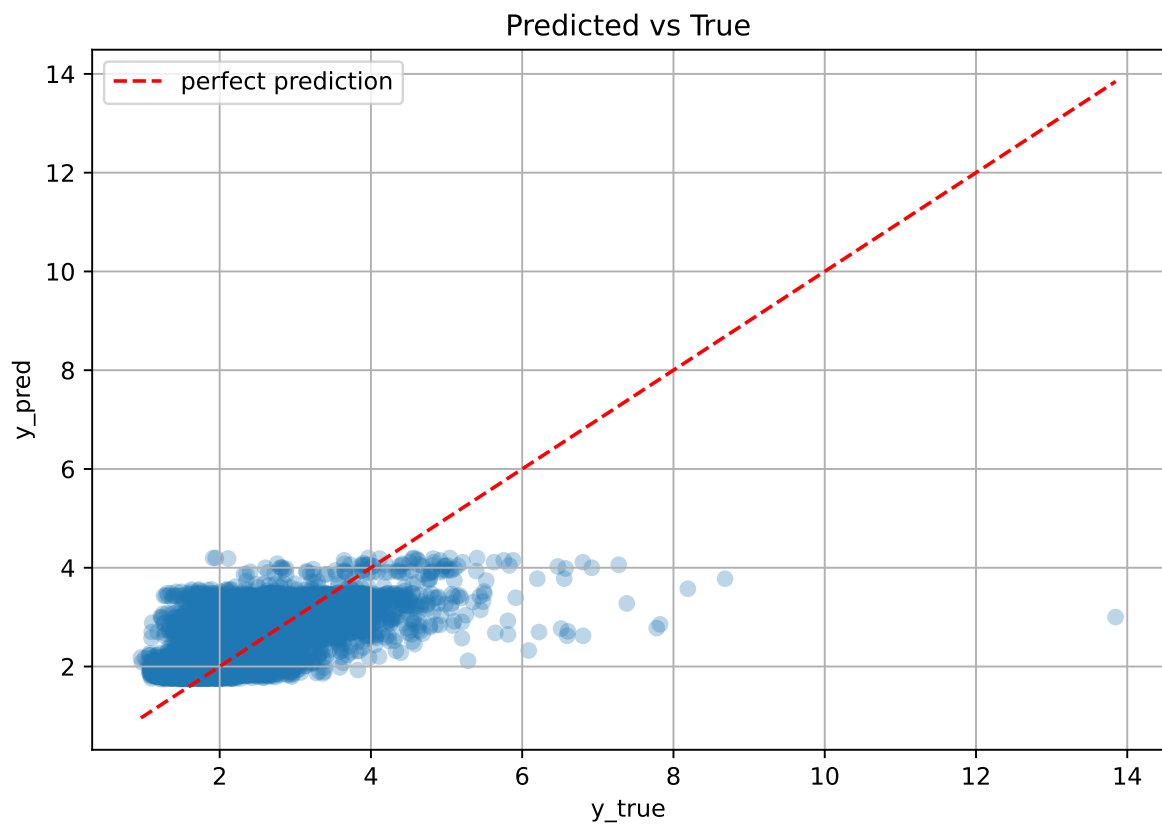
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.05097879050589461, 'reg__num_leaves': 28, 'reg__subsample': 0.7309248023203156, 'reg__colsample_bytree': 0.8406187419753106, 'reg__min_child_samples': 41, 'reg__reg_alpha': 8.539810643732796, 'reg__reg_lambda': 40.99309000605684, 'reg__min_split_gain': 2.8532722893315925, 'reg__path_smooth': 2.263830259436575}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.06726860518628824, 'reg__num_leaves': 34, 'reg__subsample': 0.6240594530880056, 'reg__colsample_bytree': 0.804687534990483, 'reg__min_child_samples': 46, 'reg__reg_alpha': 8.312292315652629, 'reg__reg_lambda': 33.21943979456037, 'reg__min_split_gain': 3.4488380776702723, 'reg__path_smooth': 1.973589548154303}

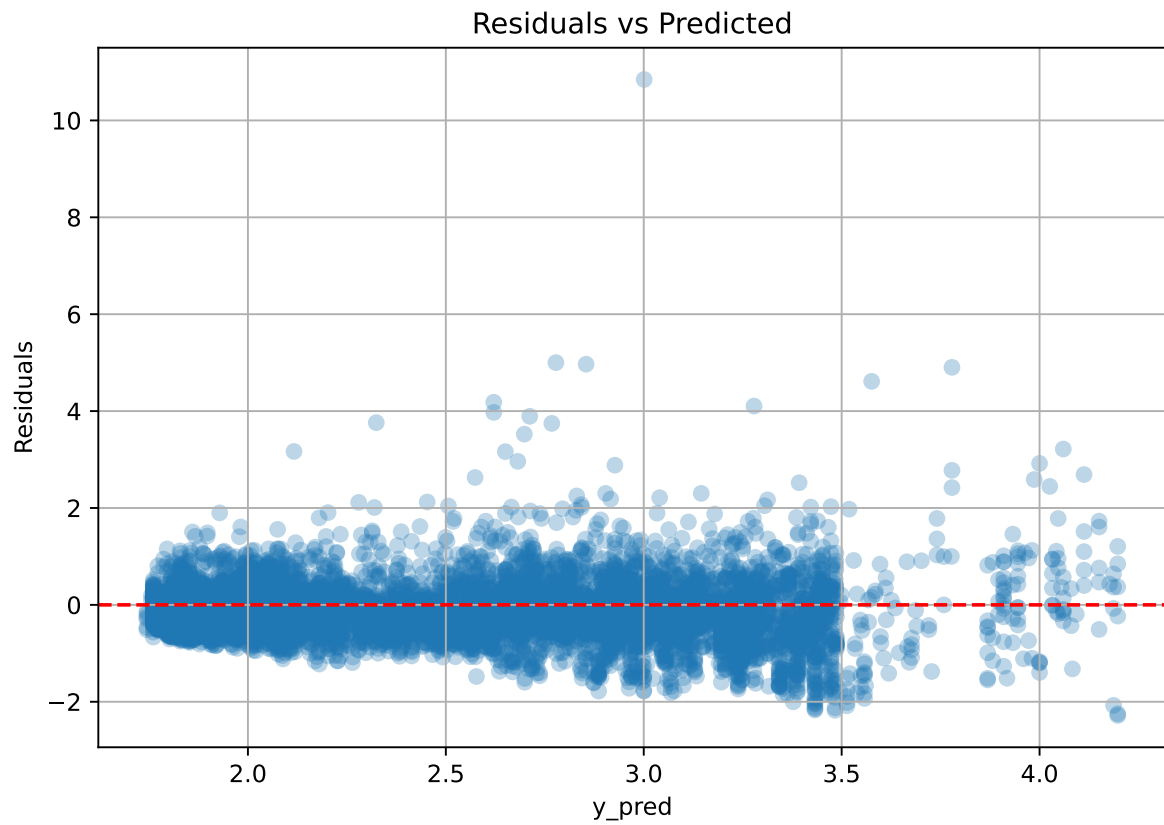
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.0585541354423821, 'reg__num_leaves': 30, 'reg__subsample': 0.613285855196445, 'reg__colsample_bytree': 0.8776692297700869, 'reg__min_child_samples': 11, 'reg__reg_alpha': 7.3673067505470975, 'reg__reg_lambda': 36.41522425326816, 'reg__min_split_gain': 3.4823740190297006, 'reg__path_smooth': 1.6957775668203916}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.054787100536455385, 'reg__num_leaves': 30, 'reg__subsample': 0.6232446409660807, 'reg__colsample_bytree': 0.8018806332242852, 'reg__min_child_samples': 42, 'reg__reg_alpha': 7.432591886106761, 'reg__reg_lambda': 35.17436334984614, 'reg__min_split_gain': 3.496845803501164, 'reg__path_smooth': 1.69523593218849}

2.4.2 Scatter



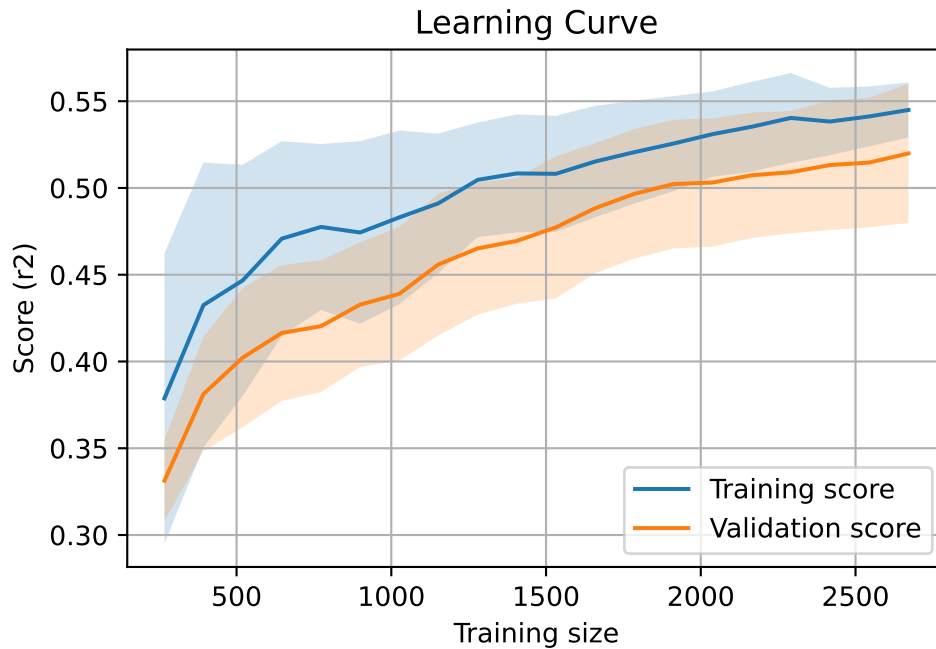
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 1

`Len(group)` : 5452, `Len(training)` : 3817,

`Len(training_real)` : 2672, `Len(test_of_training)` : 1145, `Len(test)` : 1635

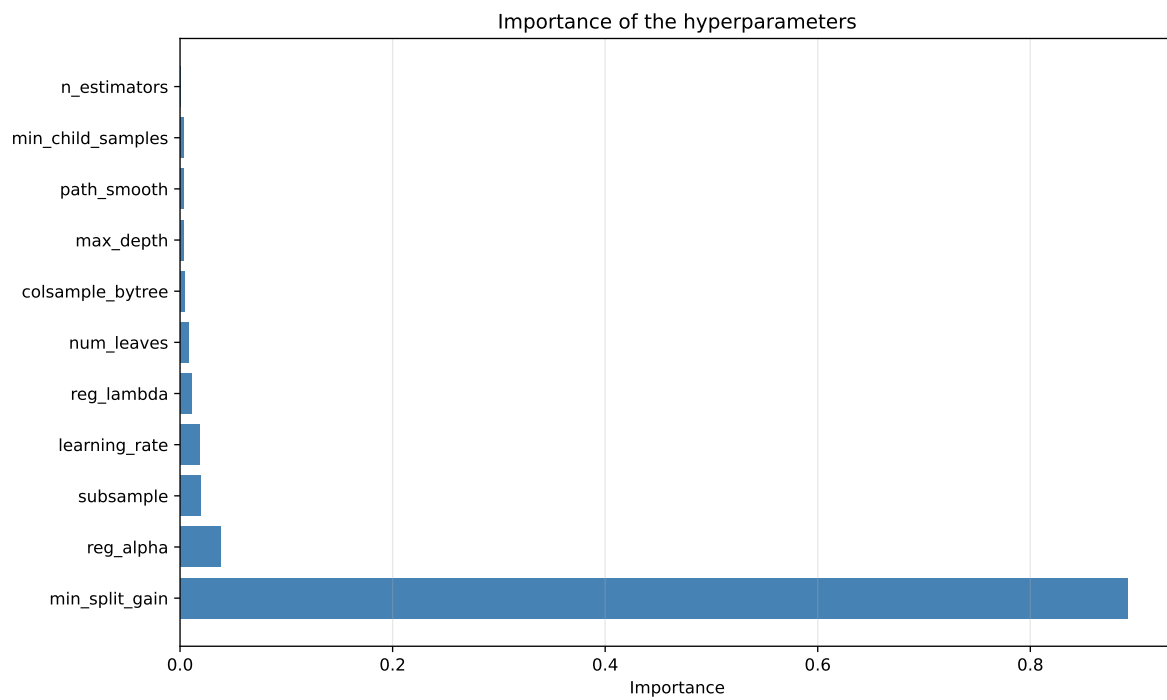
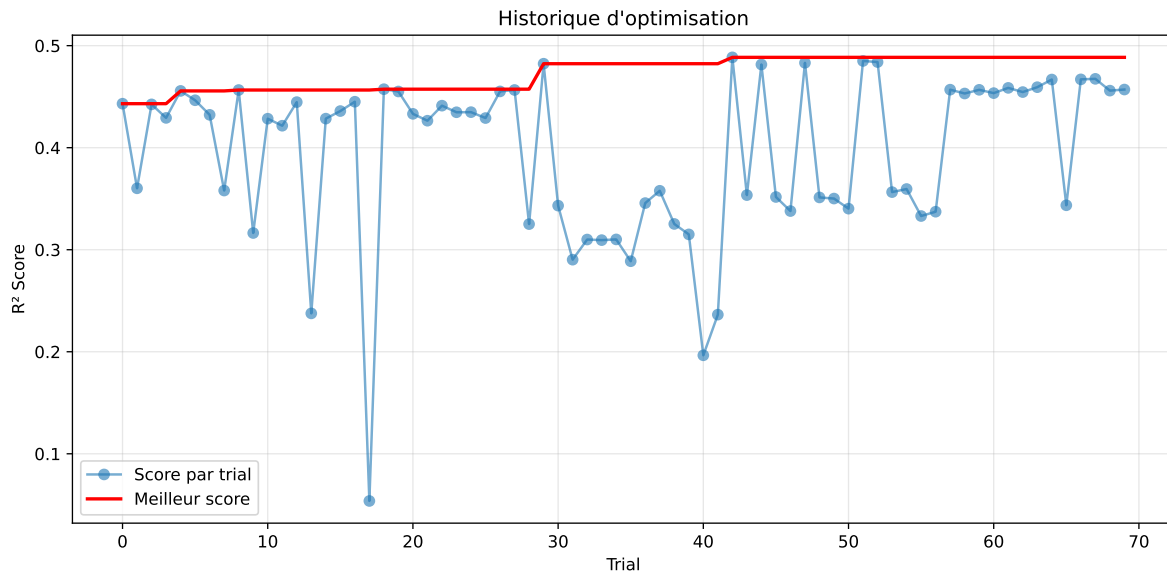


2.4.5 Gridsearch — Group: 2

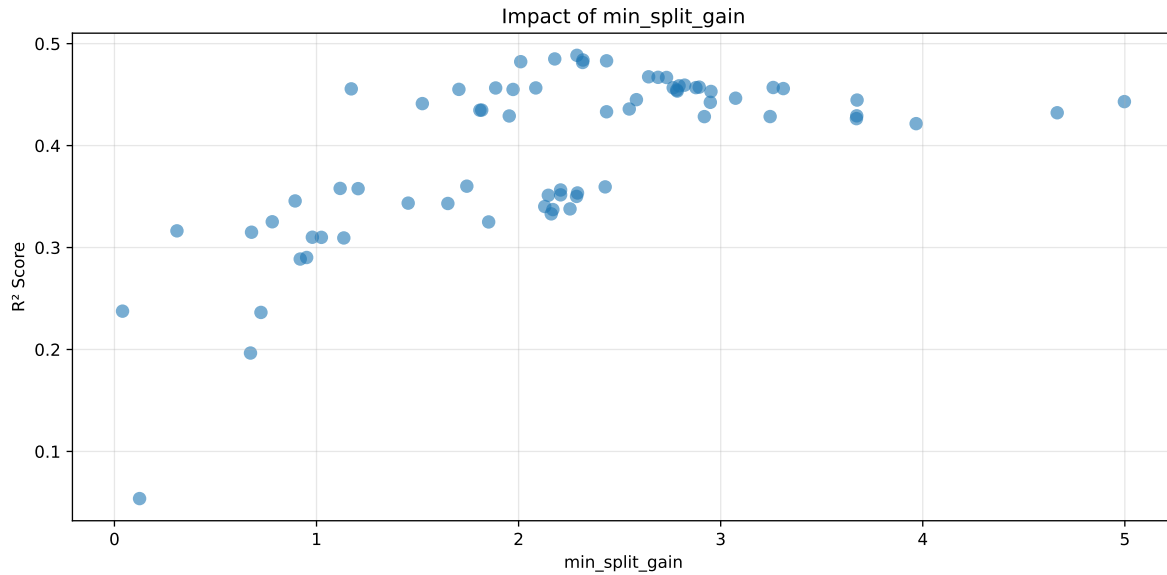
Len(group) : 2716, Len(training) : 1902, Len(test) : 814

Distribution y_train vs y_test: y_train: mean=2.260, std=0.774, min=0.959, max=7.822
y_test: mean=2.299, std=0.764, min=1.068, max=6.514

===== TOP Importance FEATURES =====
feature importance r_umidity 82 r_CO2 68 r_THI 51 r_temperature 42 r_NH3 33



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5129	0.4885	0.4408	0.0393	0.4885	0.4633	-0.0252	0.9484	1.0499
2	0.5081	0.485	0.4369	0.0391	0.485	0.4632	-0.0218	0.9551	1.0477
3	0.5074	0.484	0.4362	0.0393	0.484	0.4573	-0.0267	0.9449	1.0485
4	0.5068	0.4832	0.4361	0.0384	0.4832	0.461	-0.0222	0.9541	1.049
5	0.5058	0.4823	0.4345	0.0376	0.4823	0.4601	-0.0222	0.9539	1.0488
6	0.5034	0.4815	0.4339	0.0379	0.4815	0.456	-0.0256	0.9469	1.0455
7	0.4889	0.4675	0.4199	0.0352	0.4675	0.4456	-0.0218	0.9533	1.0458
8	0.4844	0.467	0.4169	0.0393	0.467	0.4373	-0.0296	0.9366	1.0374
9	0.4849	0.4668	0.4161	0.0371	0.4668	0.441	-0.0258	0.9446	1.0388
10	0.4735	0.4593	0.4067	0.0365	0.4593	0.4294	-0.0299	0.9349	1.0308
---	---	---	---	---	---	---	---	---	---
70	0.6439	0.5455	0.5285	0.0342	0.0538	0.5434	-0.0021	0.9961	1.1803

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.06893292823370542, 'reg__num_leaves': 52, 'reg__subsample': 0.8575510552609101, 'reg__colsample_bytree': 0.8888457789779691, 'reg__min_child_samples': 31, 'reg__reg_alpha': 0.8012115577347068, 'reg__reg_lambda': 22.907893459989857, 'reg__min_split_gain': 2.2887171516025946, 'reg__path_smooth': 3.042453760732936}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.07094185943283329, 'reg__num_leaves': 52, 'reg__subsample': 0.8562677244956024,

'reg__colsample_bytree': 0.8950418452166623, 'reg__min_child_samples': 16, 'reg__reg_alpha': 1.643588574538787, 'reg__reg_lambda': 28.00955355107353, 'reg__min_split_gain': 2.1788486927465343, 'reg__path_smooth': 1.995568216447065}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.06835716392873105, 'reg__num_leaves': 53, 'reg__subsample': 0.8439287577675616, 'reg__colsample_bytree': 0.864699525630857, 'reg__min_child_samples': 20, 'reg__reg_alpha': 1.038901831300048, 'reg__reg_lambda': 27.834732768013435, 'reg__min_split_gain': 2.3186320627124264, 'reg__path_smooth': 4.007393760197912}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.0713324315692788, 'reg__num_leaves': 54, 'reg__subsample': 0.8361830357847769, 'reg__colsample_bytree': 0.8983658445909782, 'reg__min_child_samples': 14, 'reg__reg_alpha': 0.6786449950401299, 'reg__reg_lambda': 28.393240860492508, 'reg__min_split_gain': 2.4353839029278244, 'reg__path_smooth': 4.012484712665688}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.05331373222490644, 'reg__num_leaves': 49, 'reg__subsample': 0.8848705207495606, 'reg__colsample_bytree': 0.8994227828819885, 'reg__min_child_samples': 33, 'reg__reg_alpha': 2.5107372705108473, 'reg__reg_lambda': 21.652142640354832, 'reg__min_split_gain': 2.0104286751676983, 'reg__path_smooth': 3.4252981662503035}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.07029210770010377, 'reg__num_leaves': 52, 'reg__subsample': 0.8118562308422707, 'reg__colsample_bytree': 0.8989321329965022, 'reg__min_child_samples': 20, 'reg__reg_alpha': 1.5976949656315753, 'reg__reg_lambda': 28.03024711996523, 'reg__min_split_gain': 2.31672131880007, 'reg__path_smooth': 2.9745202229518206}

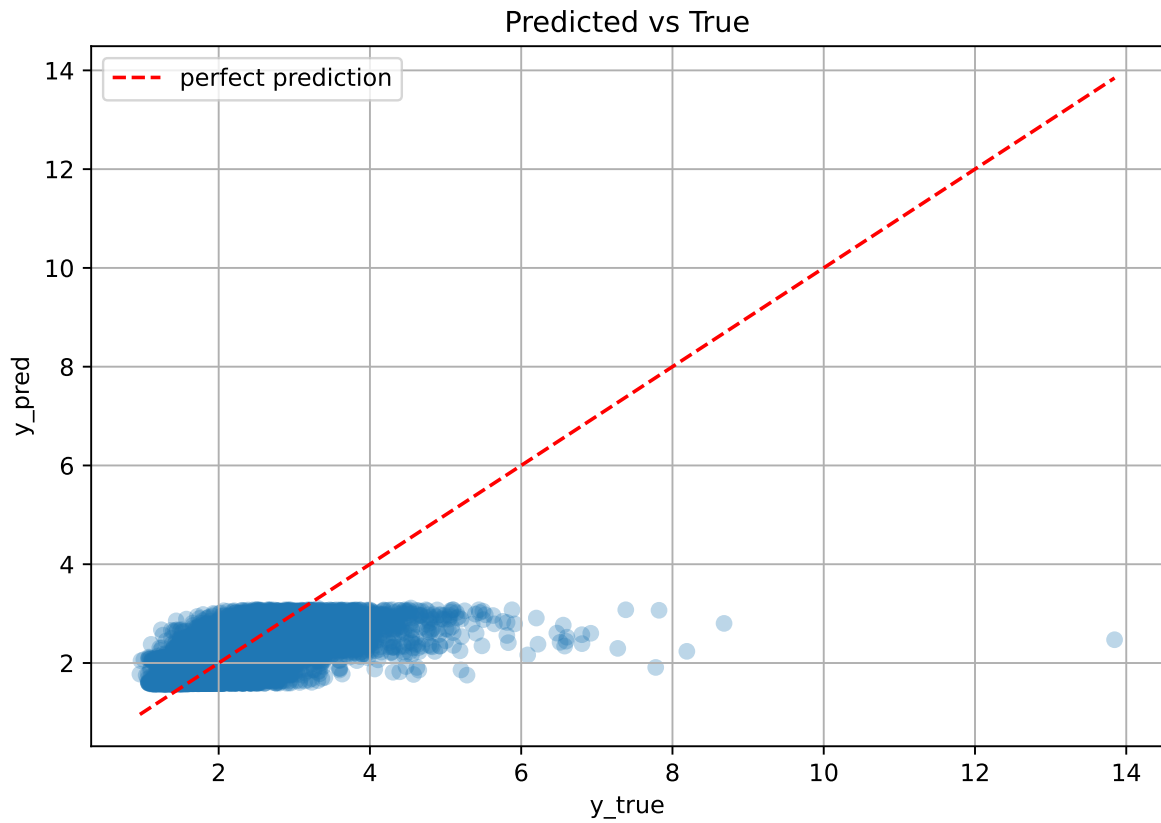
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.08141817751261428, 'reg__num_leaves': 37, 'reg__subsample': 0.7707669101595753, 'reg__colsample_bytree': 0.8268722704267988, 'reg__min_child_samples': 11, 'reg__reg_alpha': 1.7522163738224097, 'reg__reg_lambda': 25.526222827790804, 'reg__min_split_gain': 2.643385983175934, 'reg__path_smooth': 1.6237574599858875}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.08103768311207904, 'reg__num_leaves': 55, 'reg__subsample': 0.8056284082788056, 'reg__colsample_bytree': 0.8259922033315644, 'reg__min_child_samples': 11, 'reg__reg_alpha': 1.8200461765720393, 'reg__reg_lambda': 35.56717148083996, 'reg__min_split_gain': 2.6900741735941884, 'reg__path_smooth': 1.7221678000334932}

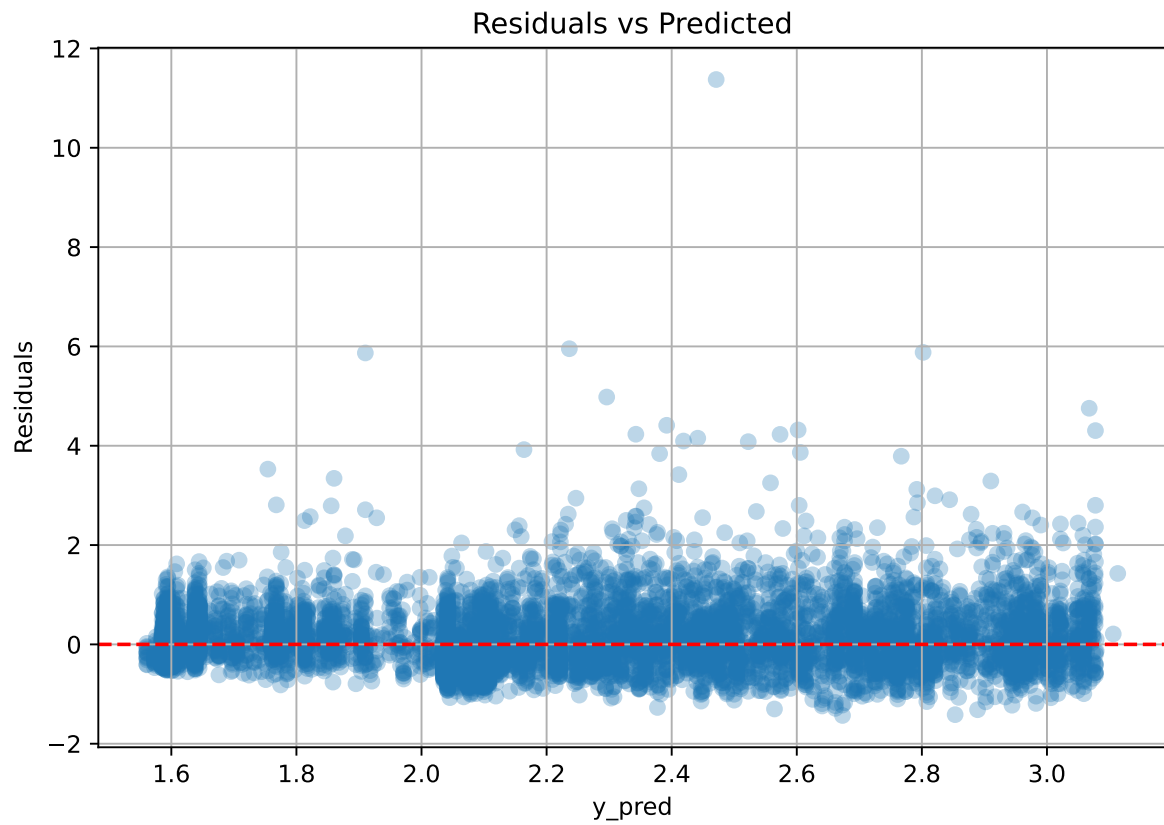
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.083442959492948, 'reg__num_leaves': 35, 'reg__subsample': 0.8115605248389137, 'reg__colsample_bytree': 0.8451799486775804, 'reg__min_child_samples': 23, 'reg__reg_alpha': 1.6184750715345568, 'reg__reg_lambda': 37.18857726441032, 'reg__min_split_gain': 2.7320635458056195, 'reg__path_smooth': 1.9331998115750575}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.0854005076800958, 'reg__num_leaves': 58, 'reg__subsample': 0.8105577980647956, 'reg__colsample_bytree': 0.8444023897220047, 'reg__min_child_samples': 11, 'reg__reg_alpha': 3.4166176697267785, 'reg__reg_lambda': 37.36376449522336, 'reg__min_split_gain': 2.8214664227839577, 'reg__path_smooth': 1.9724103338864383}

2.4.6 Scatter



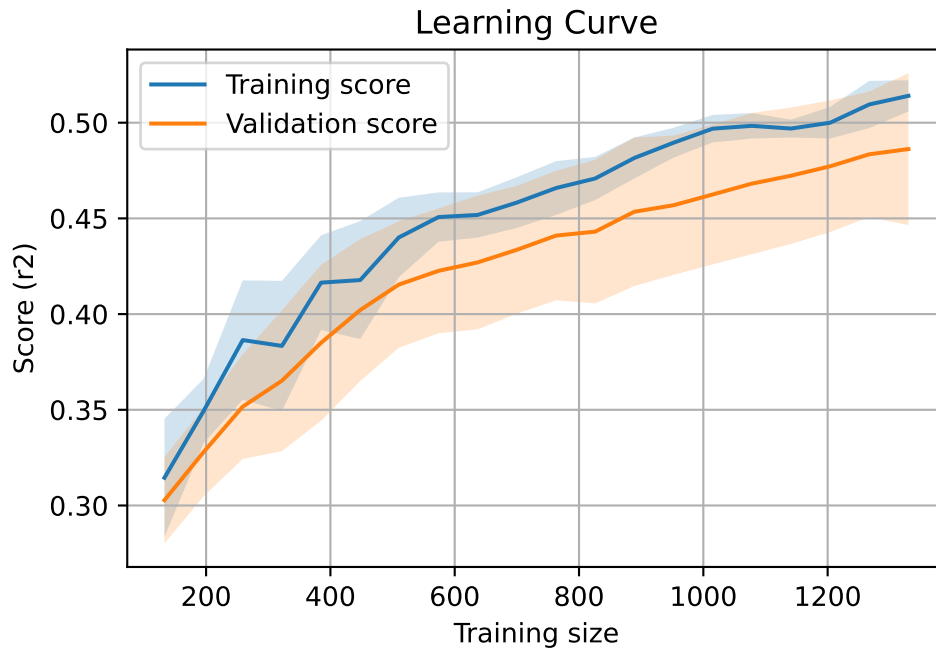
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 2

`Len(group)` : 2716, `Len(training)` : 1902,

`Len(training_real)` : 1332, `Len(test_of_training)` : 570, `Len(test)` : 814



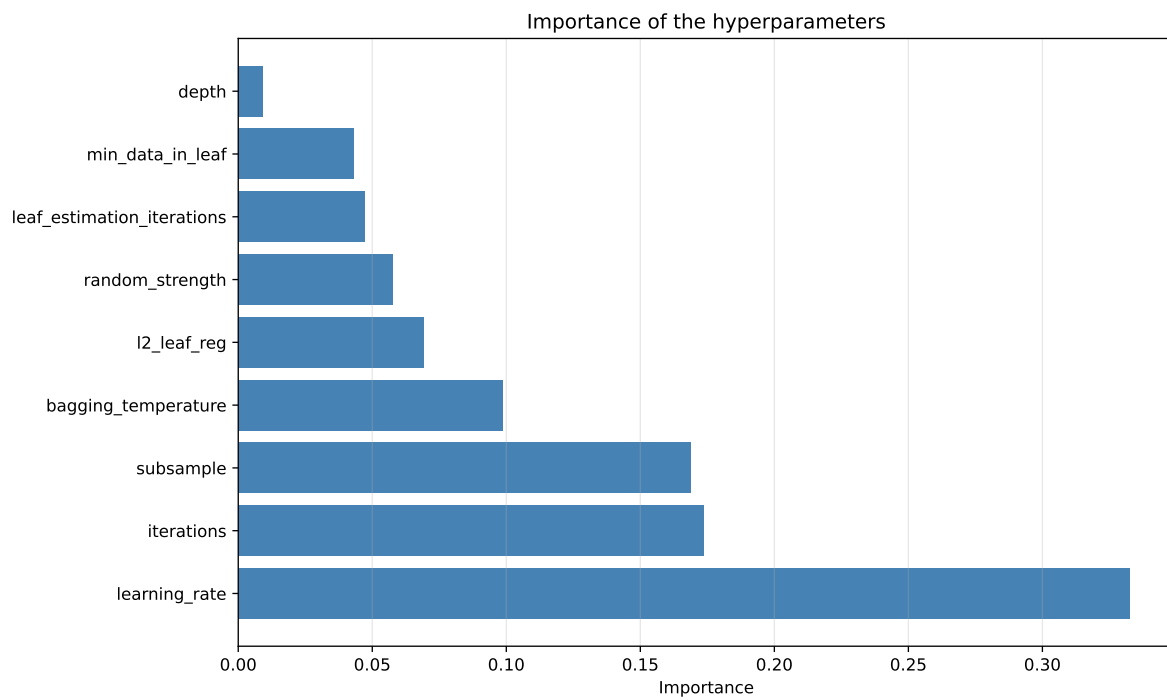
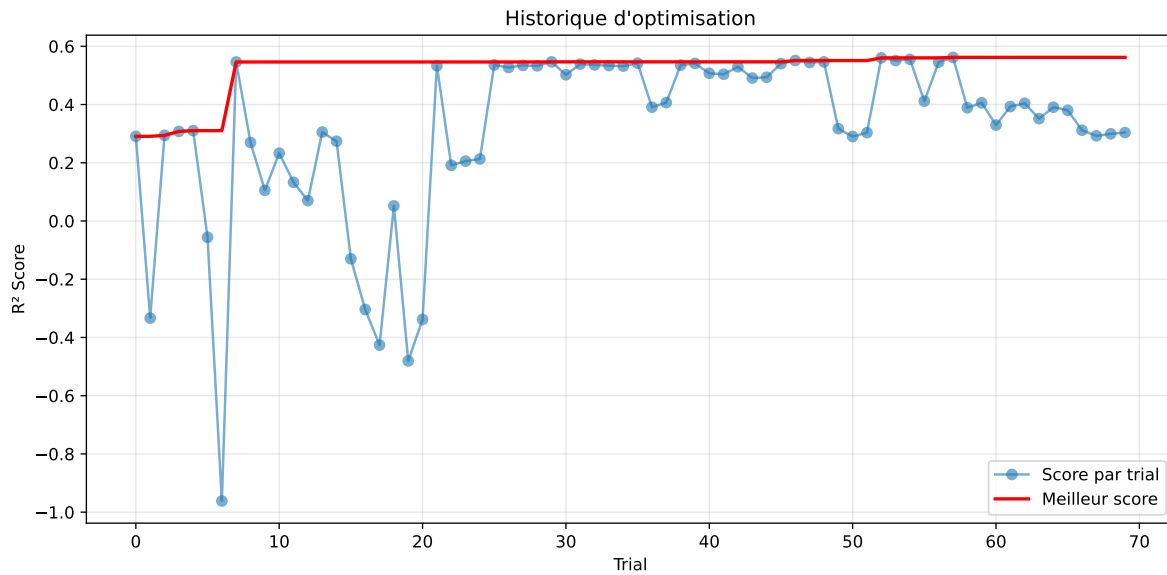
2.5 Model : CatBoost

2.5.1 Gridsearch — Group: 1

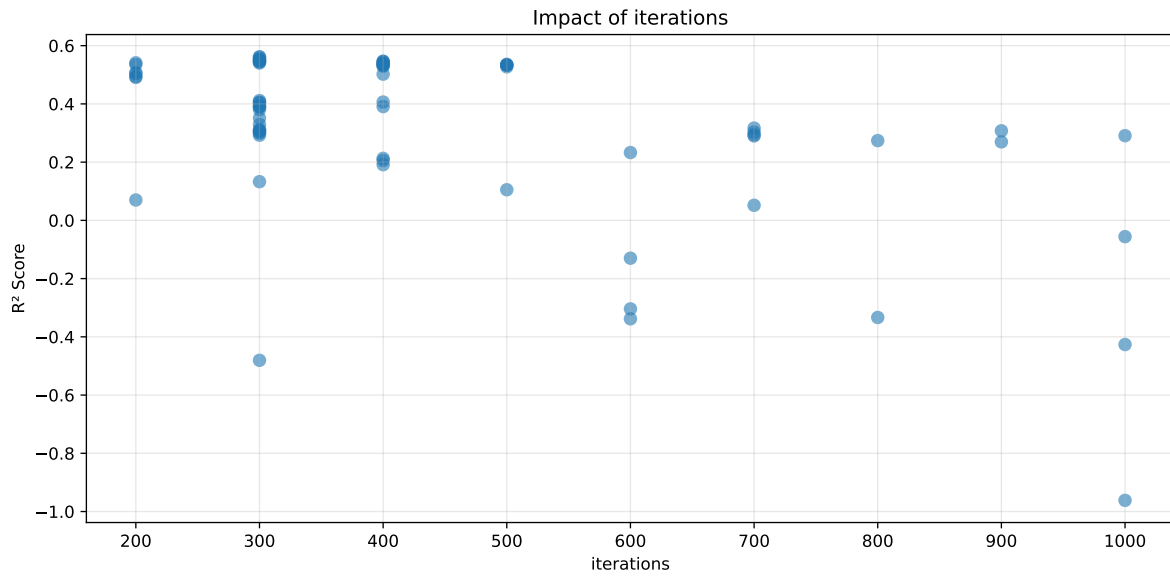
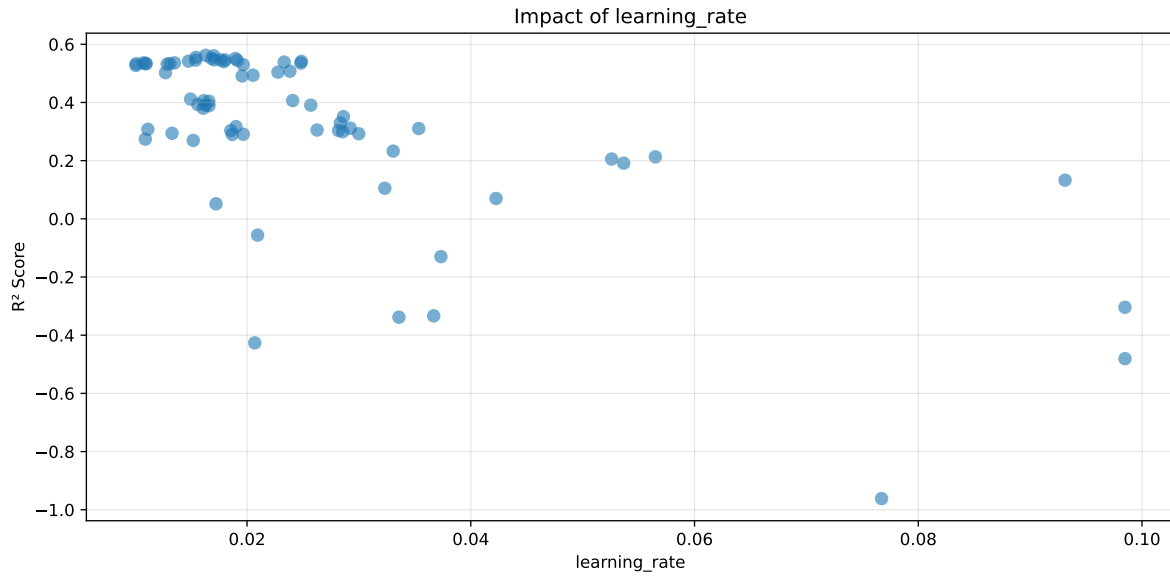
Len(group) : 5452, Len(training) : 3817, Len(test) : 1635

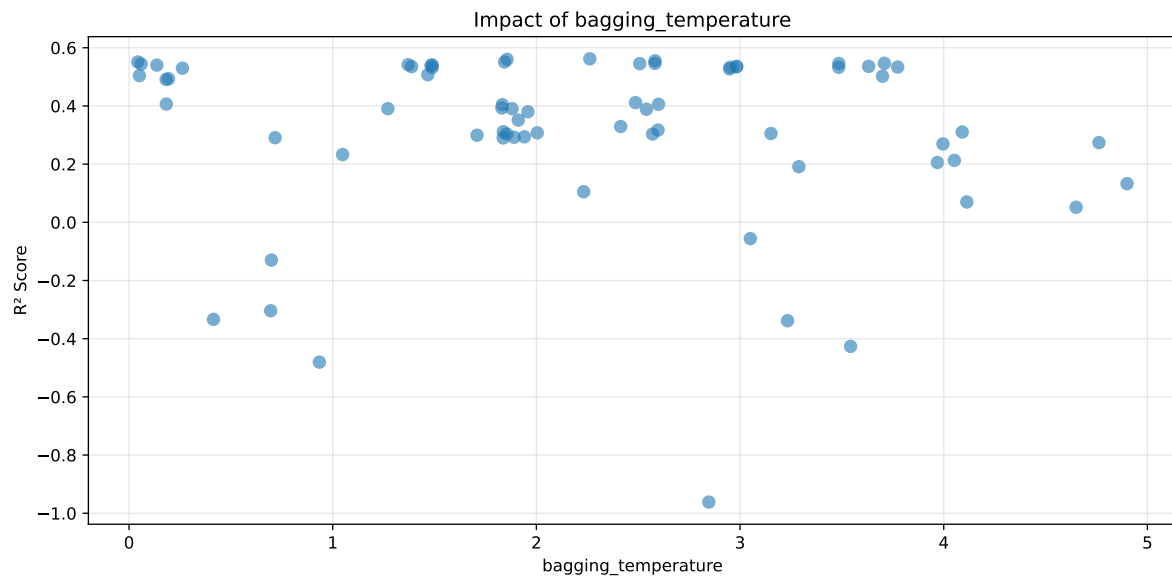
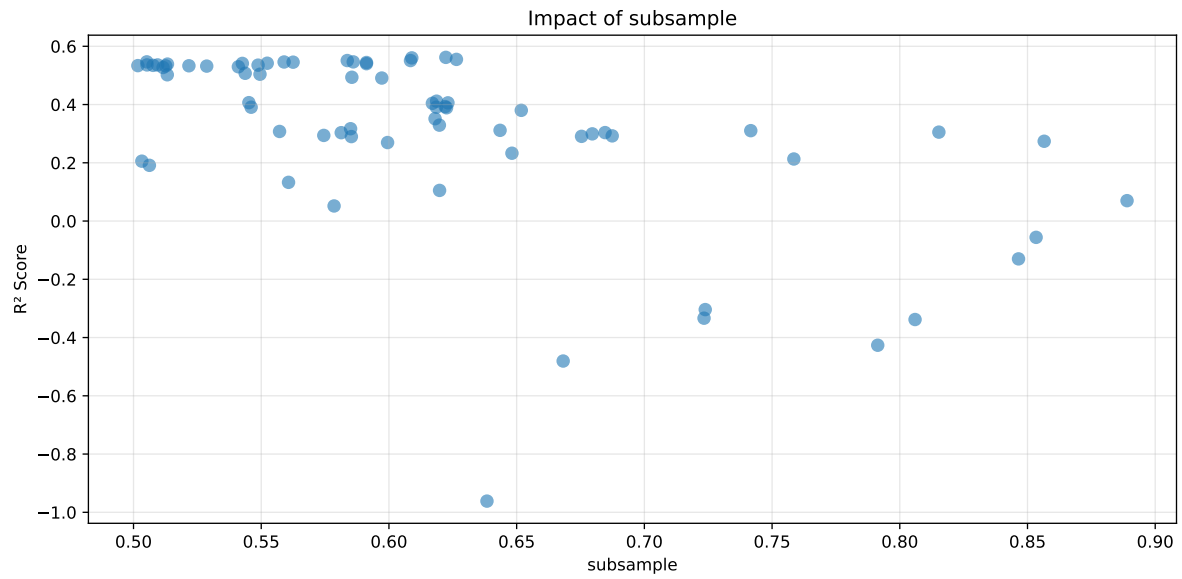
Distribution y_train vs y_test: y_train: mean=2.389, std=0.817, min=0.973, max=13.845
y_test: mean=2.385, std=0.771, min=1.088, max=7.277

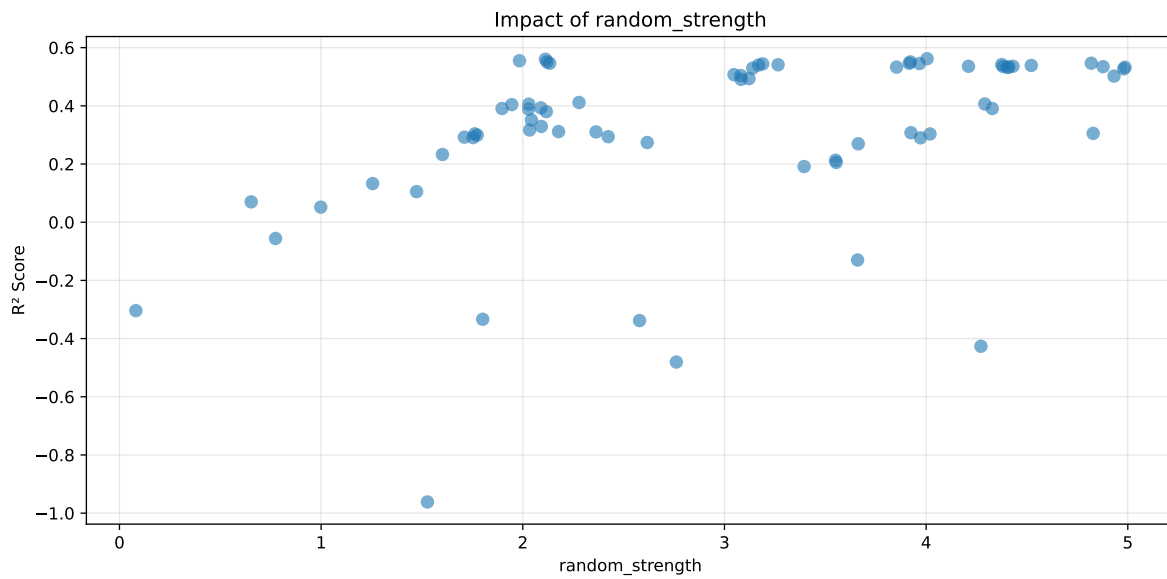
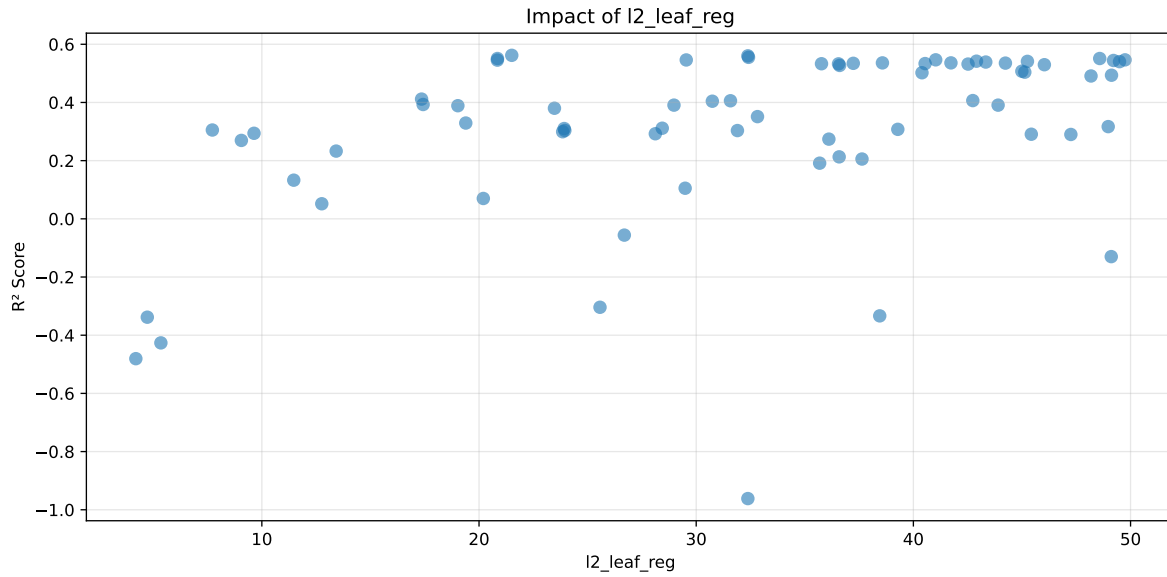
===== TOP Importance FEATURES =====
feature importance r_NH3 32.946287 r_umidity 32.766653 r_CO2 16.203904 r_temperature
9.921513 r_THI 8.161643



Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'subsample', 'bagging_temperature', 'l2_leaf_reg', 'random_strength']

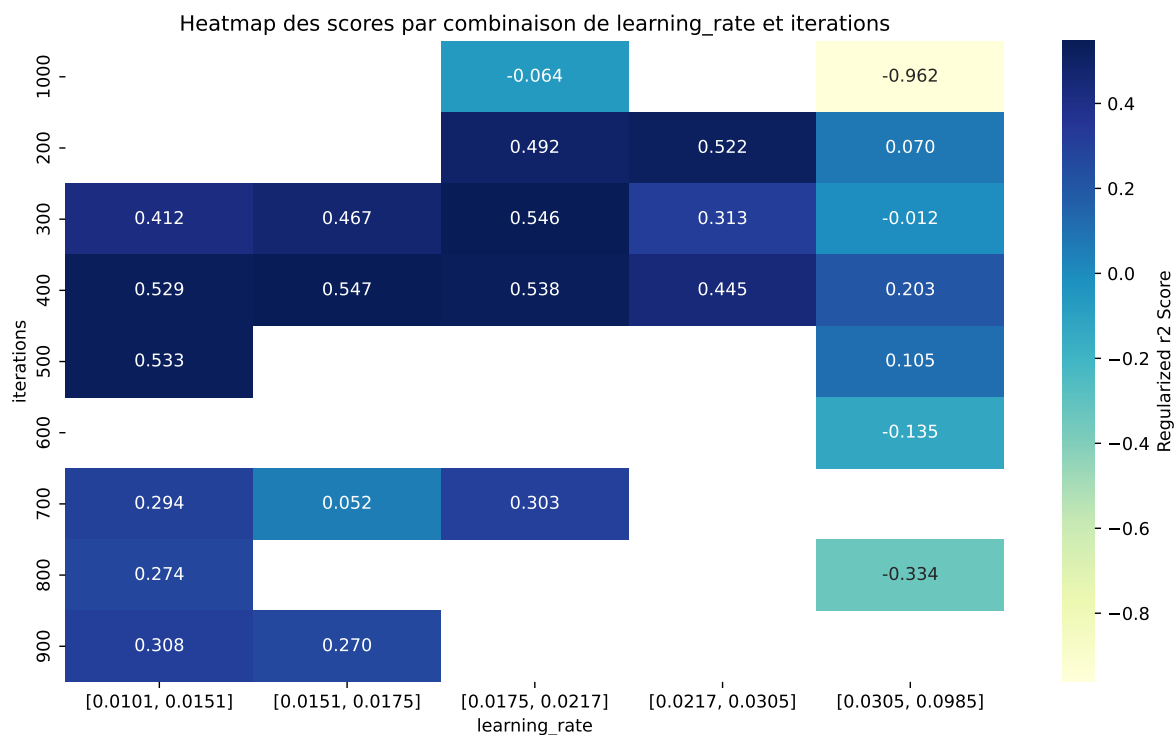






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5887	0.562	0.6102	0.0383	0.562	0.6199	0.058	1.1032	1.0475
2	0.5881	0.5602	0.6104	0.0397	0.5602	0.618	0.0578	1.1031	1.0497
3	0.5813	0.555	0.6047	0.039	0.555	0.6142	0.0592	1.1067	1.0472
4	0.5739	0.5511	0.6027	0.0377	0.5511	0.6169	0.0658	1.1194	1.0414
5	0.5747	0.551	0.5991	0.0374	0.551	0.6073	0.0563	1.1021	1.0428
6	0.5665	0.5465	0.5923	0.0394	0.5465	0.6002	0.0537	1.0982	1.0366
7	0.5681	0.5464	0.5975	0.0372	0.5464	0.6093	0.0628	1.115	1.0396
8	0.5679	0.5461	0.5972	0.0375	0.5461	0.607	0.0609	1.1115	1.0399
9	0.5671	0.5454	0.5923	0.0373	0.5454	0.6011	0.0556	1.102	1.0397
10	0.5631	0.5443	0.595	0.0365	0.5443	0.6074	0.0631	1.1159	1.0346
---	---	---	---	---	---	---	---	---	---
70	0.9421	0.6248	0.6589	0.0487	-0.9616	0.6804	0.0556	1.089	1.5078

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.016301668679146737, 'reg__l2_leaf_reg': 21.50751508580694, 'reg__bagging_temperature': 2.2622829349408864, 'reg__random_strength': 4.004724950159576, 'reg__subsample': 0.6222804816488697, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 4}

Rank 2: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.01702499175008761, 'reg__l2_leaf_reg': 32.37992748504099, 'reg__bagging_temperature': 1.8560353997616779, 'reg__random_strength': 2.1124867956177957, 'reg__subsample': 0.6089516800566582, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 4}

Rank 3: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.015430812363959783, 'reg__l2_leaf_reg': 32.4081034850778, 'reg__bagging_temperature': 2.5827717945521886, 'reg__random_strength': 1.983444001828512, 'reg__subsample': 0.6264280291891242, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 4}

Rank 4: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.018960084907365934, 'reg__l2_leaf_reg': 48.57965639959339, 'reg__bagging_temperature': 0.04314191140661339, 'reg__random_strength': 2.1200301413053344, 'reg__subsample': 0.5836782559737748, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 2}

Rank 5: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.016827767763258752, 'reg__l2_leaf_reg': 20.84953522488785, 'reg__bagging_temperature': 1.8438194966669532, 'reg__random_strength': 3.9216049126399093, 'reg__subsample': 0.60844449409026, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 4}

Rank 6: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.017073931400579924, 'reg__l2_leaf_reg': 41.027931875503064, 'reg__bagging_temperature': 3.7089799586040195, 'reg__random_strength': 4.818850533485142, 'reg__subsample': 0.5051994763595675, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 10}

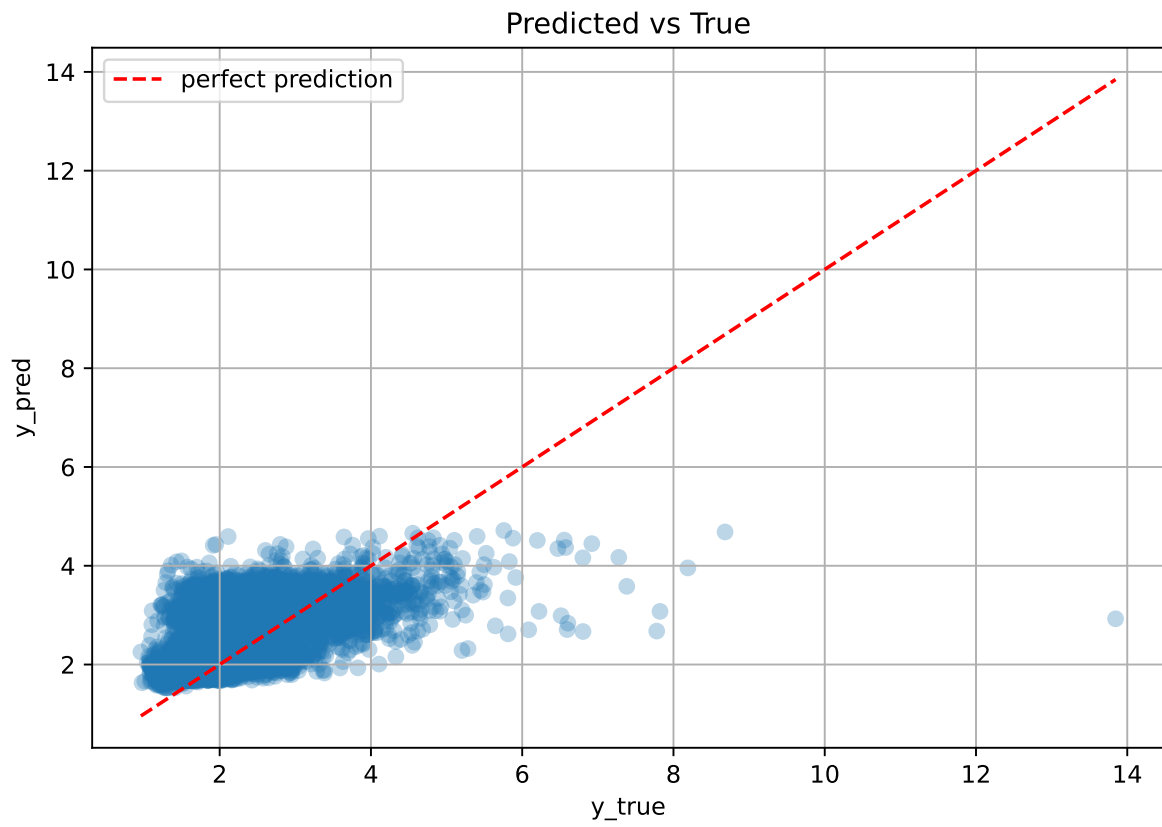
Rank 7: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.017656604056325013, 'reg__l2_leaf_reg': 49.745857729160065, 'reg__bagging_temperature': 2.58189720207527, 'reg__random_strength': 2.1335446180909114, 'reg__subsample': 0.5860250092737784, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 2}

Rank 8: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.018054816314517677, 'reg__l2_leaf_reg': 29.54198876072114, 'reg__bagging_temperature': 3.4850165877834325, 'reg__random_strength': 3.9160841395023036, 'reg__subsample': 0.5589554247319288, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 1}

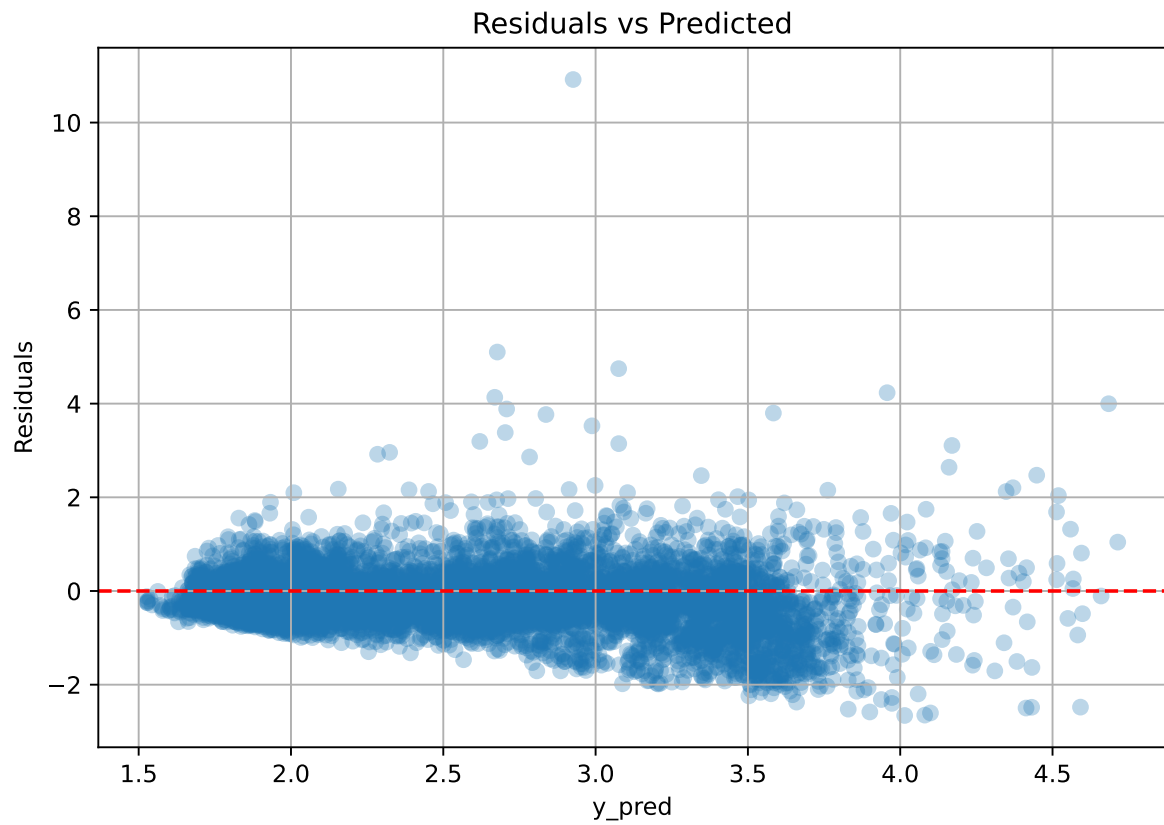
Rank 9: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.015406501038717388, 'reg__l2_leaf_reg': 20.84537782963081, 'reg__bagging_temperature': 2.5072893347690965, 'reg__random_strength': 3.966838661736141, 'reg__subsample': 0.5624812619411286, 'reg__min_data_in_leaf': 32, 'reg__leaf_estimation_iterations': 4}

Rank 10: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.0191417583118441, 'reg__l2_leaf_reg': 49.215808976092866, 'reg__bagging_temperature': 0.05951941898965174, 'reg__random_strength': 3.1896068558497386, 'reg__subsample': 0.5912114380873442, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 2}

2.5.2 Scatter



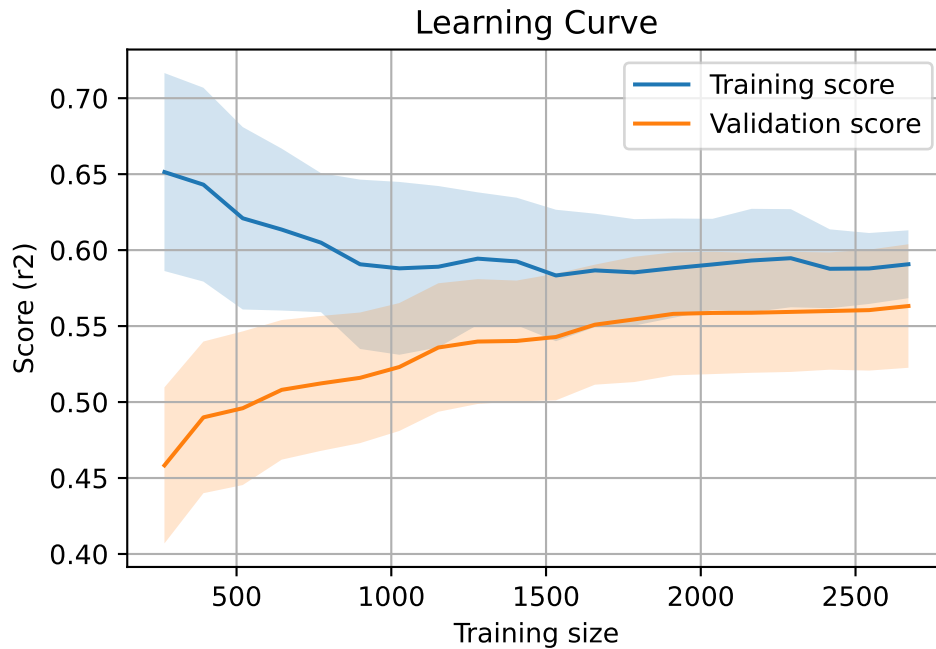
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 1

`Len(group)` : 5452, `Len(training)` : 3817,

`Len(training_real)` : 2672, `Len(test_of_training)` : 1145, `Len(test)` : 1635

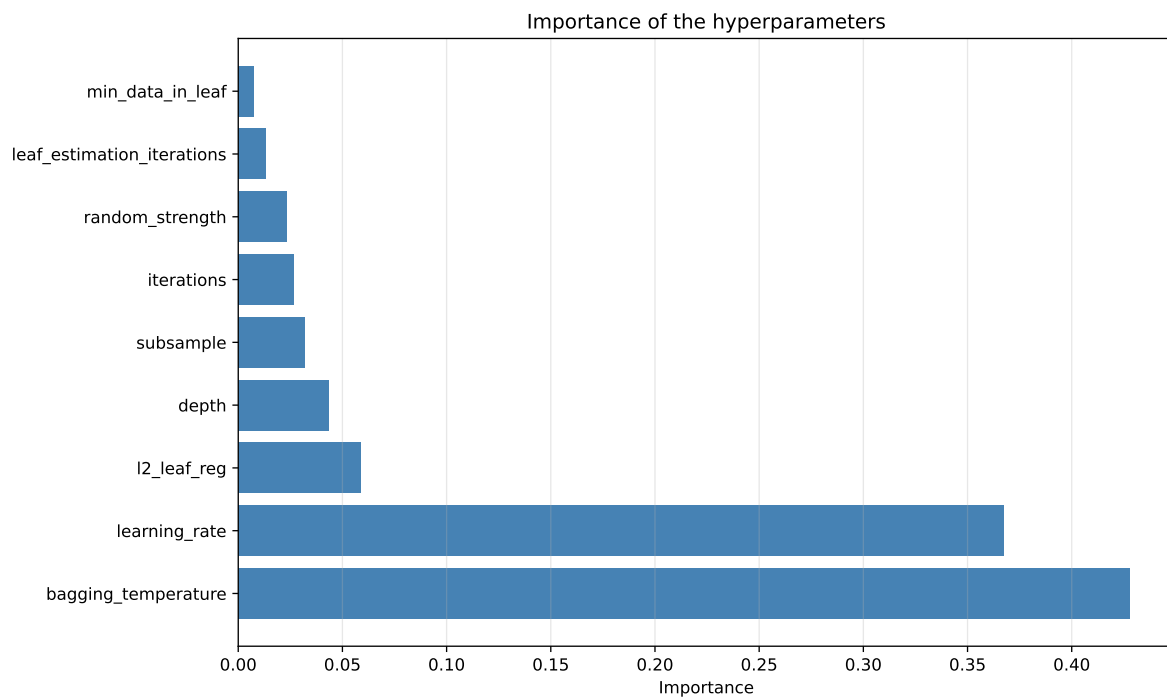
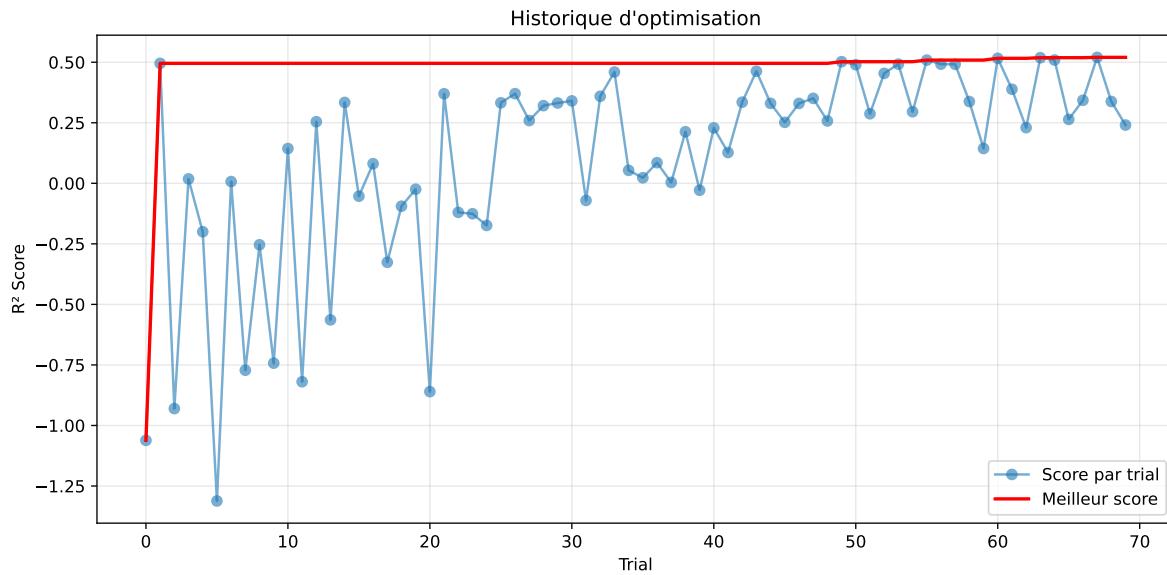


2.5.5 Gridsearch — Group: 2

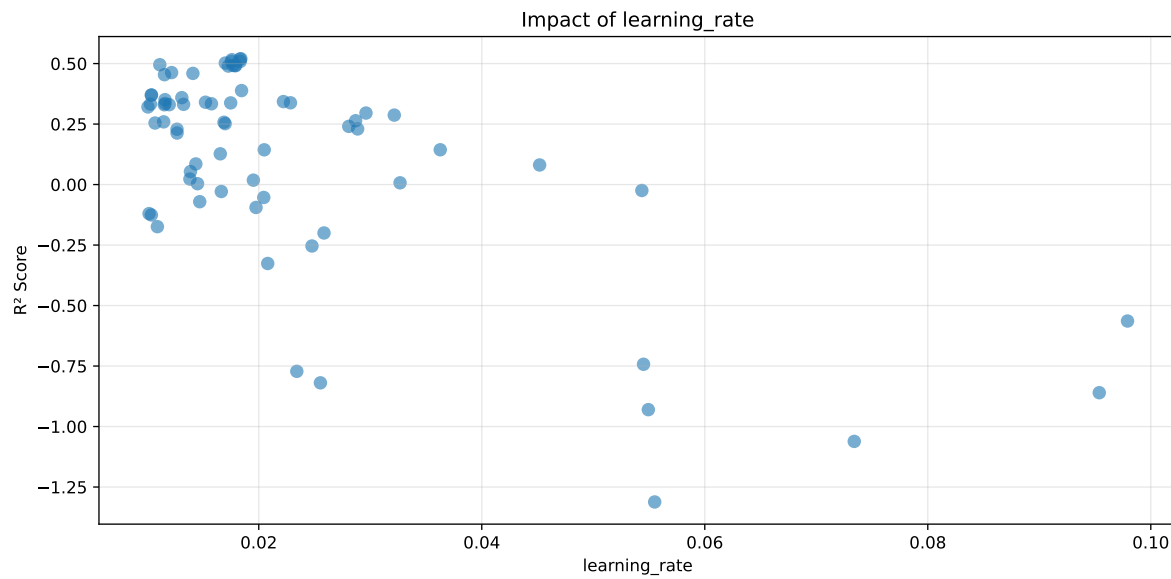
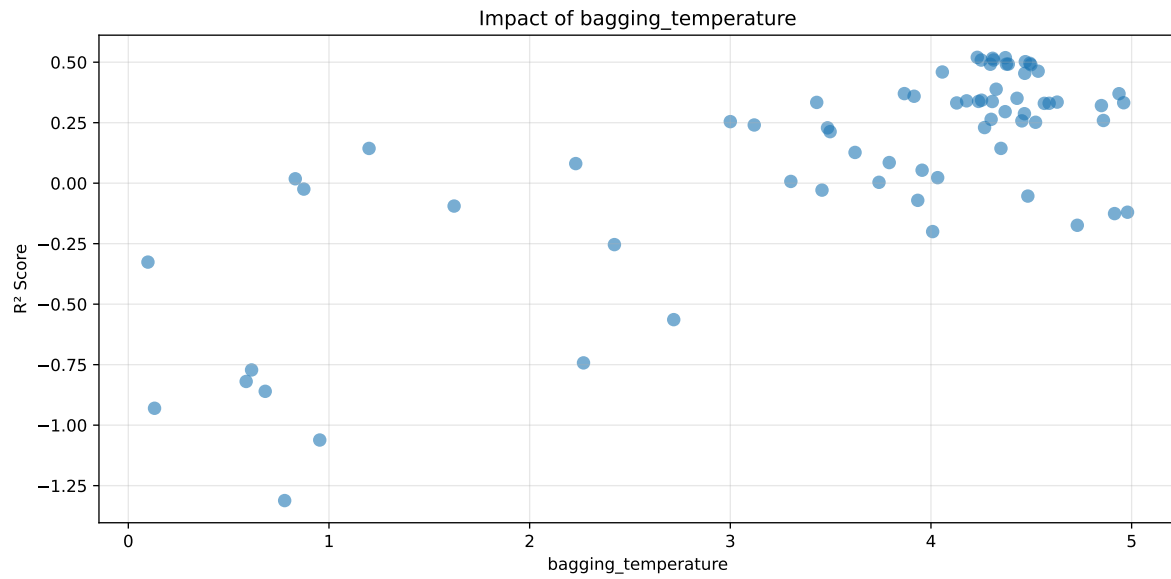
Len(group) : 2716, Len(training) : 1902, Len(test) : 814

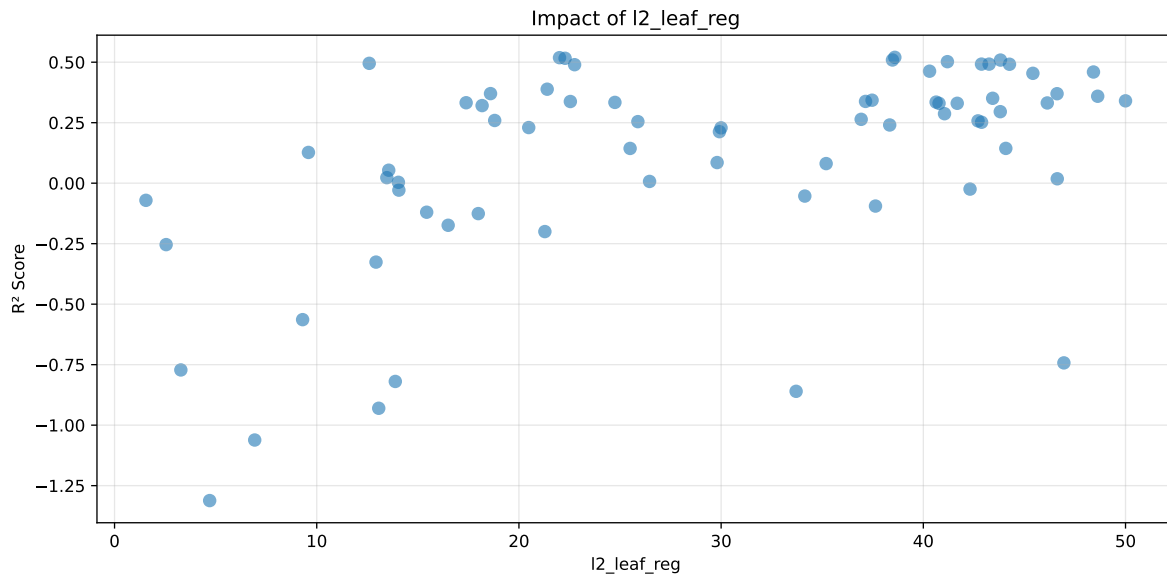
Distribution y_train vs y_test: y_train: mean=2.260, std=0.774, min=0.959, max=7.822
y_test: mean=2.299, std=0.764, min=1.068, max=6.514

===== TOP Importance FEATURES =====
feature importance r_umidity 55.523096 r_THI 14.199703 r_CO2 12.529713 r_NH3 11.202837
r_temperature 6.544650



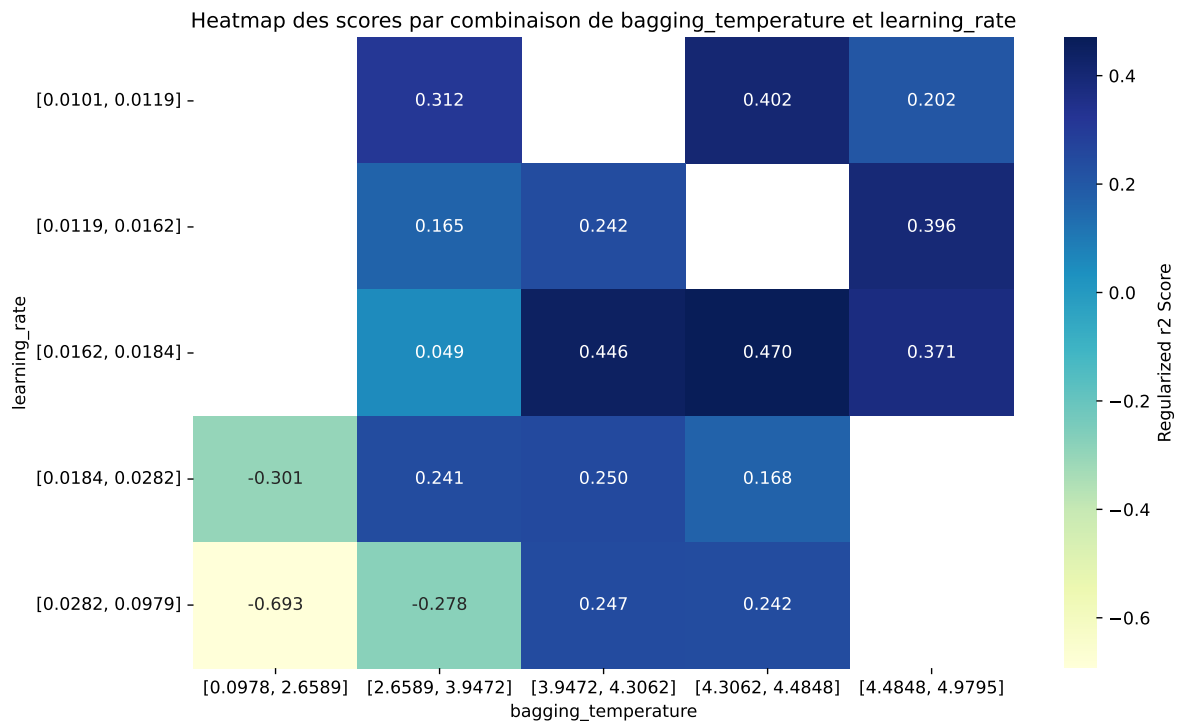
Paramètres avec >5% d'importance : ['bagging_temperature', 'learning_rate', 'l2_leaf_reg']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5426	0.5201	0.4978	0.0335	0.5201	0.5016	-0.0185	0.9645	1.0433
2	0.5443	0.5187	0.5001	0.0337	0.5187	0.5033	-0.0154	0.9703	1.0494
3	0.5386	0.5161	0.4949	0.0326	0.5161	0.5004	-0.0157	0.9696	1.0435
4	0.5257	0.5091	0.4866	0.032	0.5091	0.4914	-0.0177	0.9652	1.0326
5	0.5288	0.5087	0.4845	0.0323	0.5087	0.4915	-0.0172	0.9661	1.0394
6	0.5205	0.5022	0.4693	0.0331	0.5022	0.4761	-0.0261	0.9481	1.0366
7	0.5087	0.4953	0.4655	0.0321	0.4953	0.4701	-0.0252	0.9491	1.027
8	0.5021	0.4921	0.46	0.0326	0.4921	0.4614	-0.0306	0.9378	1.0204
9	0.5038	0.492	0.4607	0.0321	0.492	0.4599	-0.0321	0.9347	1.024
10	0.503	0.4915	0.4595	0.0336	0.4915	0.4624	-0.0291	0.9408	1.0233
---	---	---	---	---	---	---	---	---	---
70	0.9805	0.5985	0.5939	0.0238	-1.3117	0.6171	0.0186	1.0311	1.6383

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.01839566745515397, 'reg__l2_leaf_reg': 38.59036837378155, 'reg__bagging_temperature': 4.230812049675452, 'reg__random_strength': 2.255621661226444, 'reg__subsample': 0.7114117632591499, 'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 9}

Rank 2: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.01831612566789107, 'reg__l2_leaf_reg': 22.007158483711276, 'reg__bagging_temperature': 4.37084482557805, 'reg__random_strength': 2.2955624034718882, 'reg__subsample': 0.8162626542846014, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 9}

Rank 3: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.017593788030480402, 'reg__l2_leaf_reg': 22.28277617590092, 'reg__bagging_temperature': 4.307219930540958, 'reg__random_strength': 2.317303025342273, 'reg__subsample': 0.8201694457670149, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 9}

Rank 4: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.01834564021875643, 'reg__l2_leaf_reg': 38.474340423781236, 'reg__bagging_temperature': 4.3122051812511035, 'reg__random_strength': 3.4404298141058174, 'reg__subsample': 0.7070854821050288, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 9}

Rank 5: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.01755104087434947, 'reg__l2_leaf_reg': 43.81530875664545, 'reg__bagging_temperature': 4.250535172300963, 'reg__random_strength': 2.628257818793262, 'reg__subsample': 0.8221668101845546, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 10}

Rank 6: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01699863003270027, 'reg__l2_leaf_reg': 41.19087365479176, 'reg__bagging_temperature': 4.470132345639243, 'reg__random_strength': 1.2602168072588331, 'reg__subsample': 0.721717254173937, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 10}

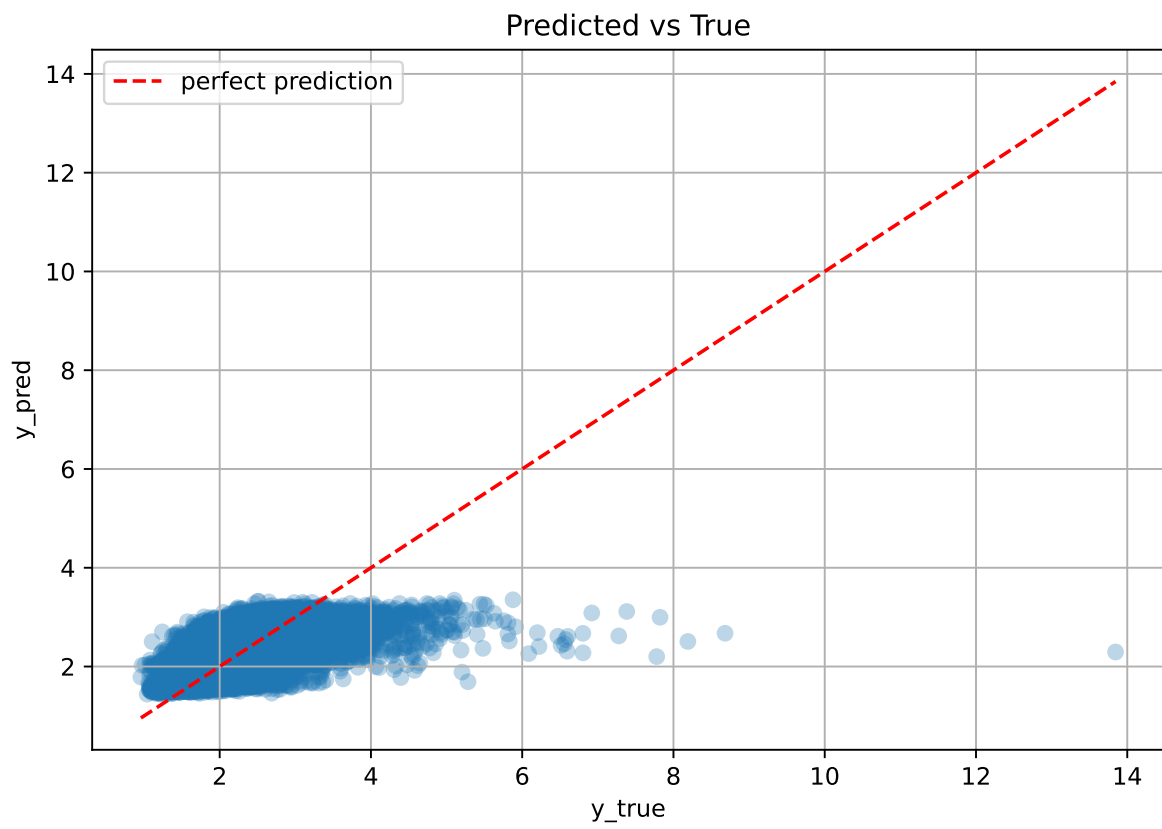
Rank 7: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.011125187538319986, 'reg__l2_leaf_reg': 12.595599823182772, 'reg__bagging_temperature': 4.492774470036819, 'reg__random_strength': 2.5259005124130325, 'reg__subsample': 0.709319370875075, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 10}

Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.017691901435691156, 'reg__l2_leaf_reg': 42.877902964275634, 'reg__bagging_temperature': 4.385427468358155, 'reg__random_strength': 2.533468960244413, 'reg__subsample': 0.7115099268243356, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 10}

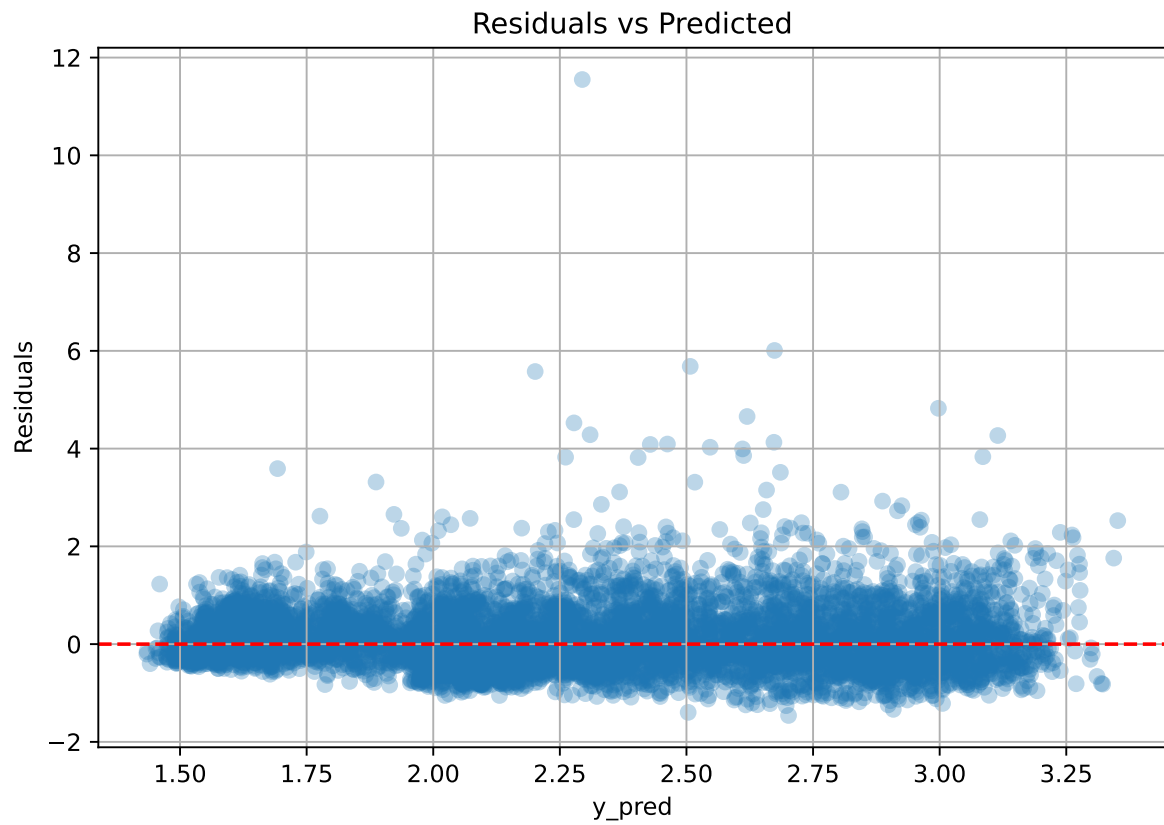
Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01787787984248267, 'reg__l2_leaf_reg': 43.25329372845556, 'reg__bagging_temperature': 4.375948879391668, 'reg__random_strength': 2.4771972103890403, 'reg__subsample': 0.8239757170616564, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 10}

Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.017927664854465033, 'reg__l2_leaf_reg': 44.261069253380526, 'reg__bagging_temperature': 4.295975256192741, 'reg__random_strength': 2.560953834706423, 'reg__subsample': 0.8200972432055151, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 10}

2.5.6 Scatter



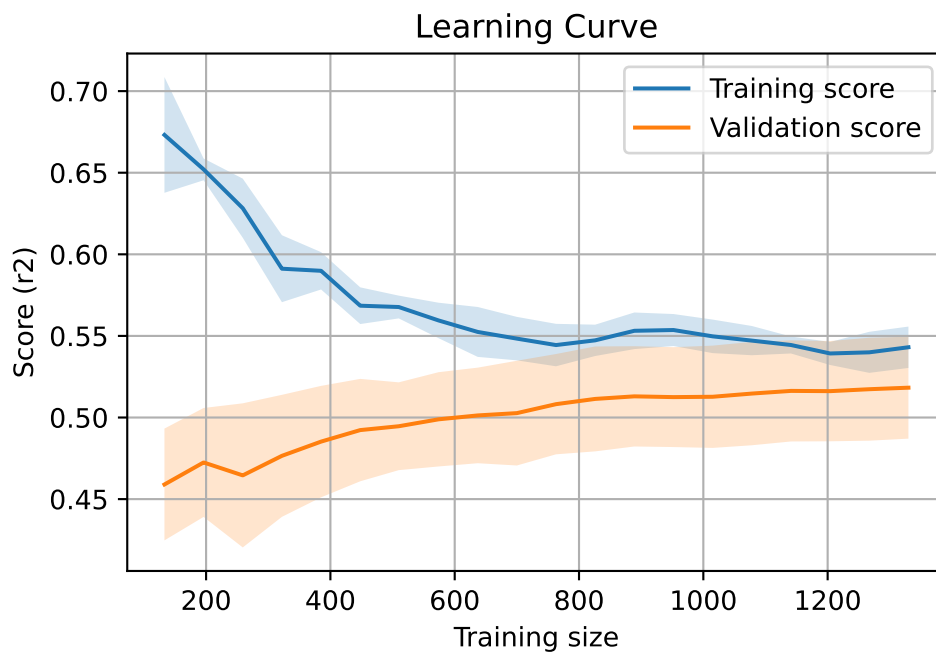
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 2

`Len(group)` : 2716, `Len(training)` : 1902,

`Len(training_real)` : 1332, `Len(test_of_training)` : 570, `Len(test)` : 814



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	1	3817	1635	5452	0.441	0.43	0.035	0.478	0.478	0.048	1.112	1.025	nan
Linear	2	1902	814	2716	0.358	0.378	0.032	0.322	0.323	-0.055	0.856	0.947	nan
Linear	MEAN	5719	2449	8168	0.399	0.404	0.033	0.4	0.401	-0.003	0.984	0.986	nan
LGBM	1	3817	1635	5452	0.544	0.519	0.038	0.568	0.589	0.071	1.136	1.05	0.5186
LGBM	2	1902	814	2716	0.513	0.488	0.039	0.441	0.463	-0.025	0.948	1.05	0.4885
LGBM	MEAN	5719	2449	8168	0.529	0.504	0.039	0.505	0.526	0.023	1.042	1.05	0.504
CatBoost	1	3817	1635	5452	0.589	0.562	0.038	0.61	0.62	0.058	1.103	1.048	0.562
CatBoost	2	1902	814	2716	0.543	0.52	0.034	0.498	0.502	-0.018	0.964	1.043	0.5201
CatBoost	MEAN	5719	2449	8168	0.566	0.541	0.036	0.554	0.561	0.02	1.034	1.045	0.541

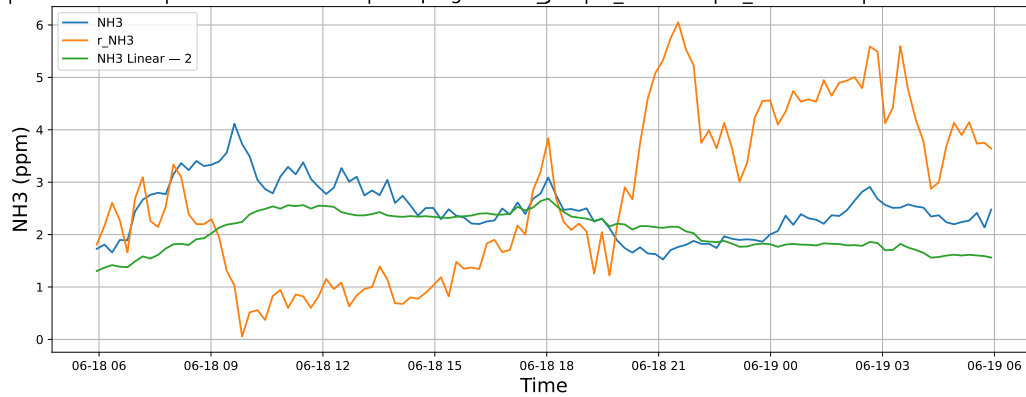
3 Plot

3.1 Standalone plots

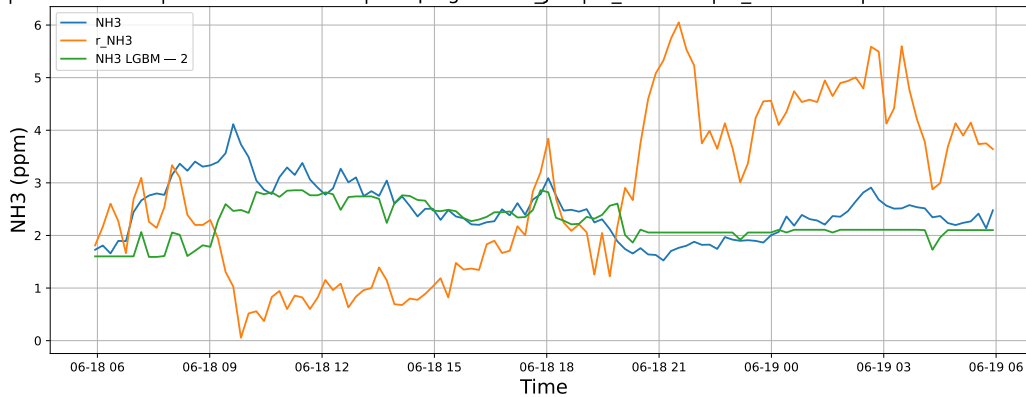
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

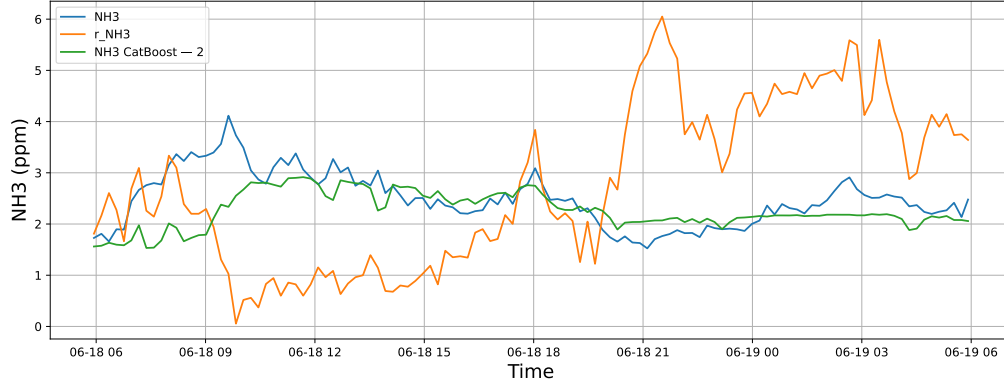
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

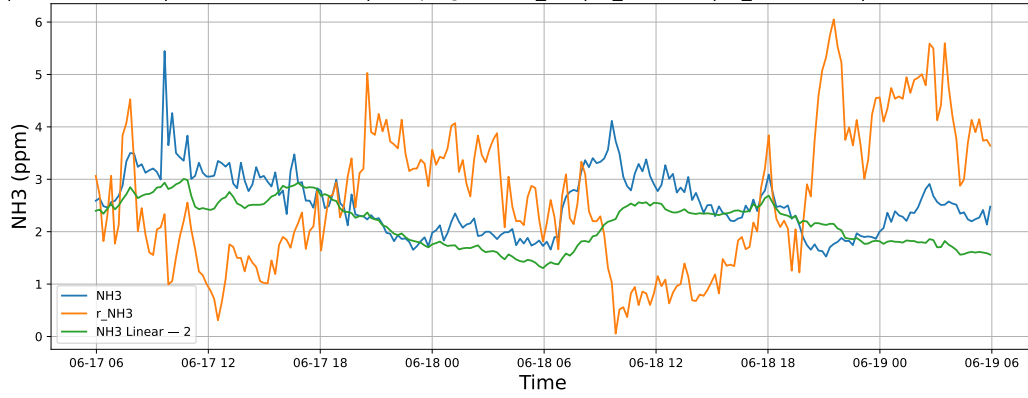


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

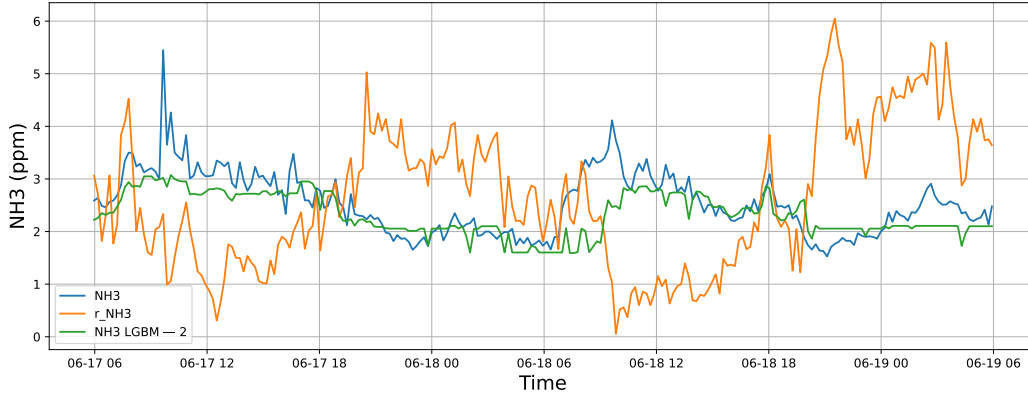


3.1.1.2 Time window : 48

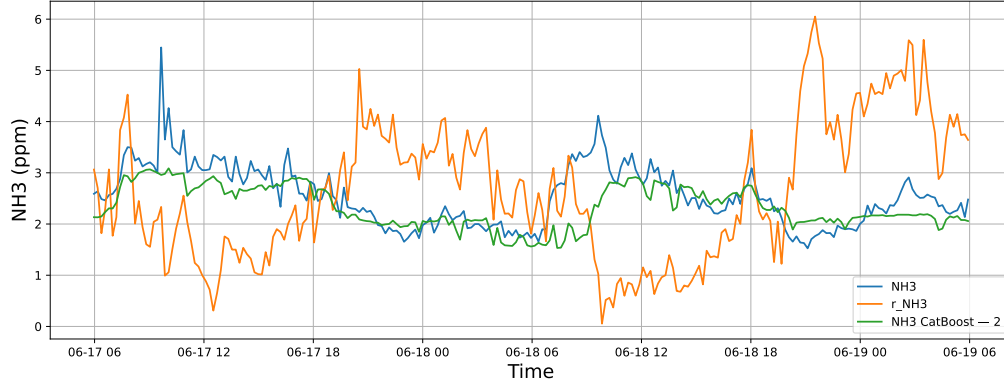
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

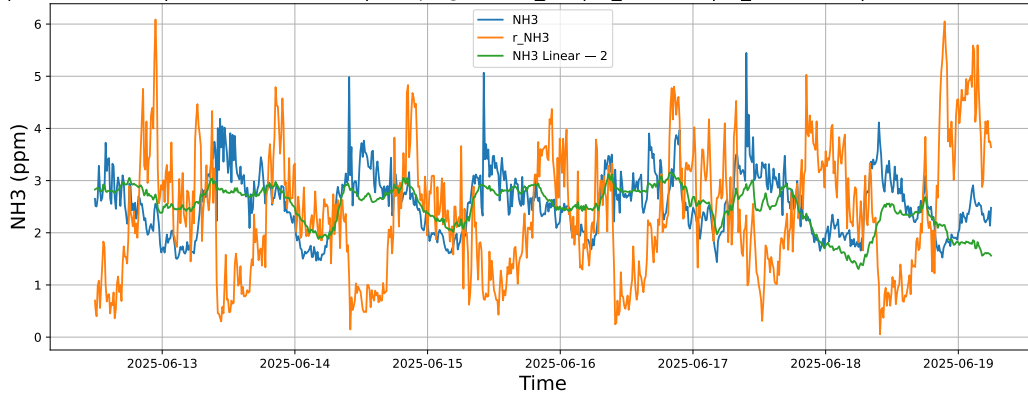


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

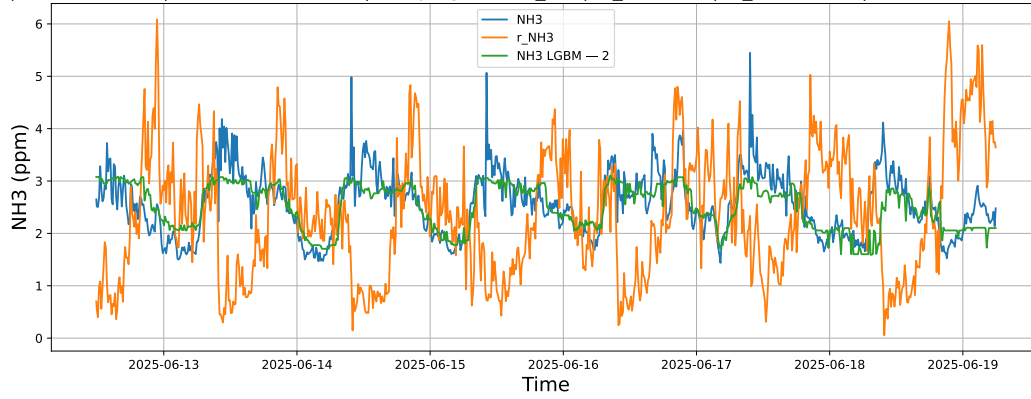


3.1.1.3 Time window : ALL

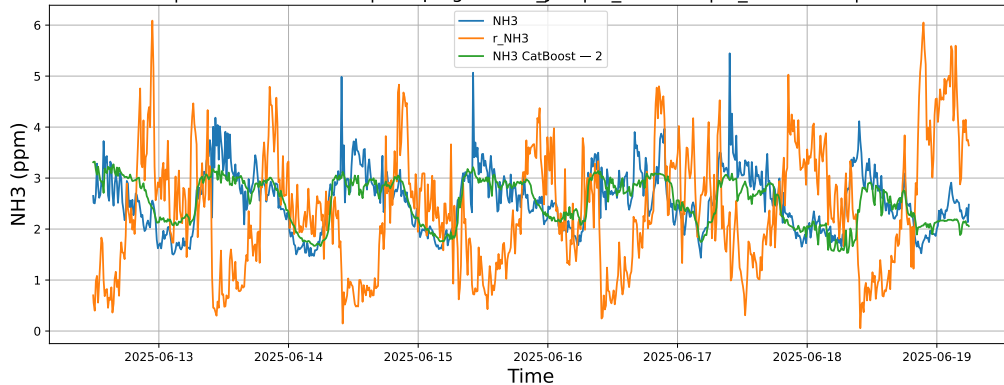
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



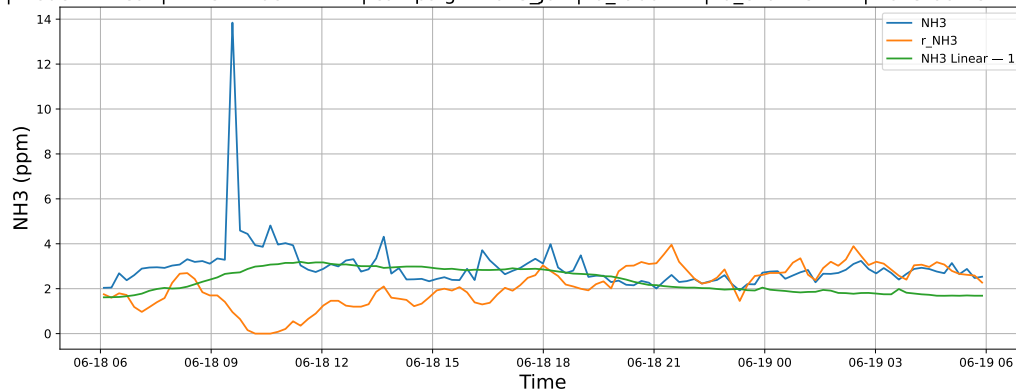
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



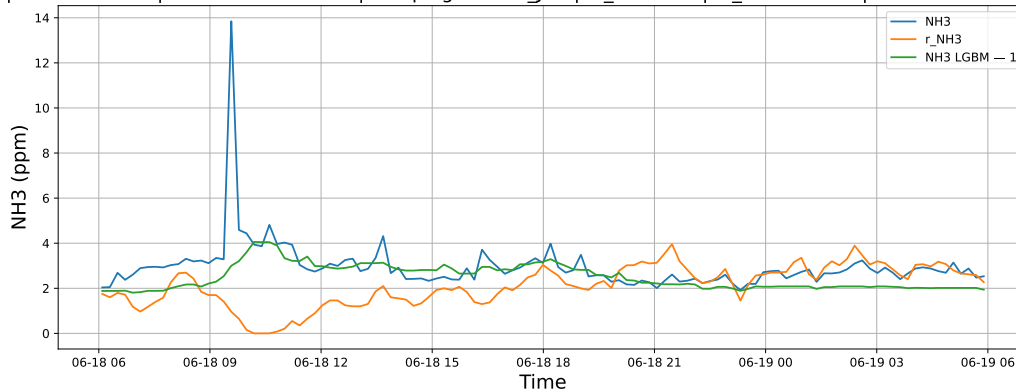
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

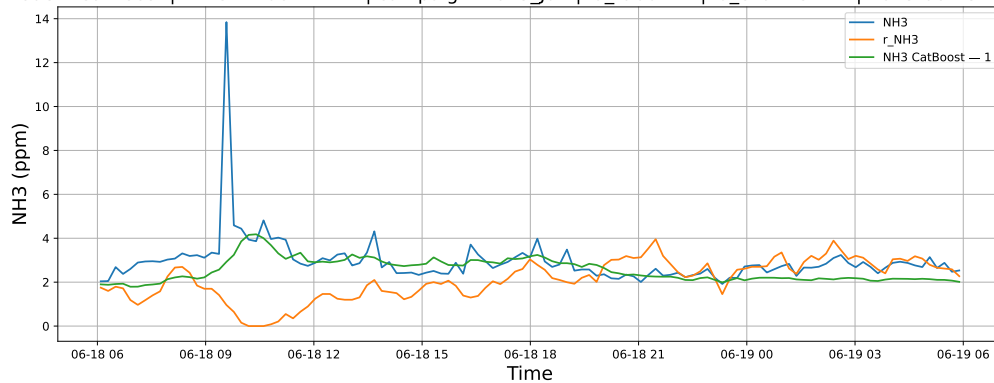
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

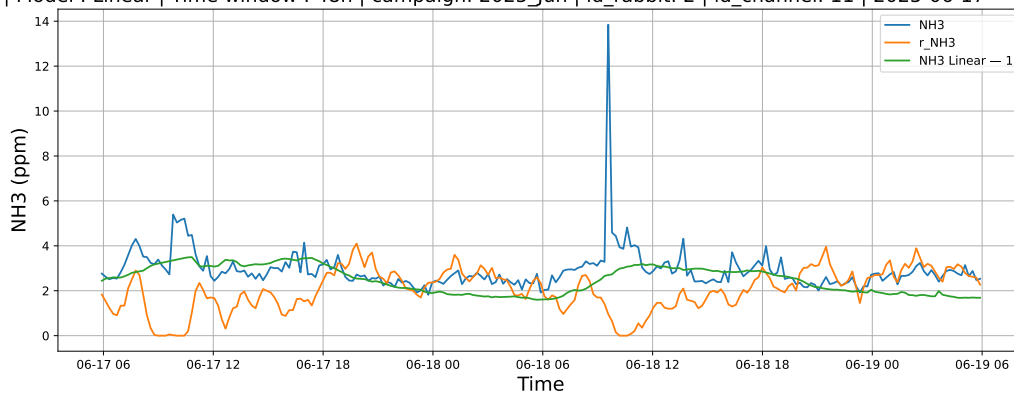


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

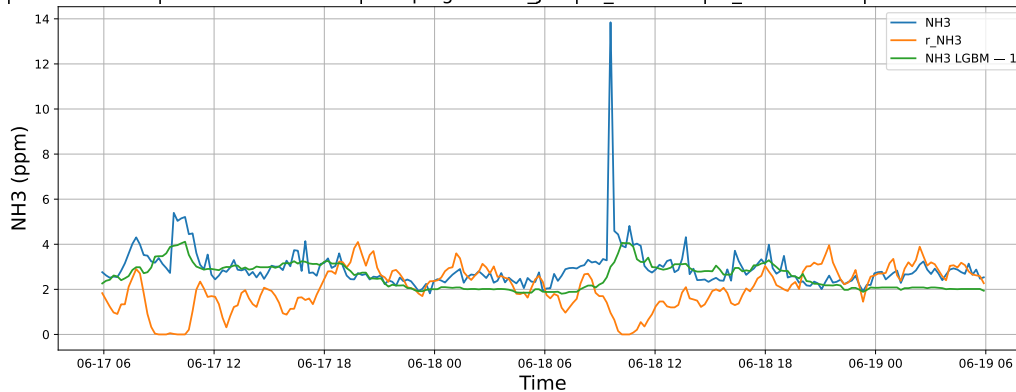


3.1.2.2 Time window : 48

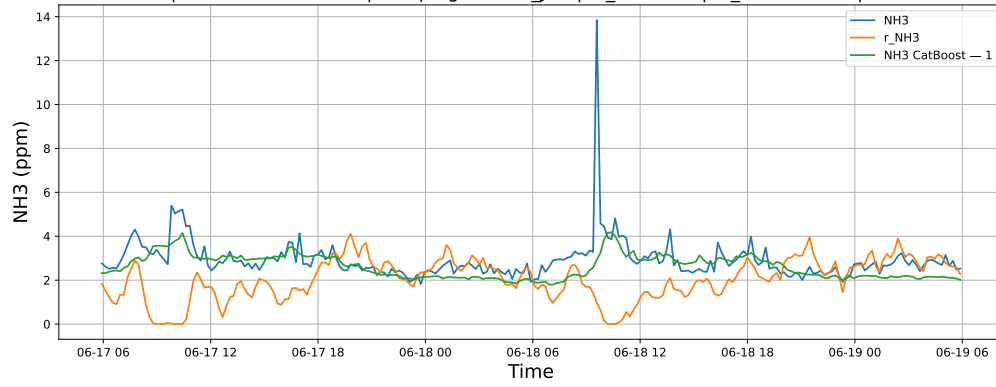
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

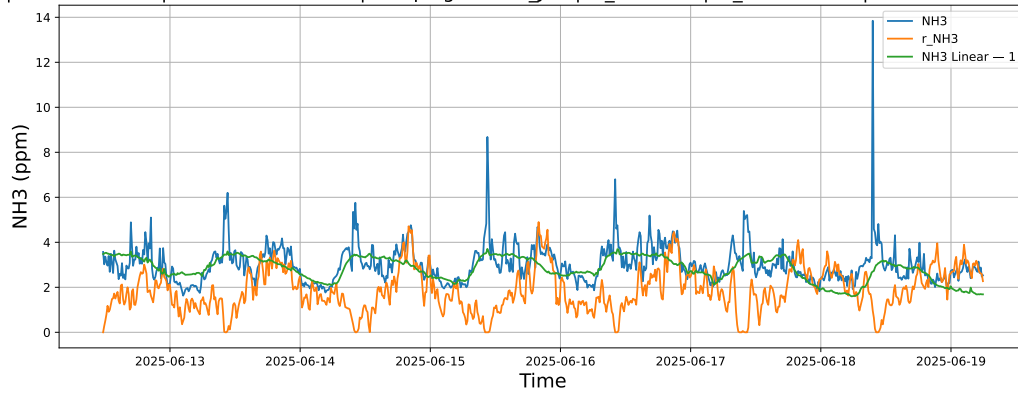


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

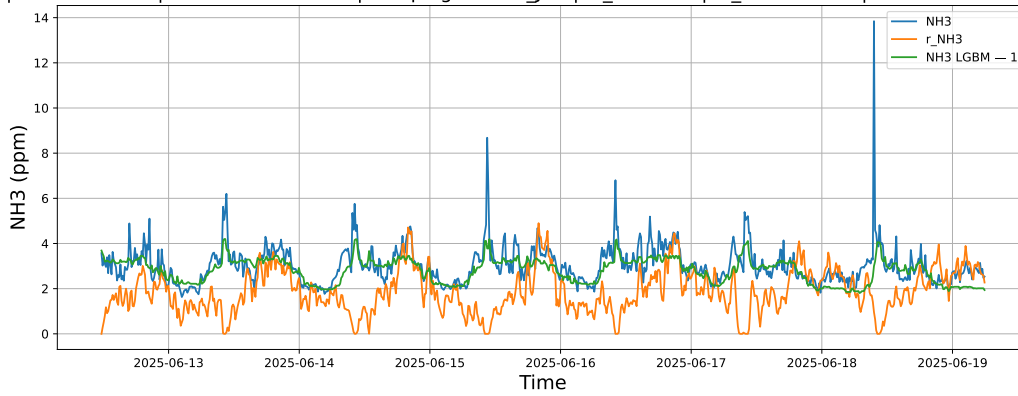


3.1.2.3 Time window : ALL

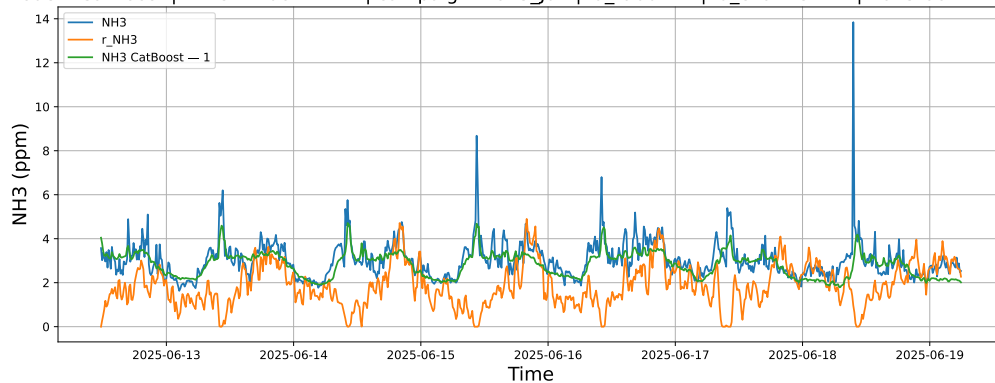
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



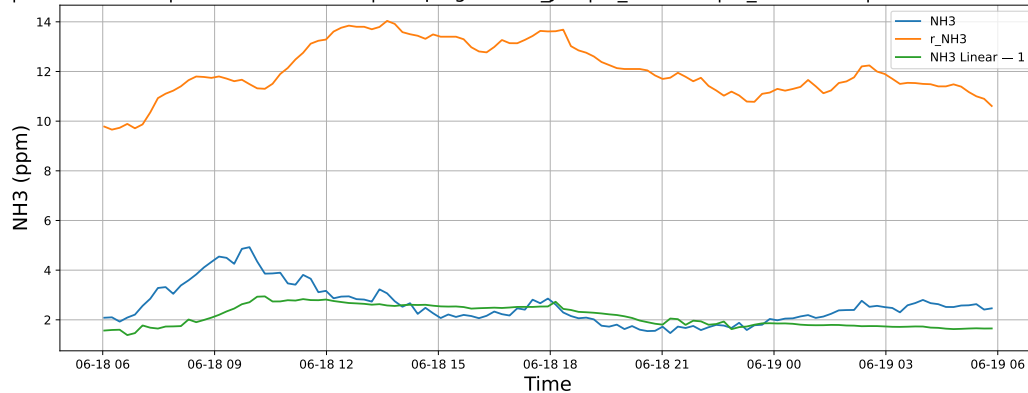
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



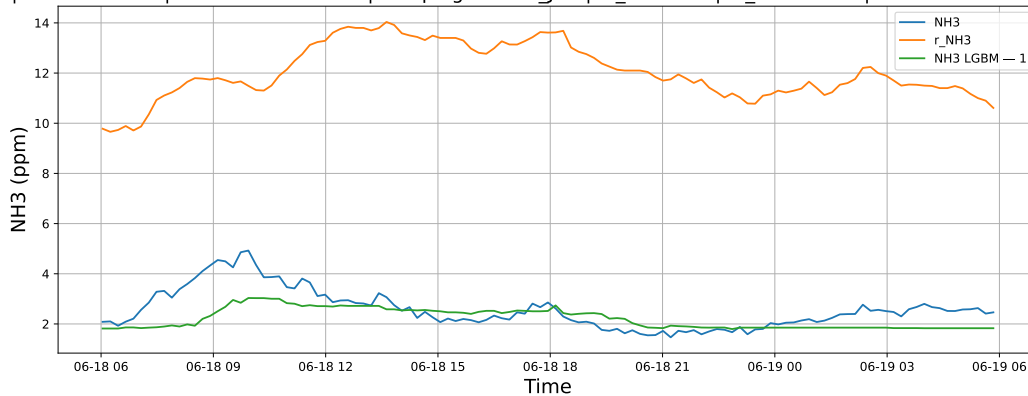
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

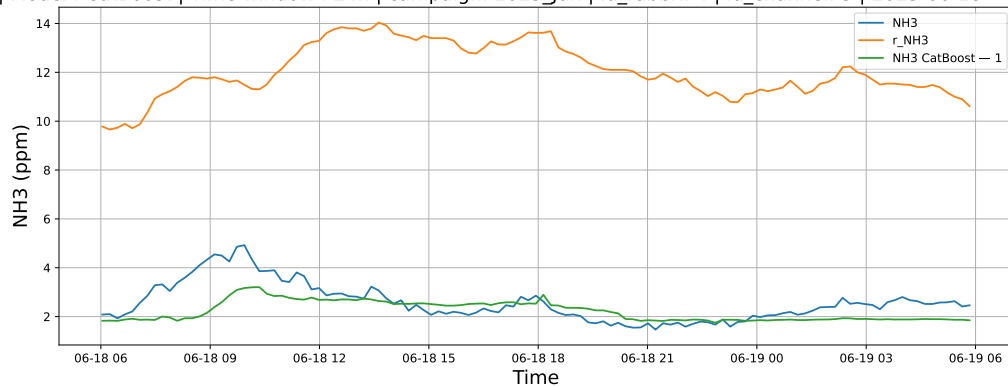
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

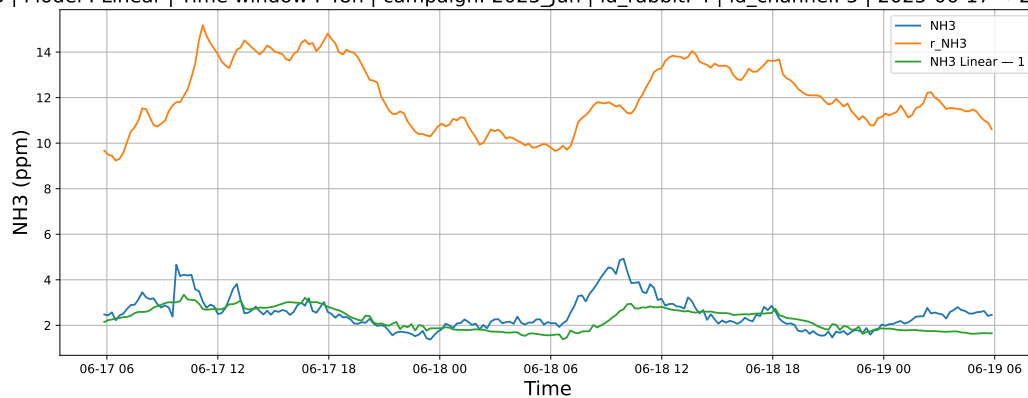


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

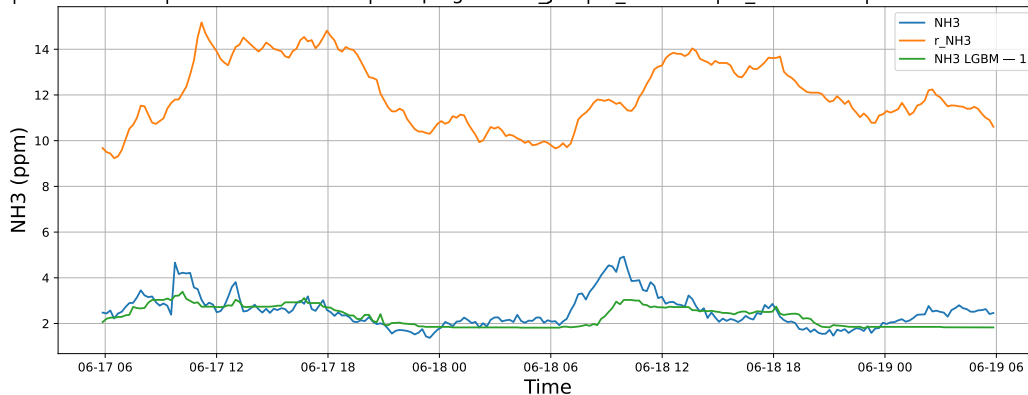


3.1.3.2 Time window : 48

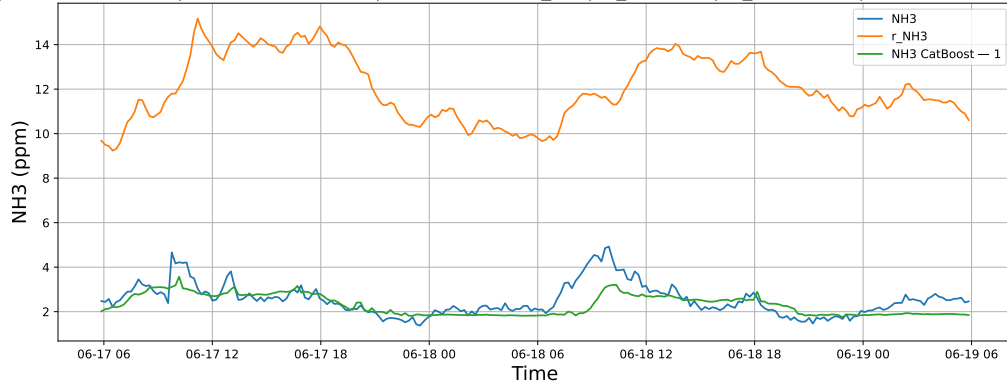
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

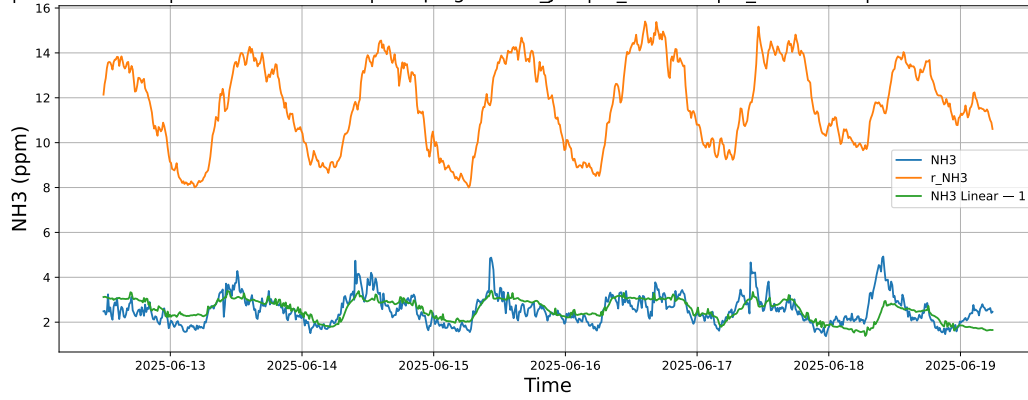


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

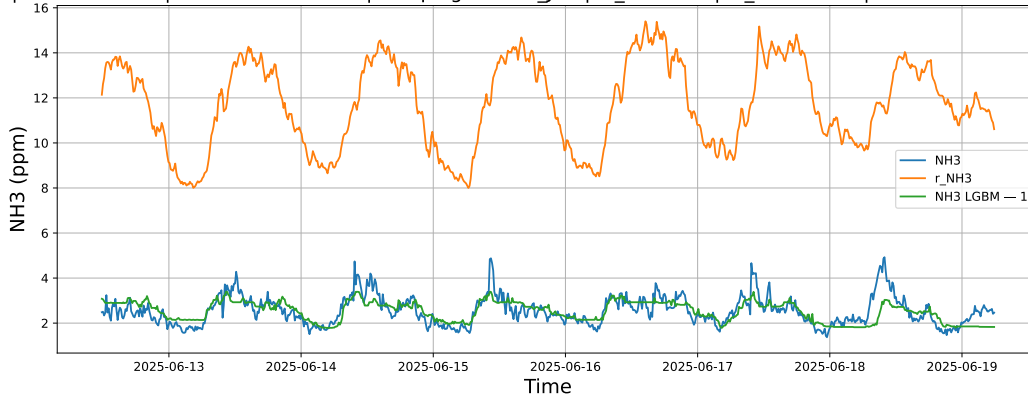


3.1.3.3 Time window : ALL

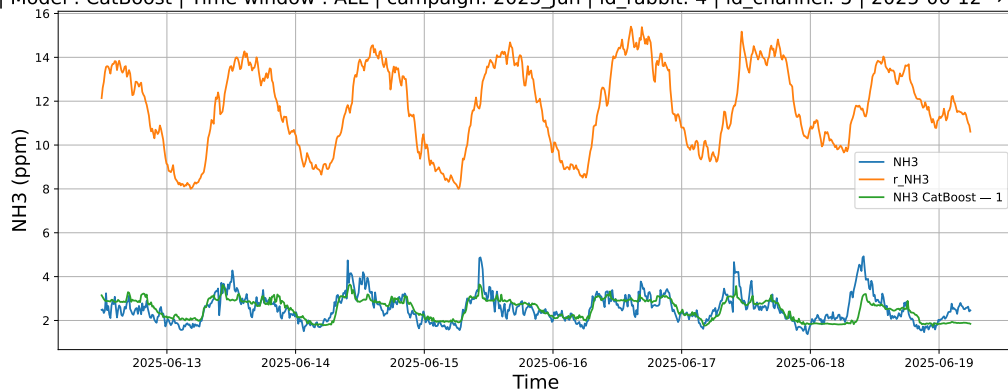
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



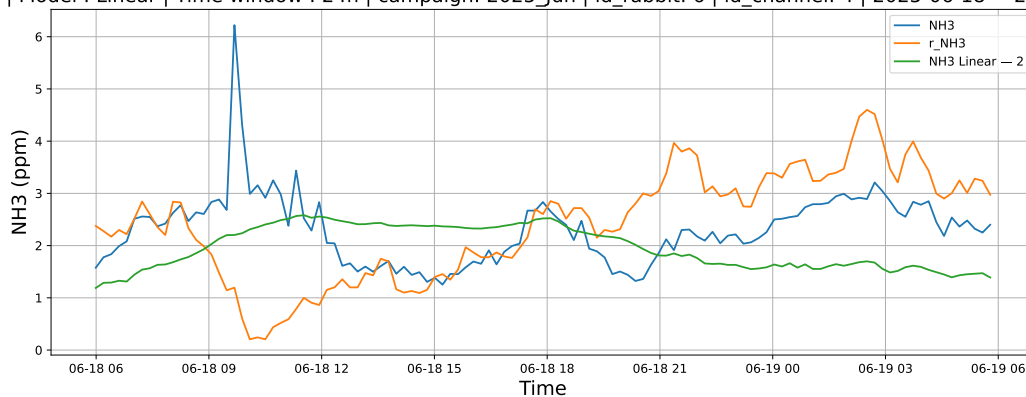
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



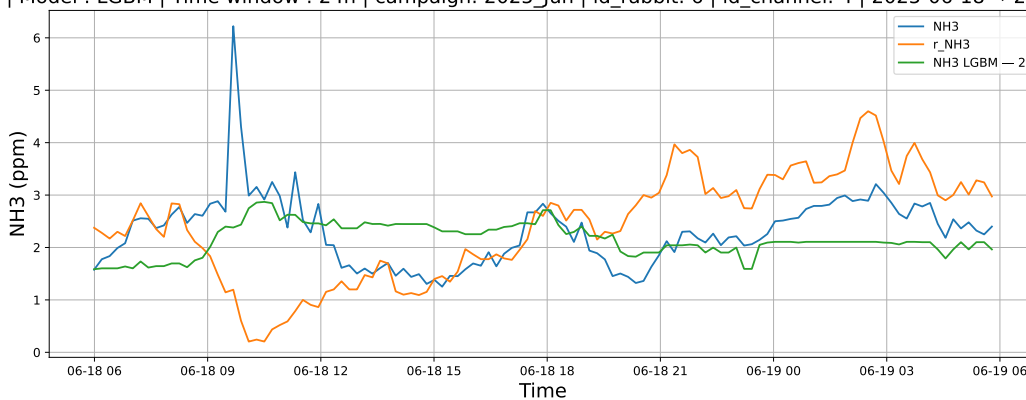
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

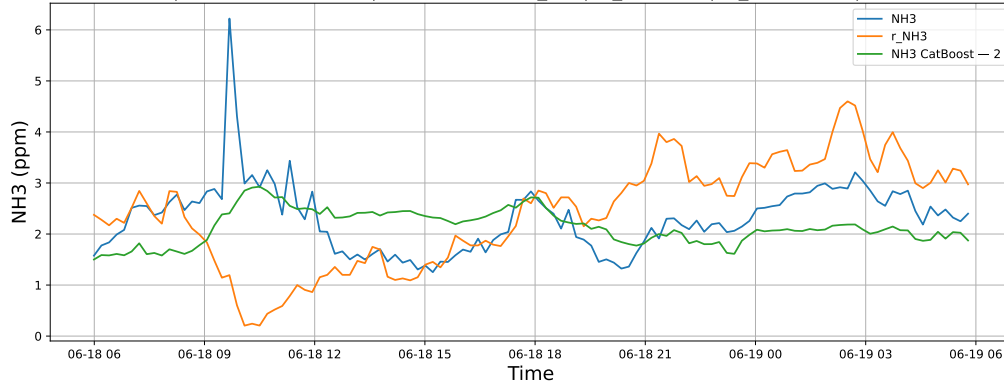
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

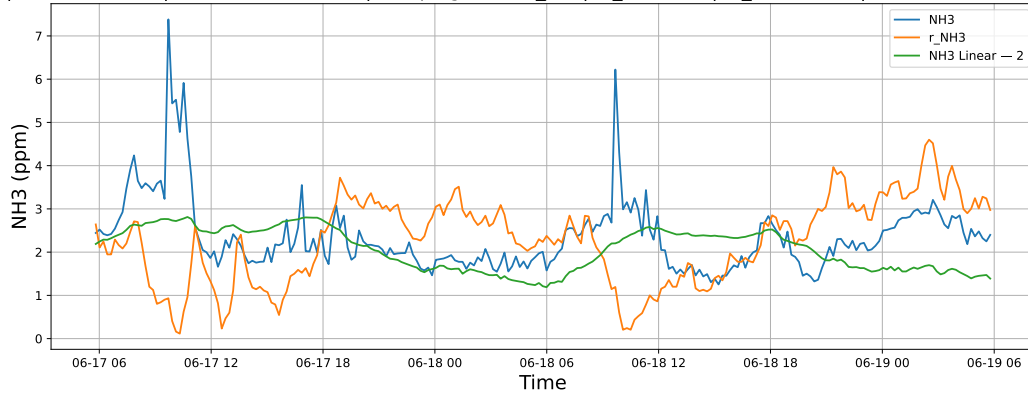


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

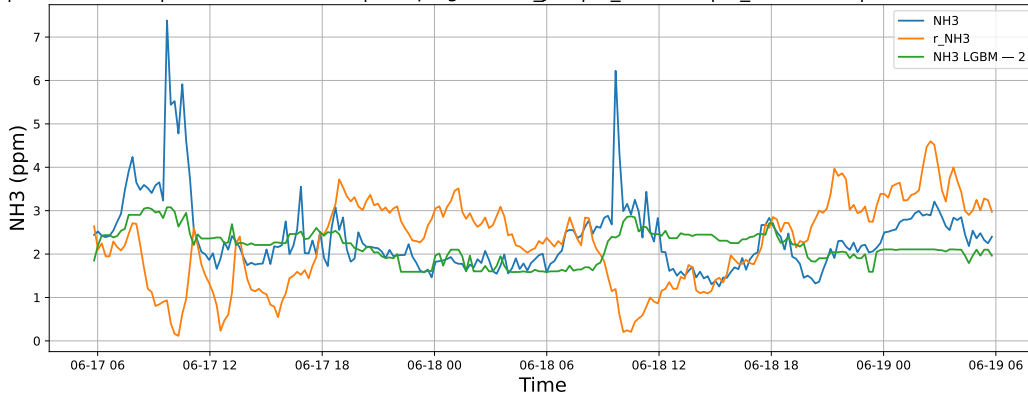


3.1.4.2 Time window : 48

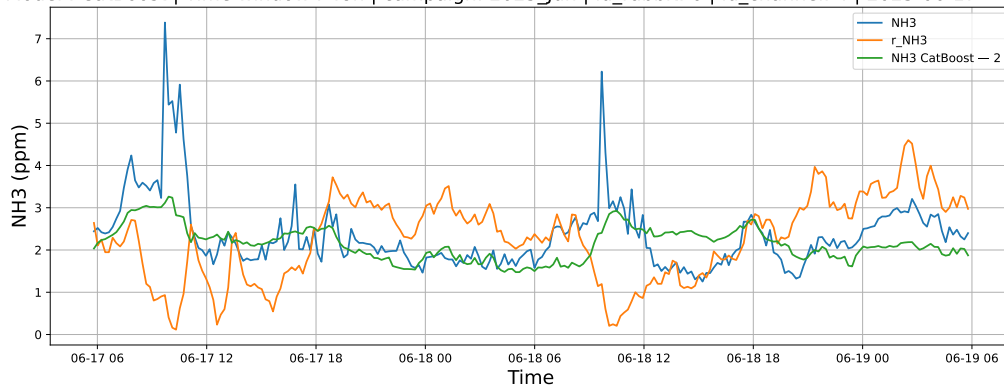
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

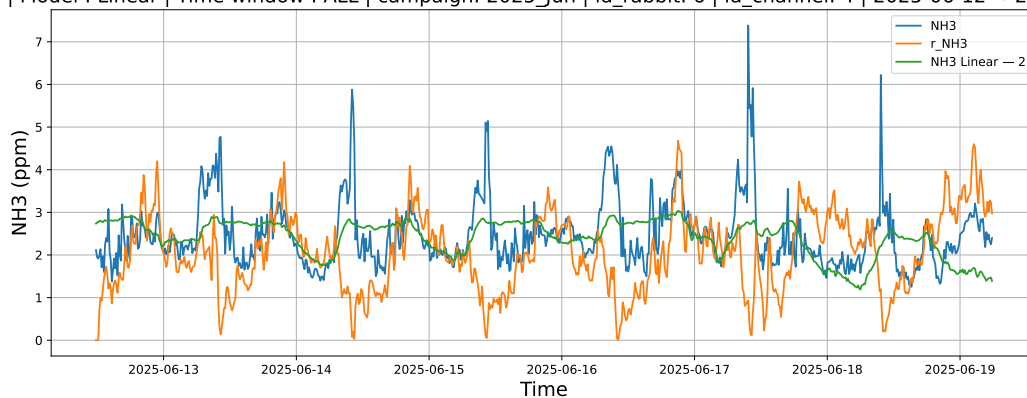


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

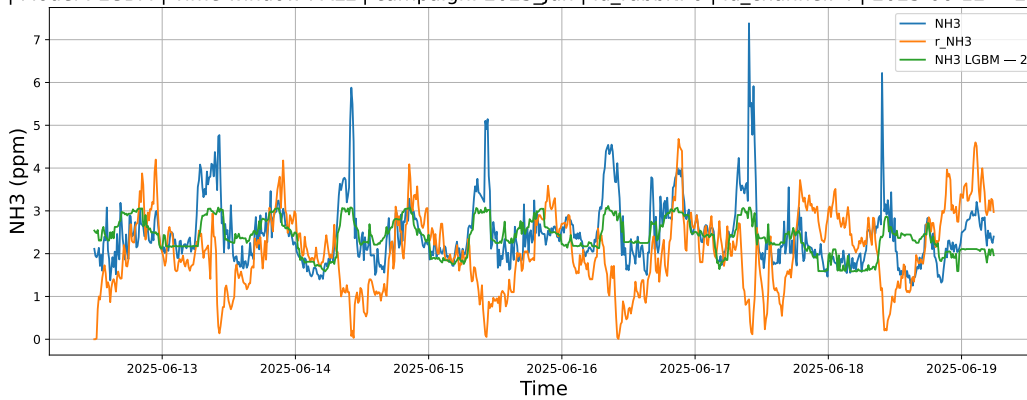


3.1.4.3 Time window : ALL

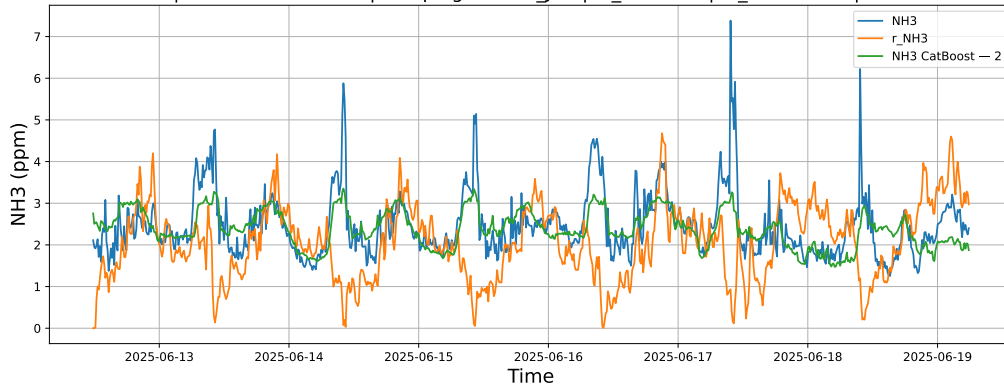
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



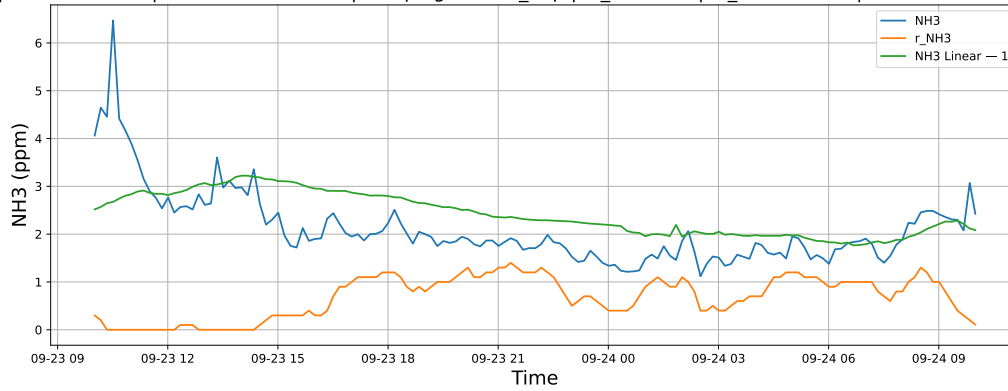
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



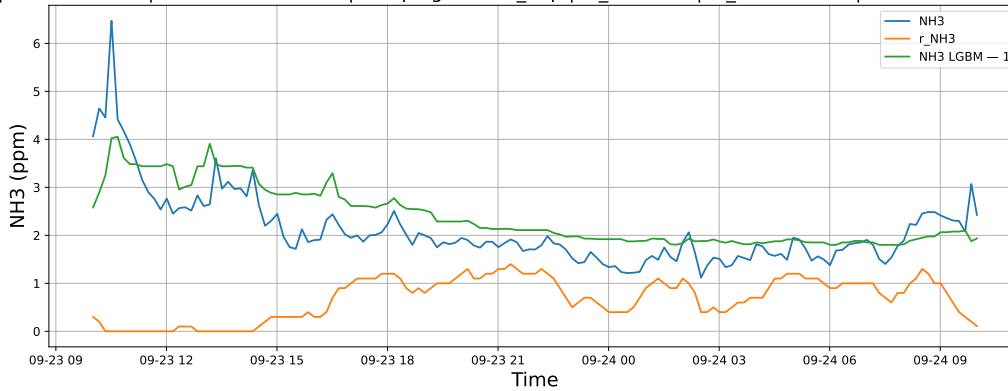
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

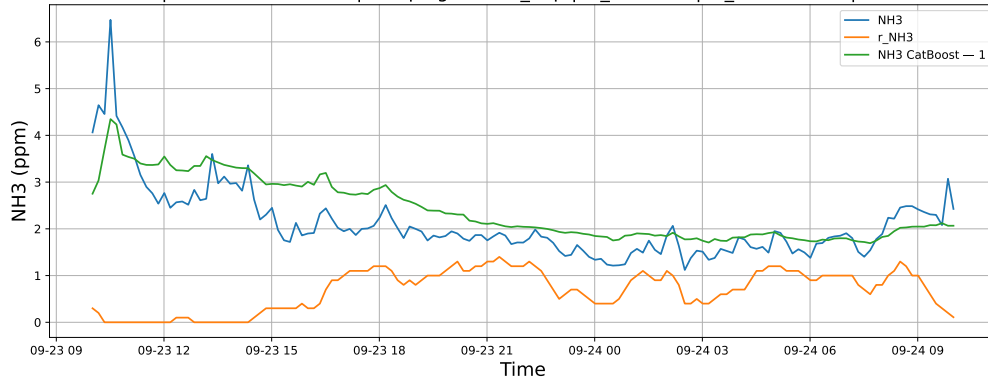
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

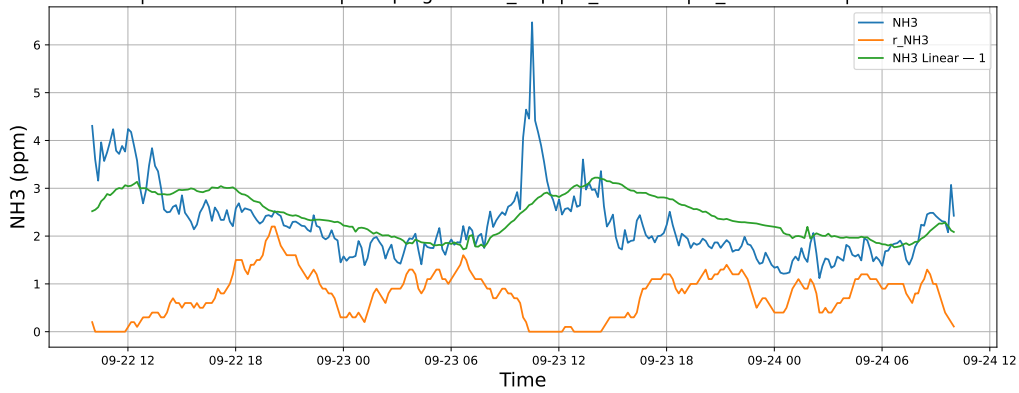


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

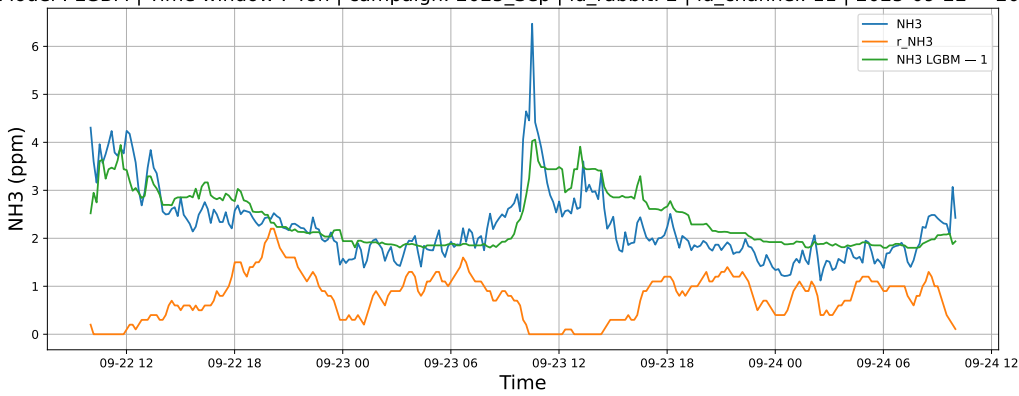


3.1.5.2 Time window : 48

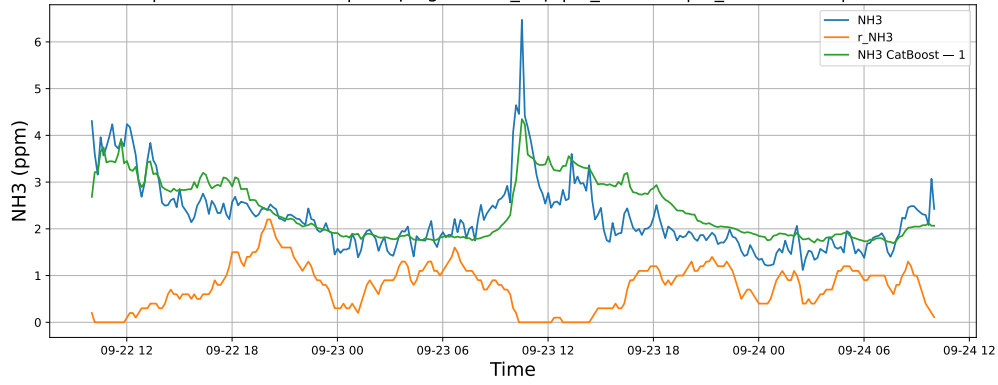
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

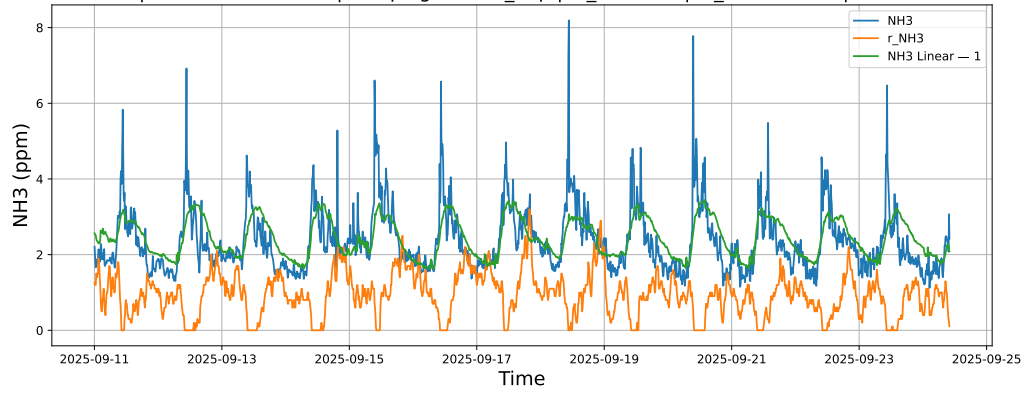


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

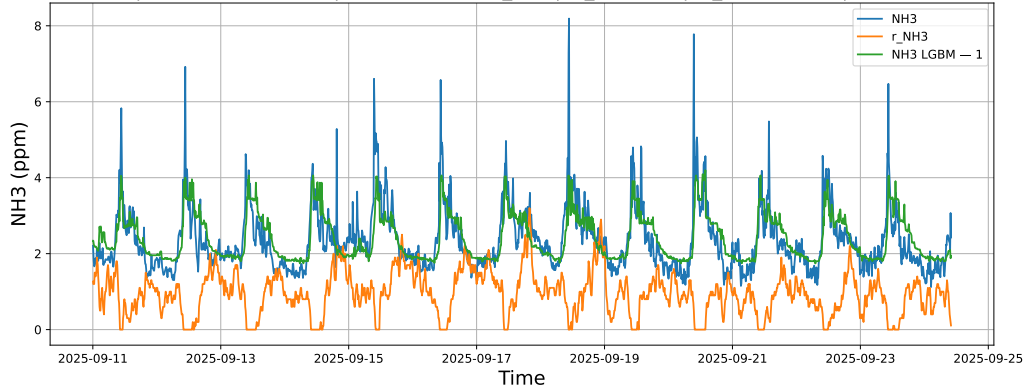


3.1.5.3 Time window : ALL

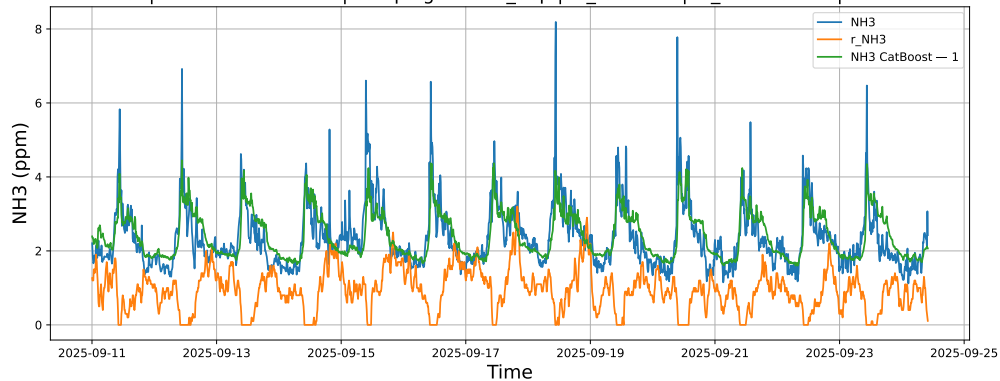
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



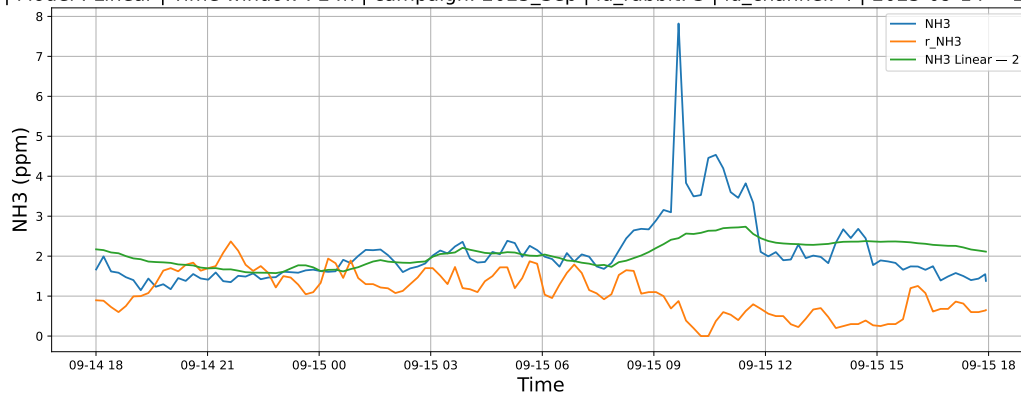
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



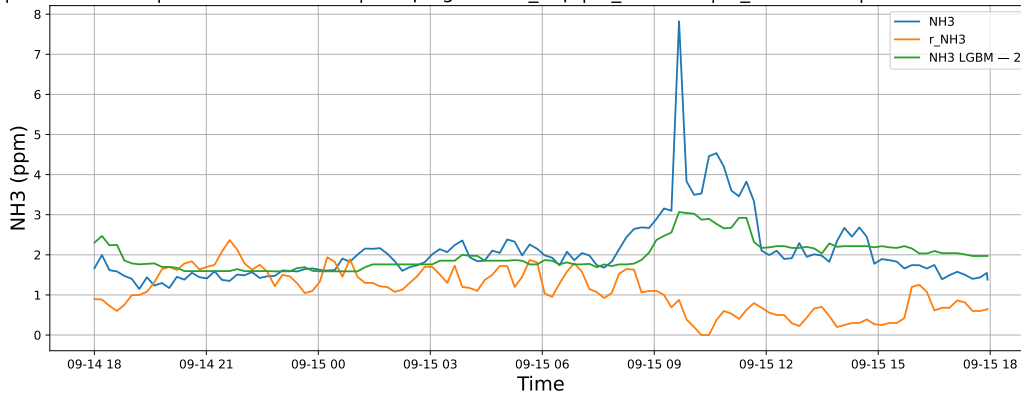
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

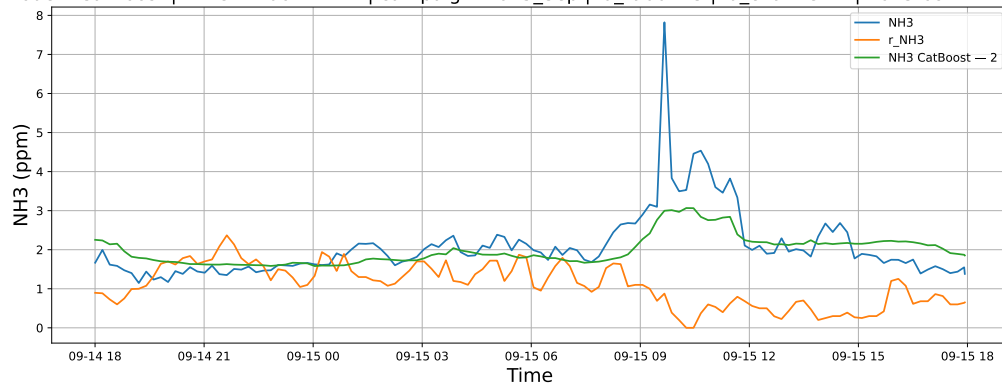
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

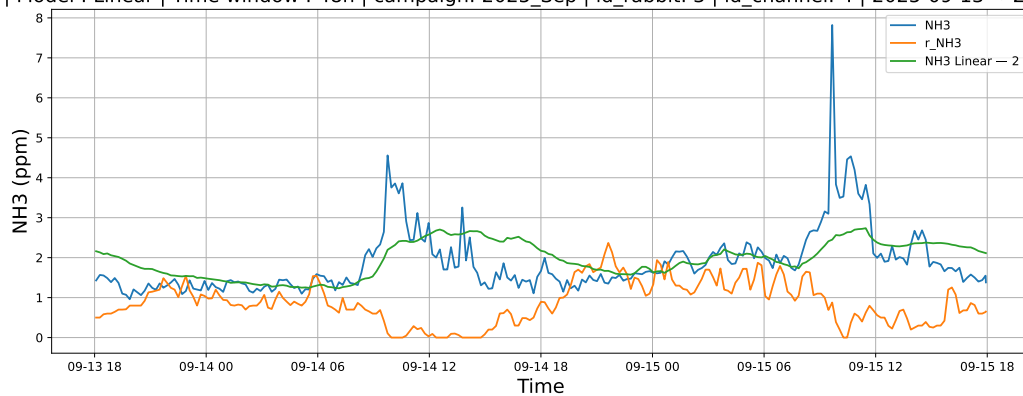


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

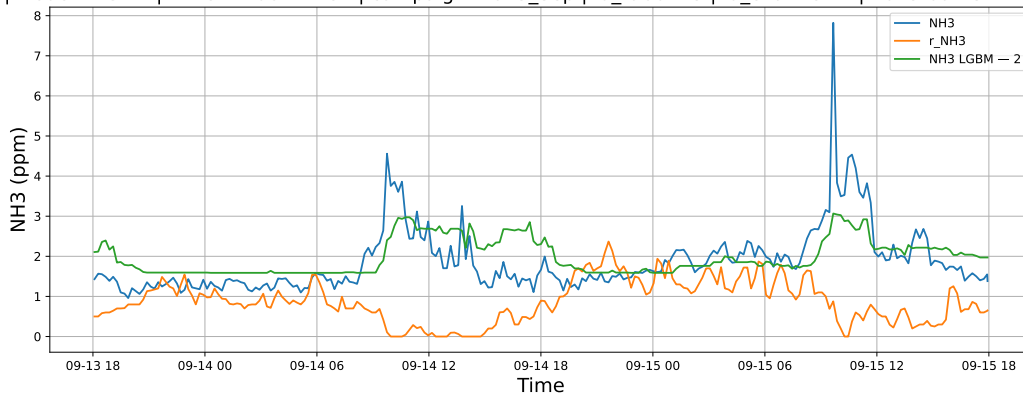


3.1.6.2 Time window : 48

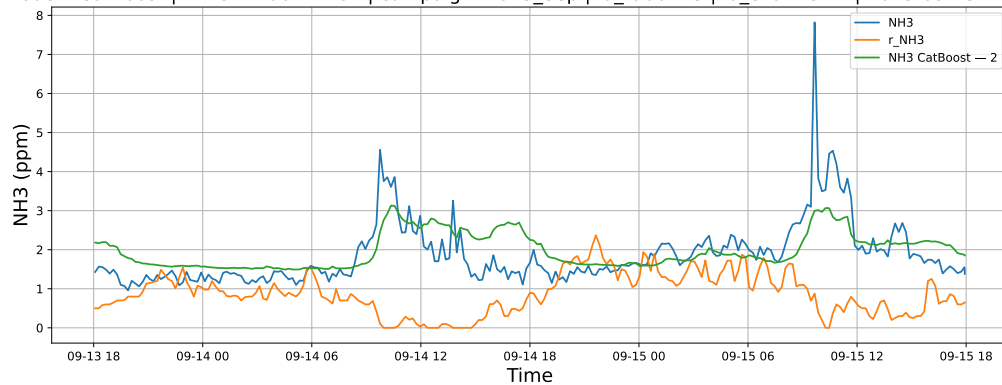
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

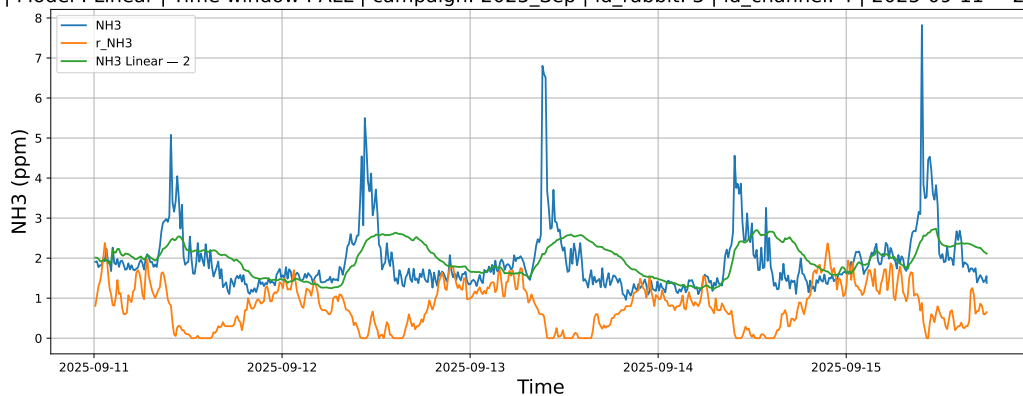


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

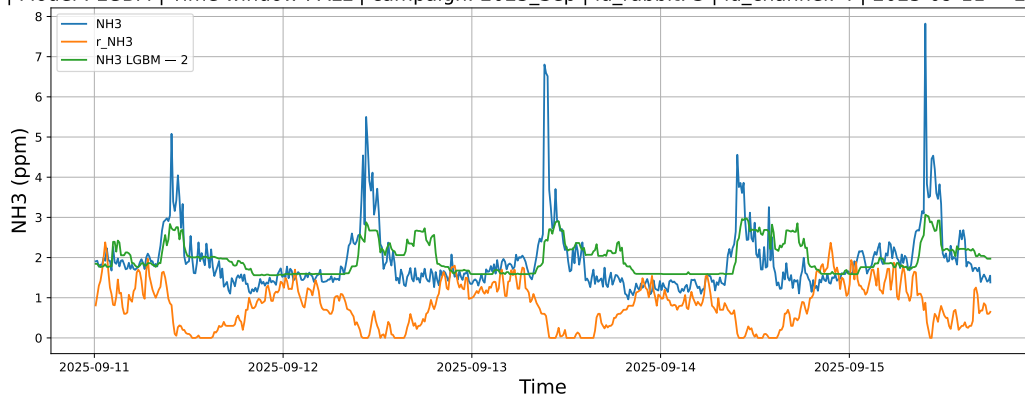


3.1.6.3 Time window : ALL

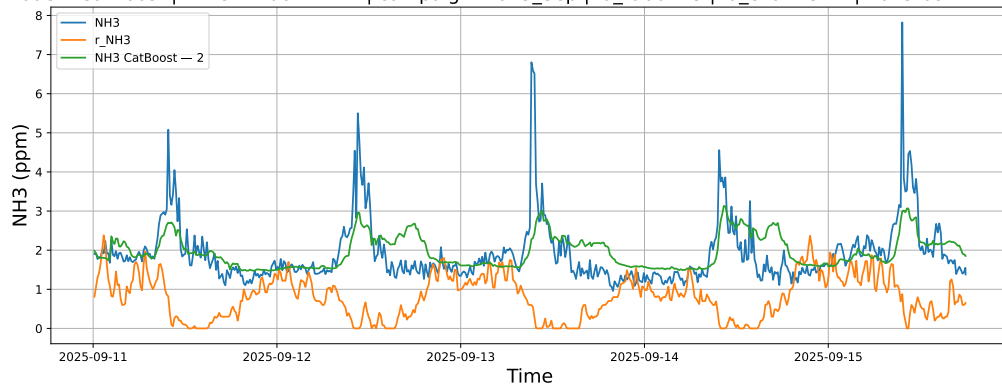
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



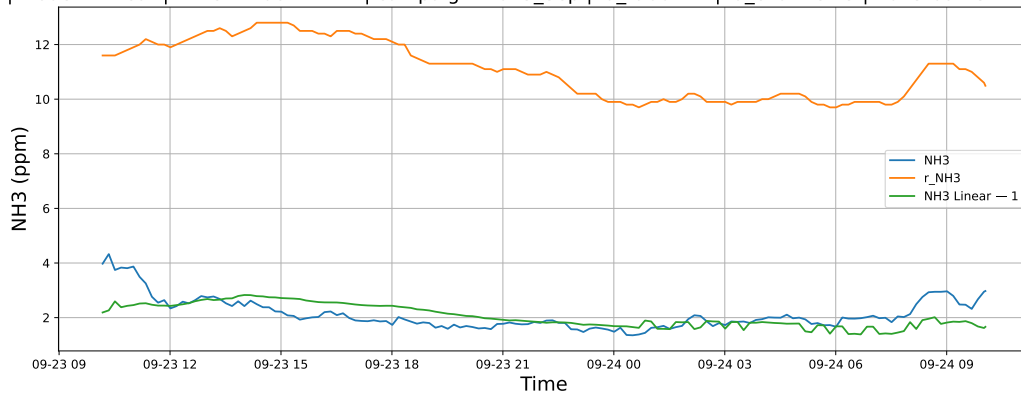
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



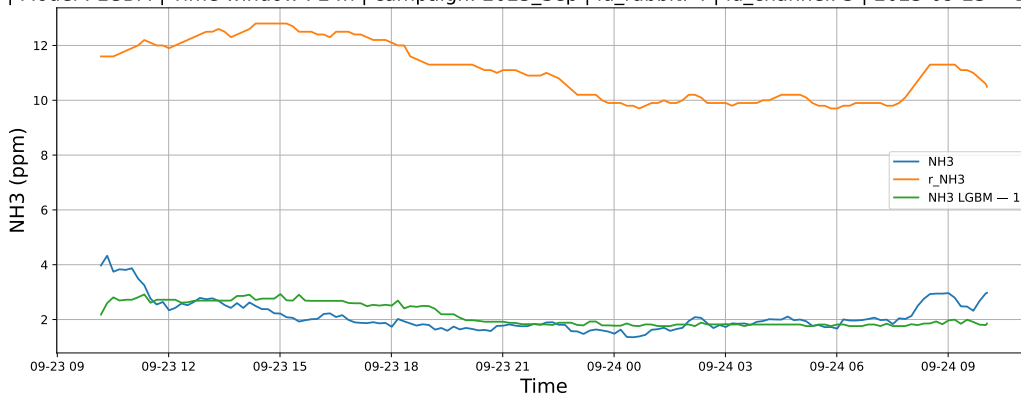
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

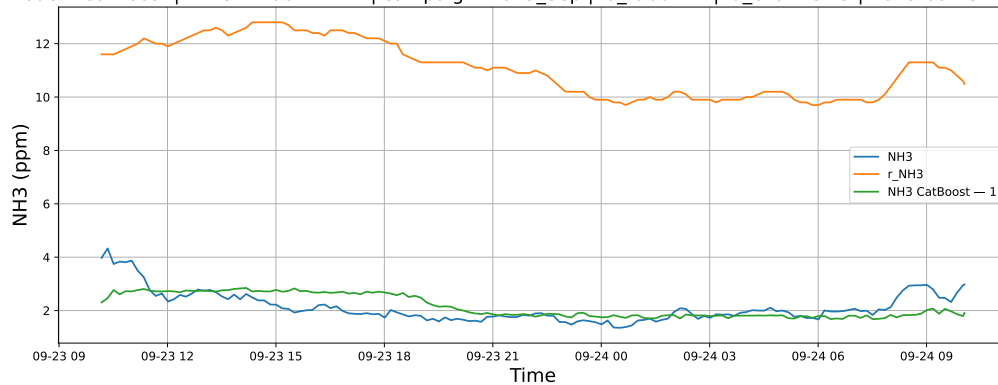
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

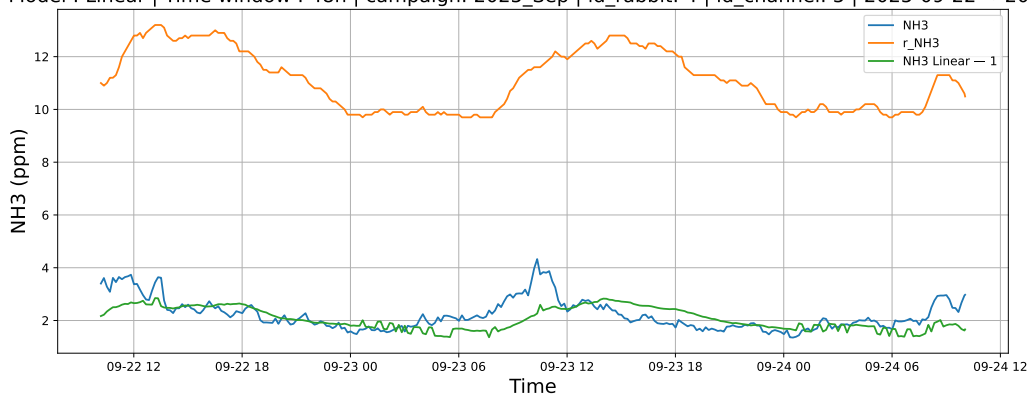


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

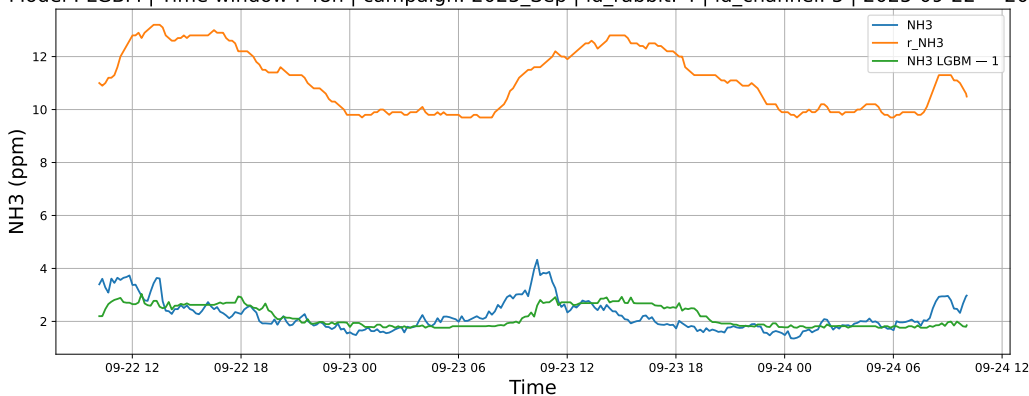


3.1.7.2 Time window : 48

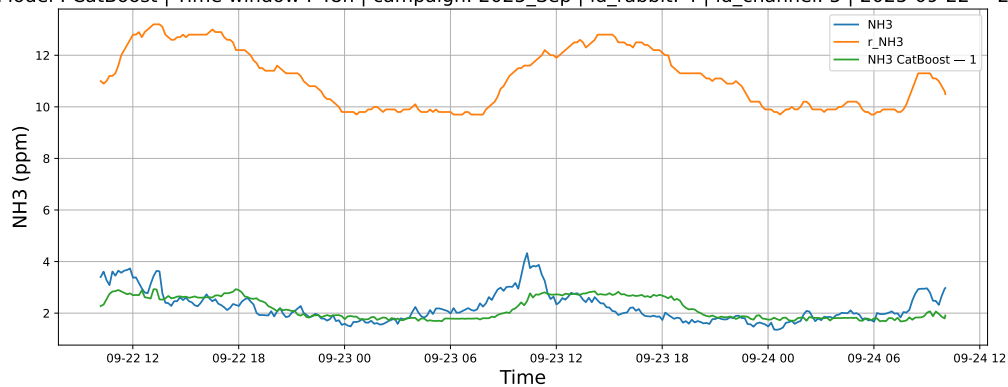
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

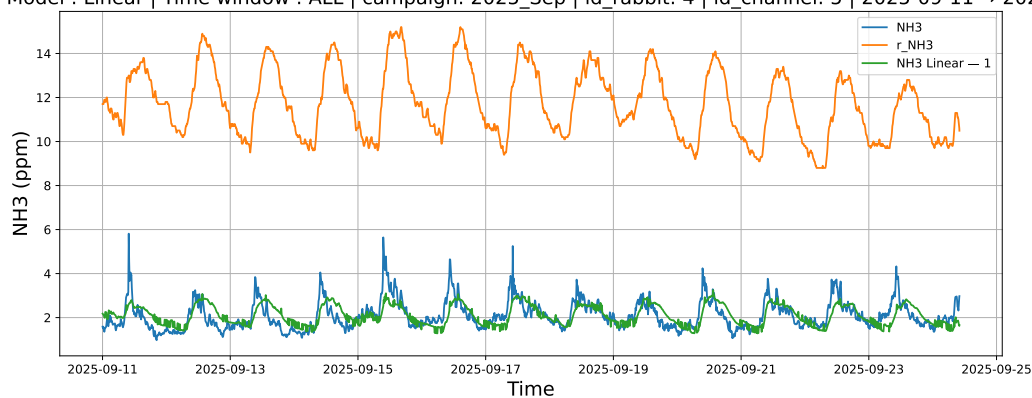


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

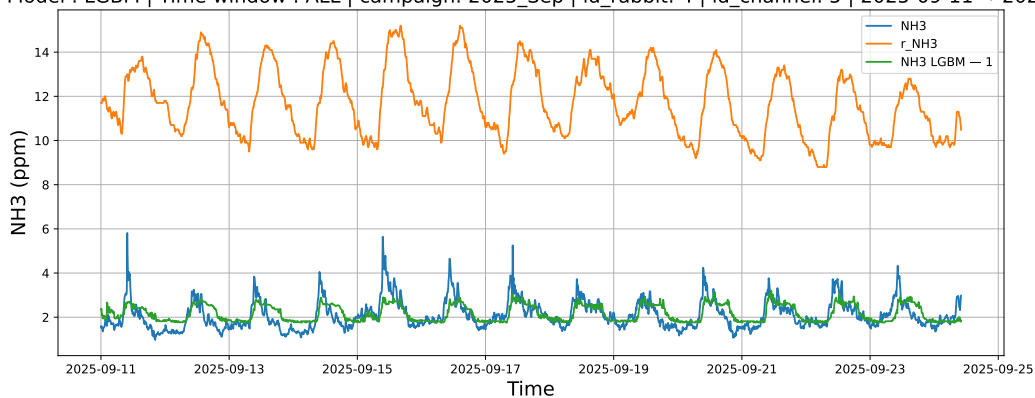


3.1.7.3 Time window : ALL

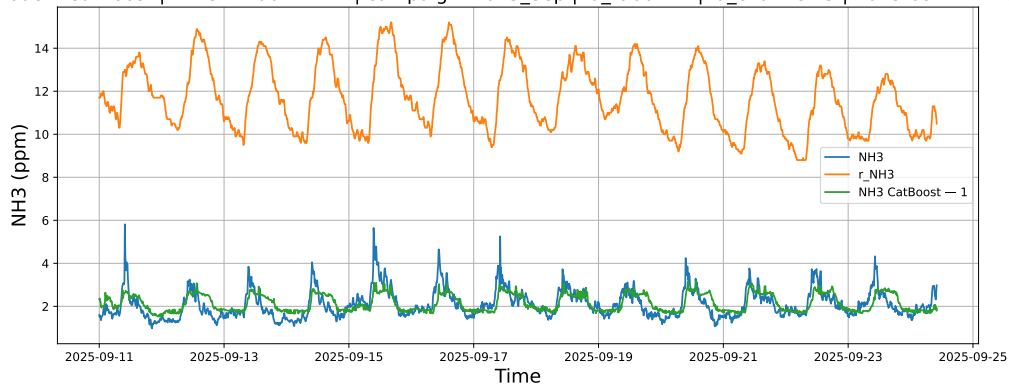
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



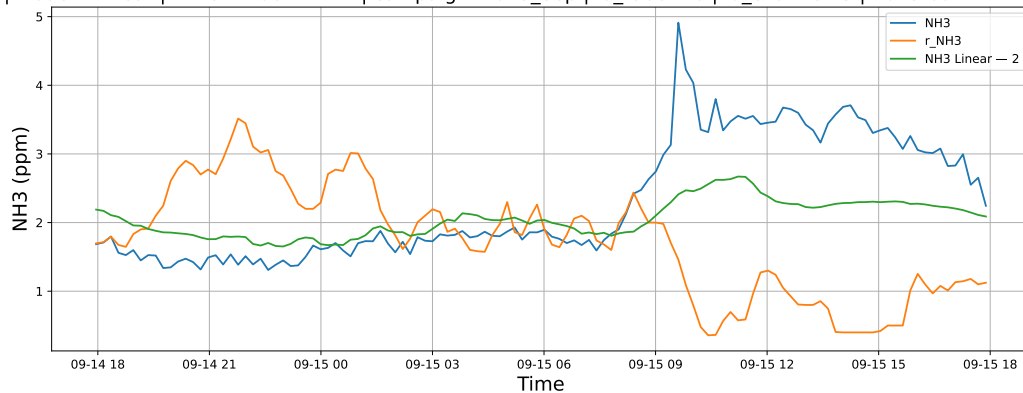
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



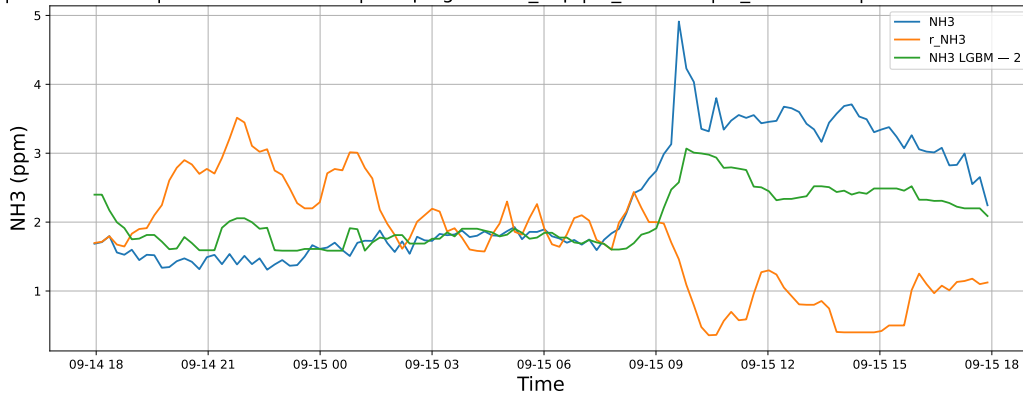
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

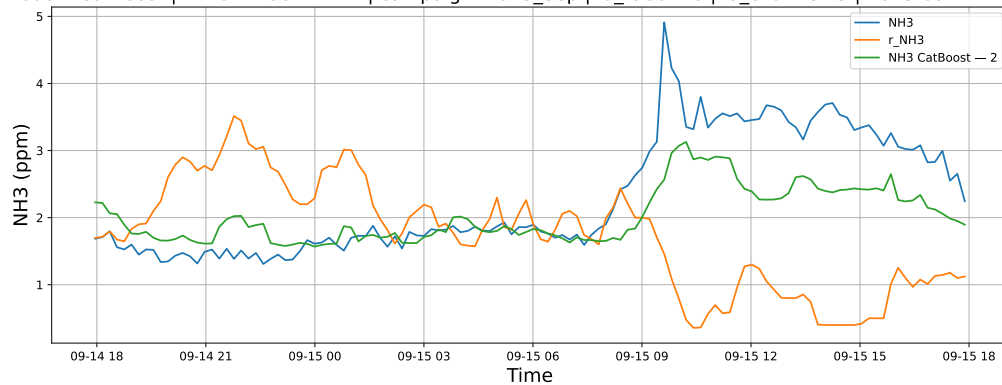
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

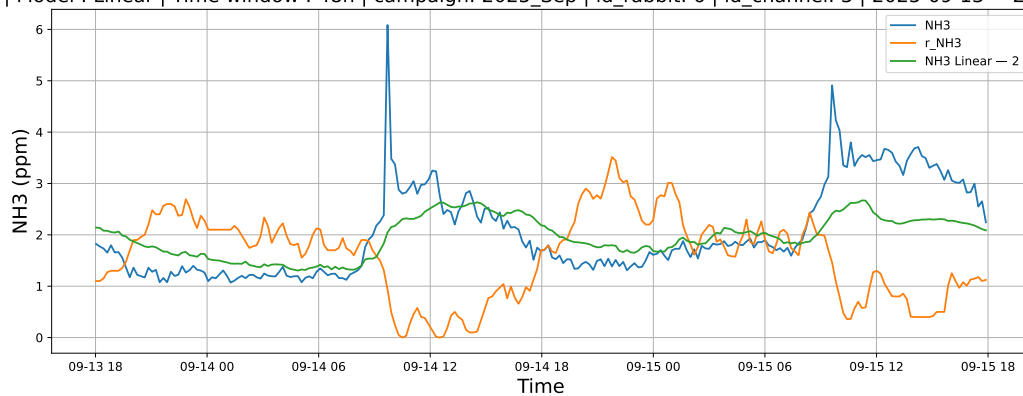


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

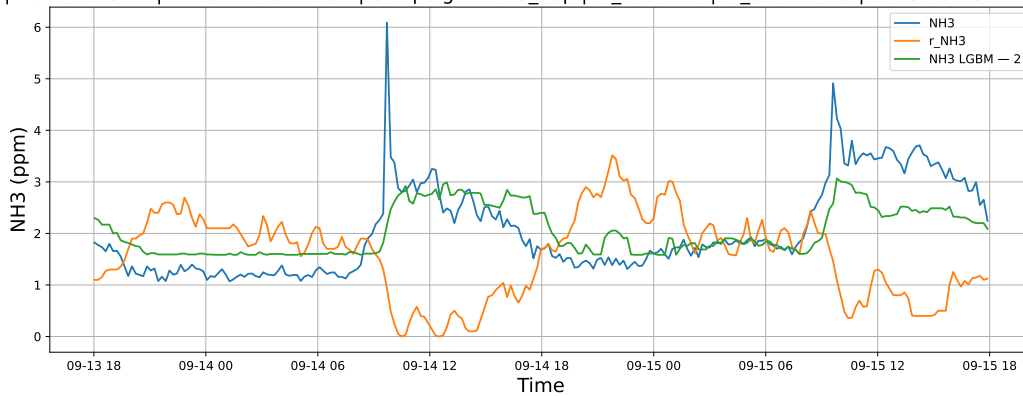


3.1.8.2 Time window : 48

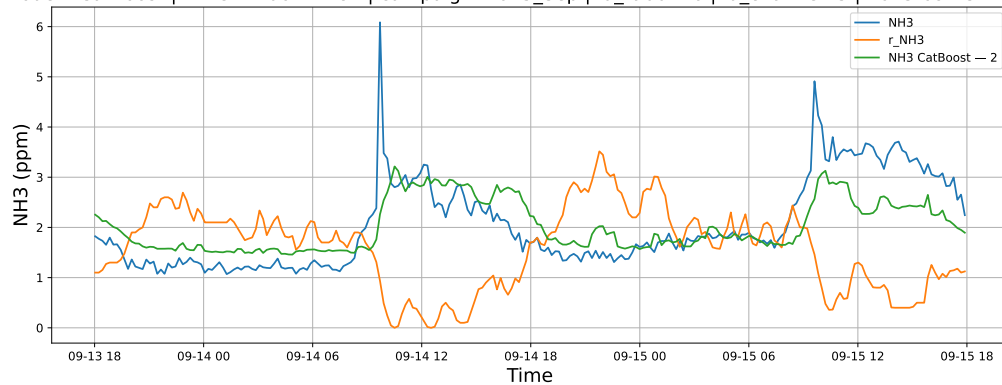
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

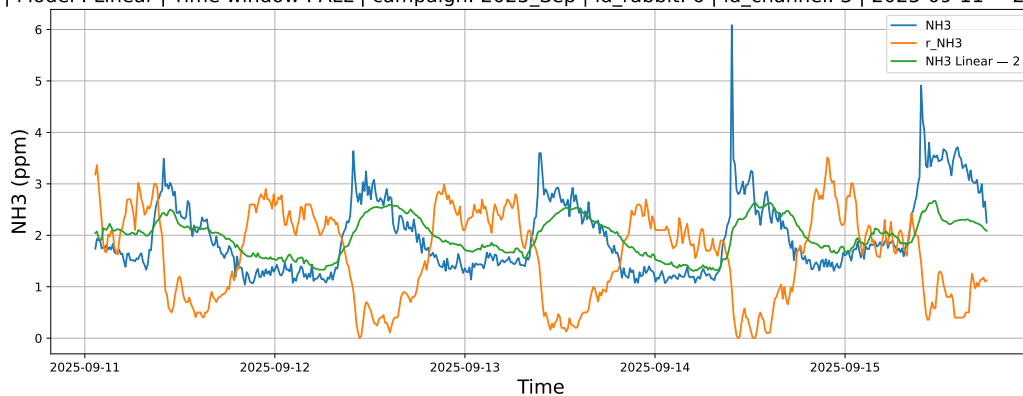


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

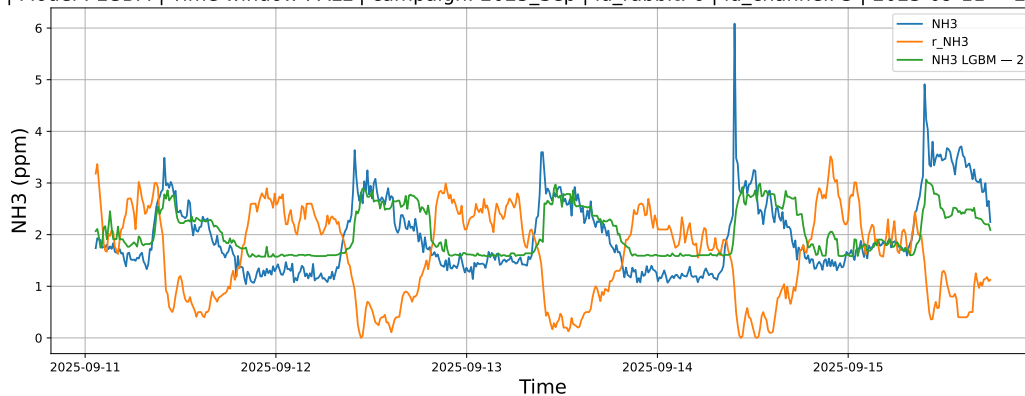


3.1.8.3 Time window : ALL

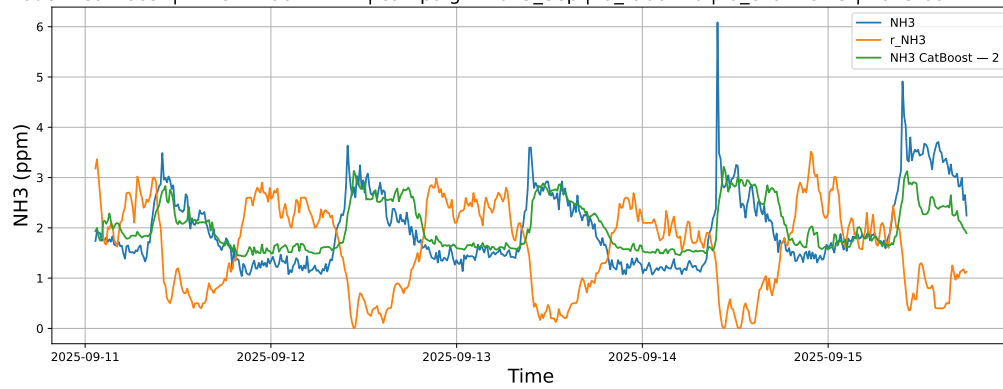
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

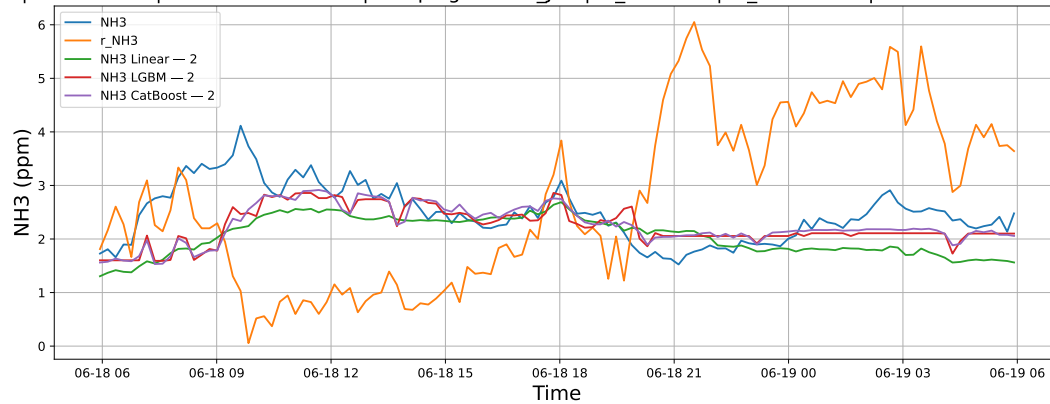


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

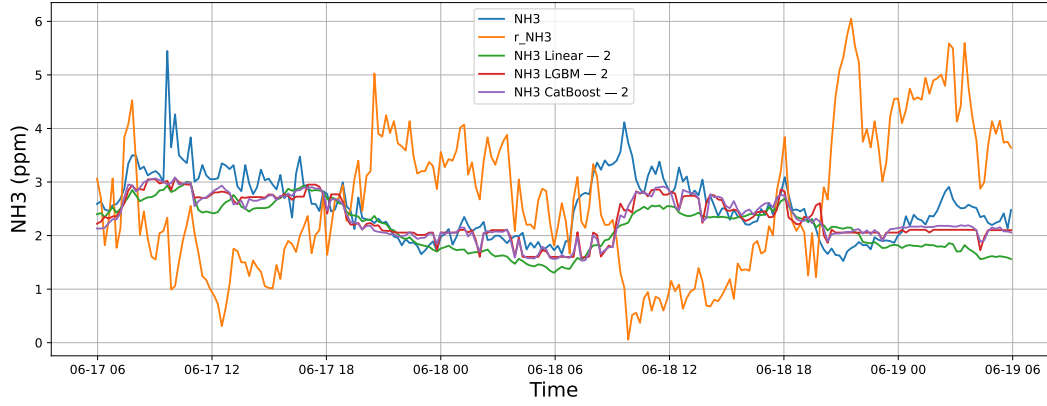
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



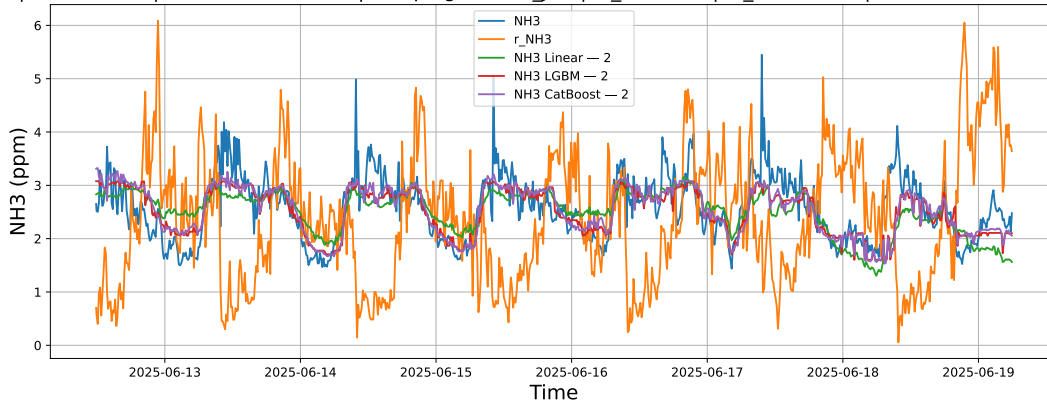
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

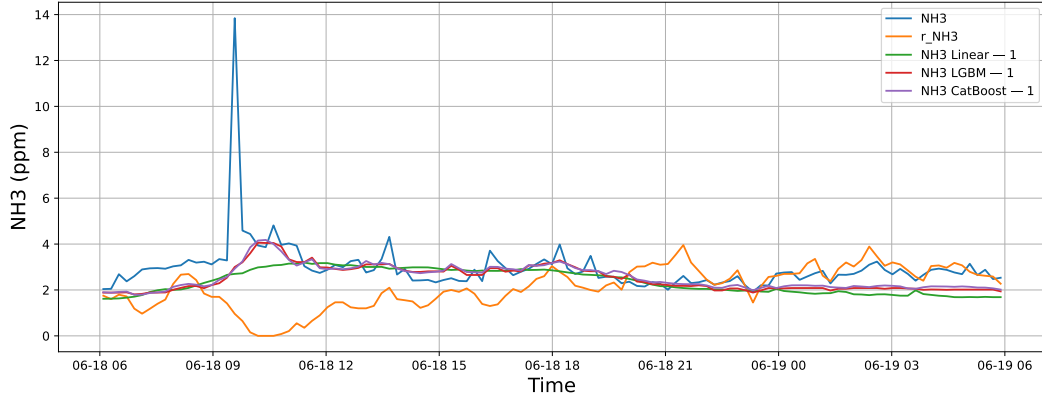
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

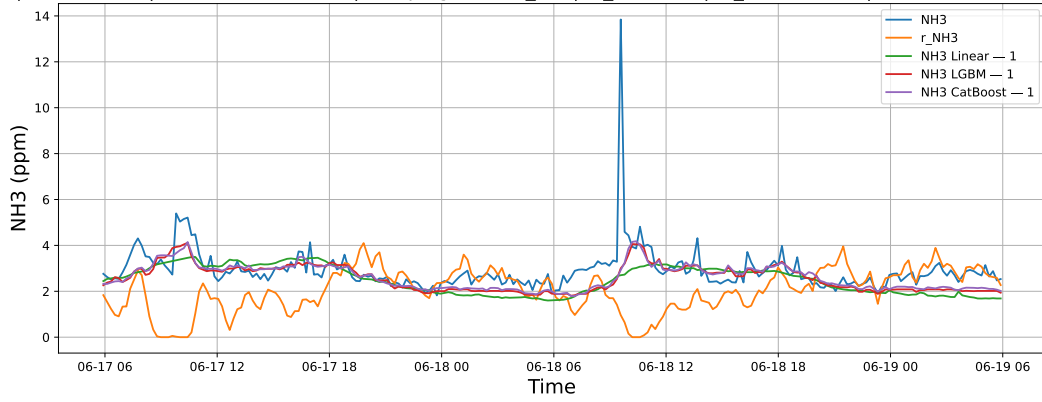
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



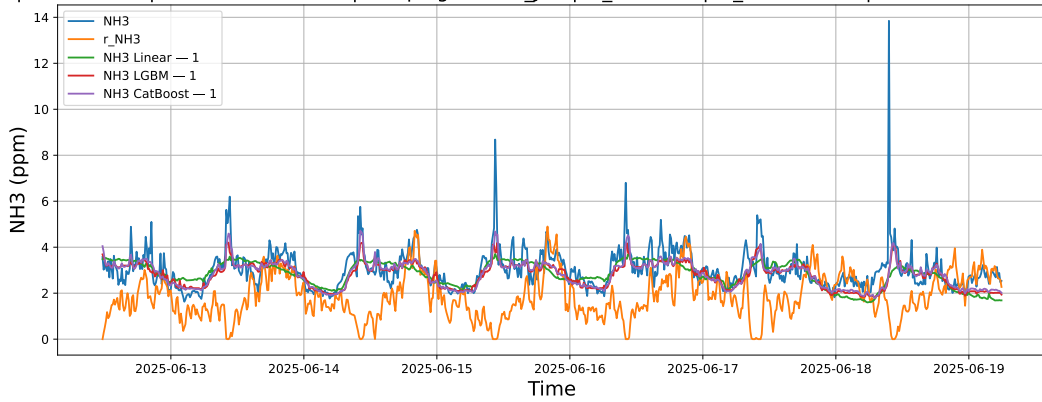
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

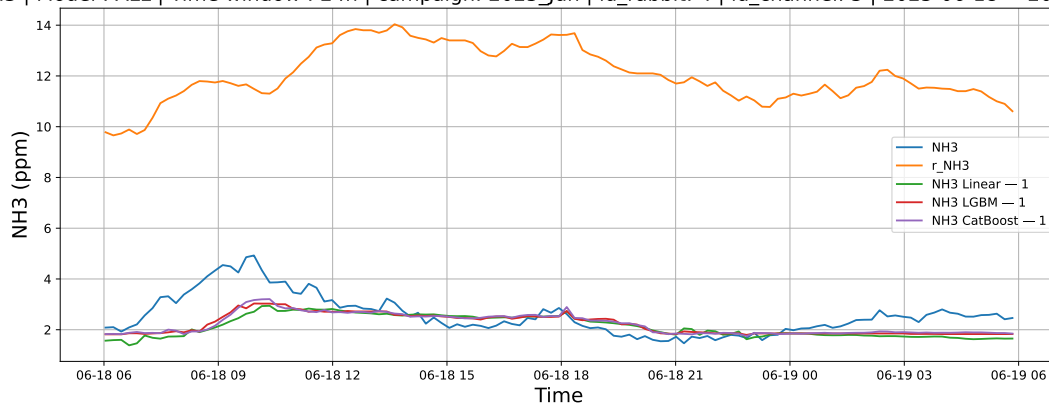
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

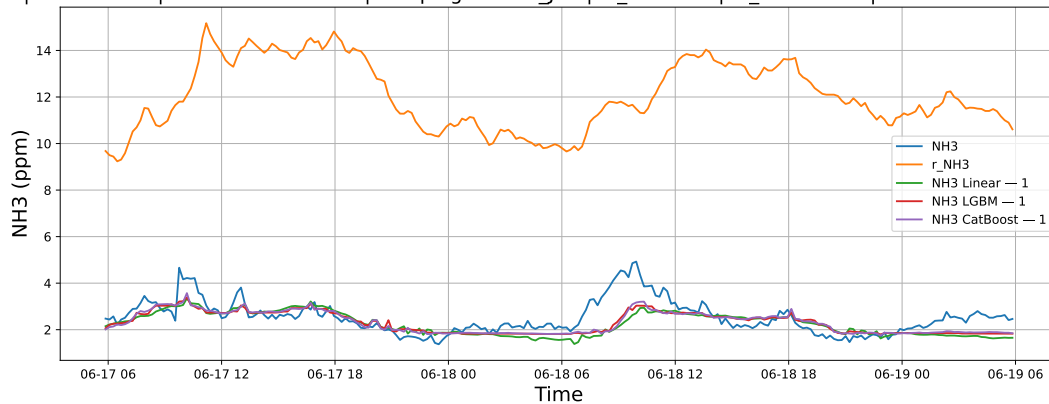
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



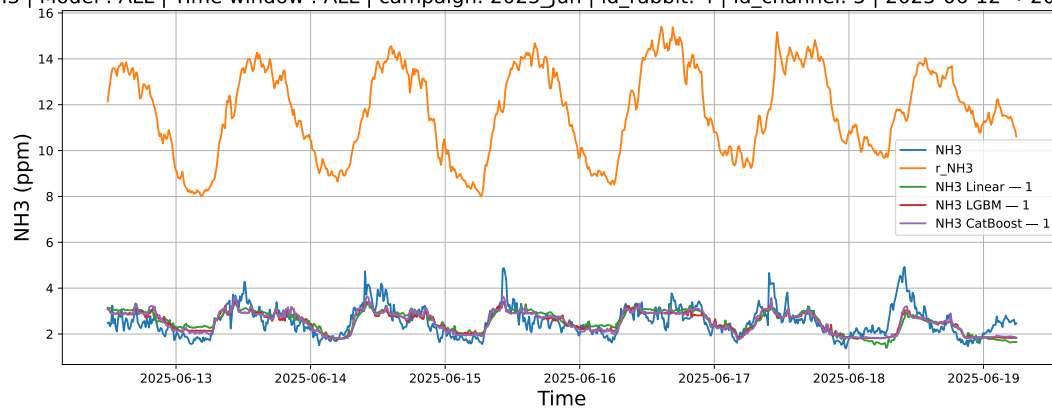
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

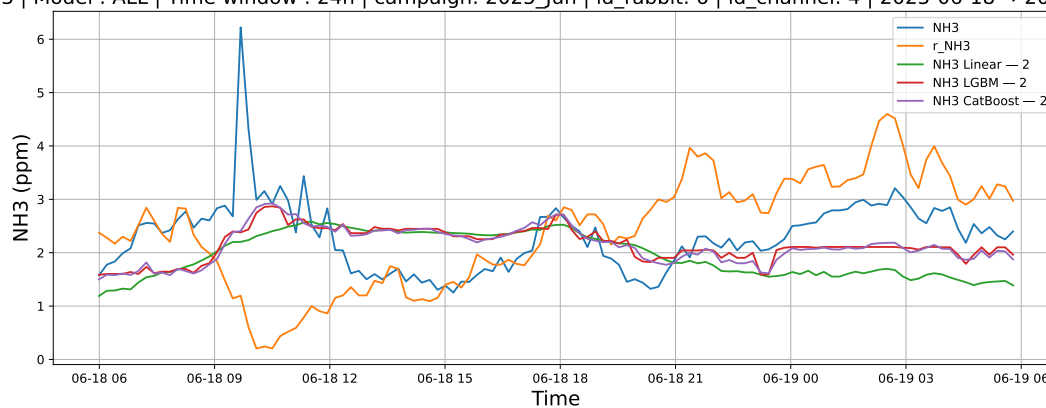
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

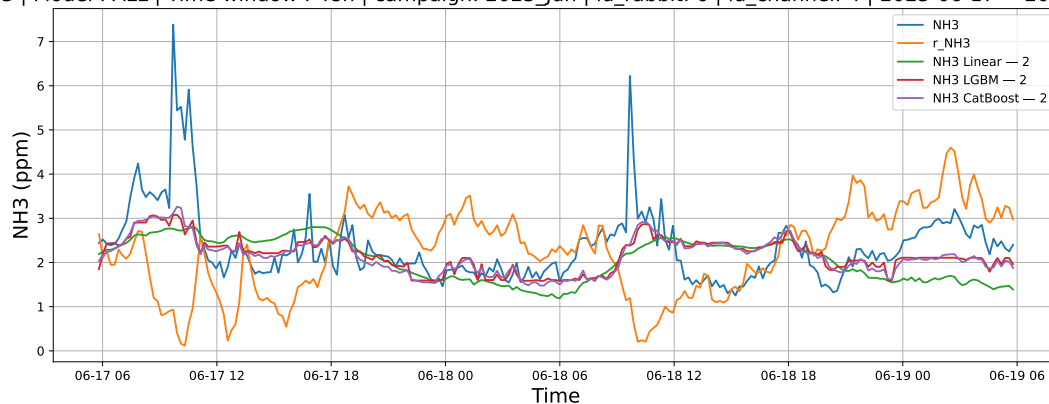
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



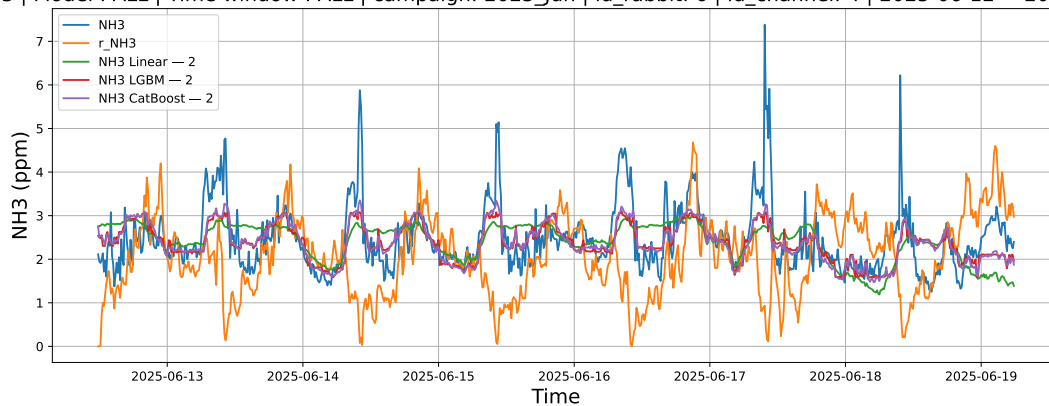
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

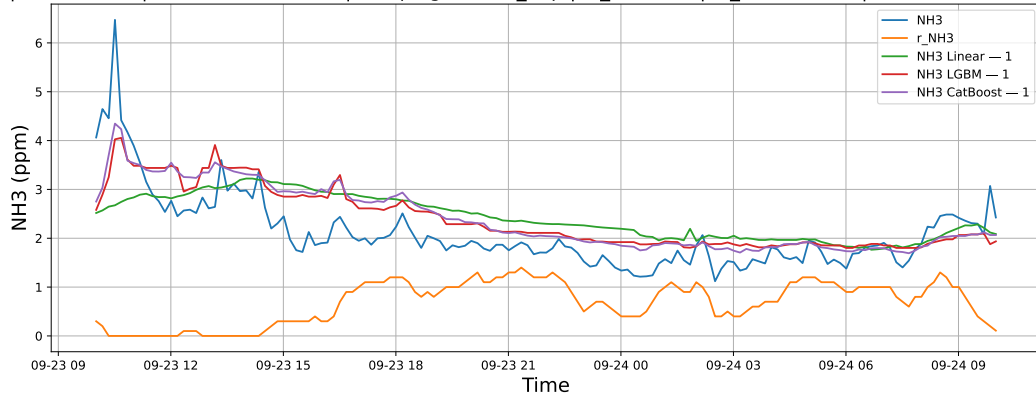
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

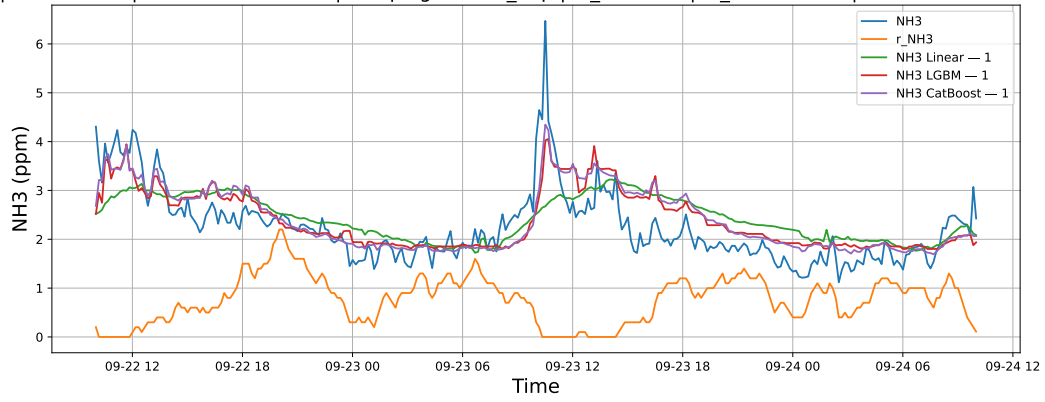
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



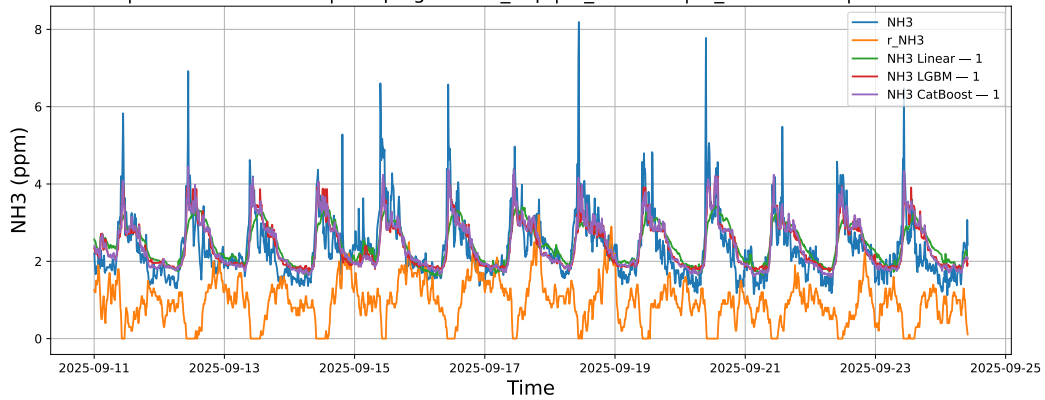
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

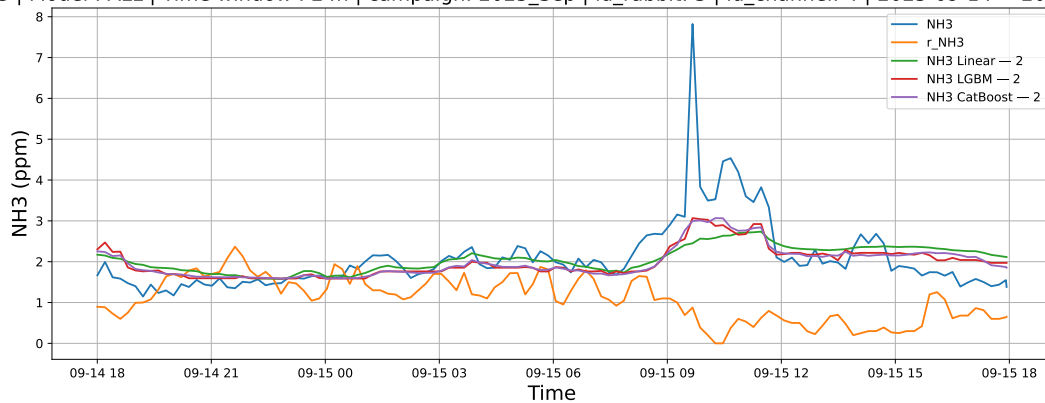
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

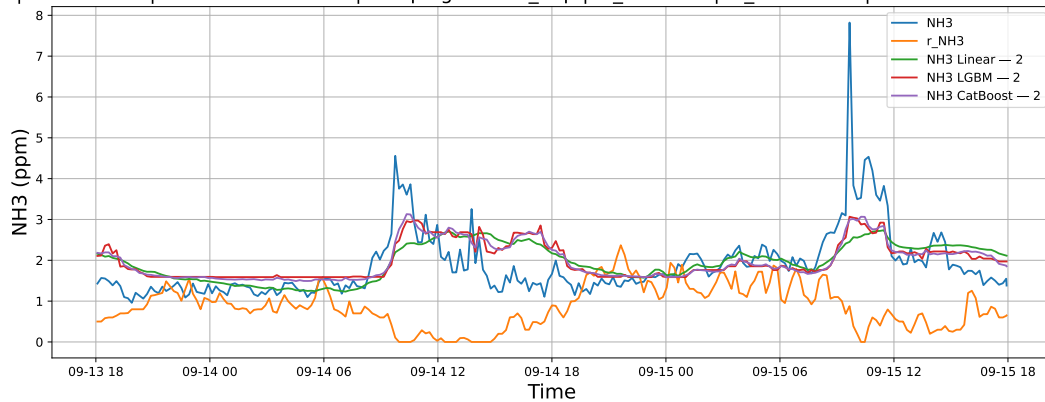
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



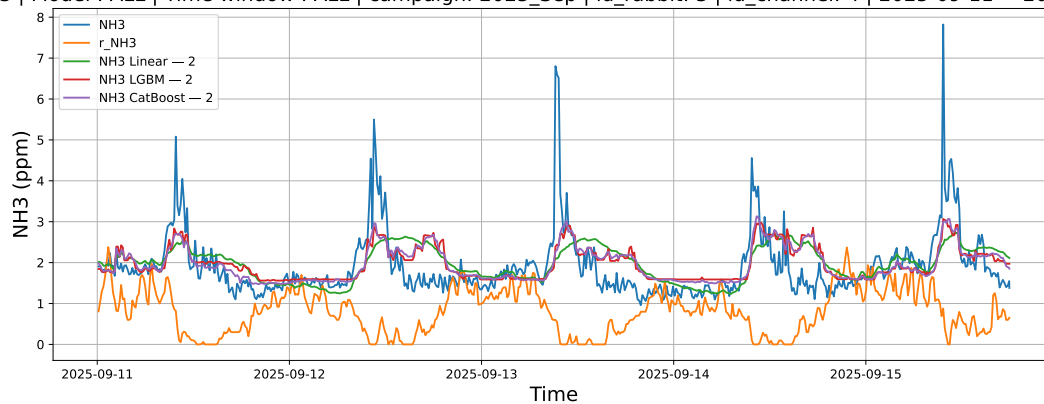
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

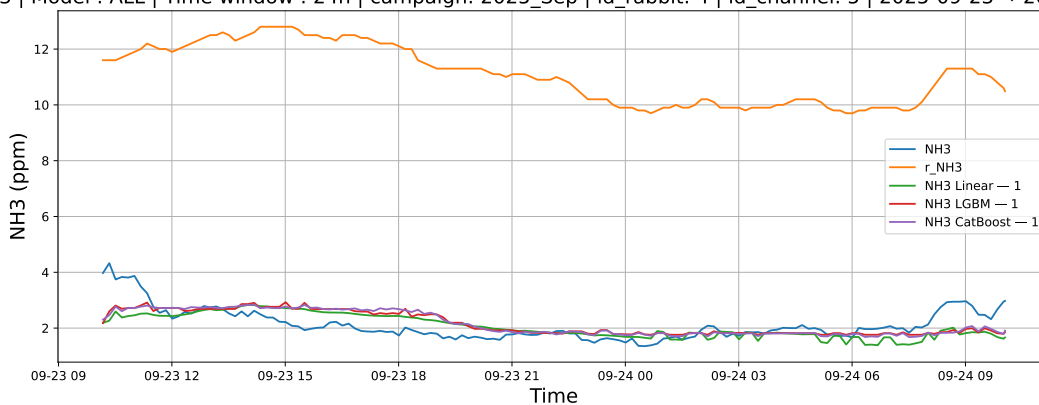
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

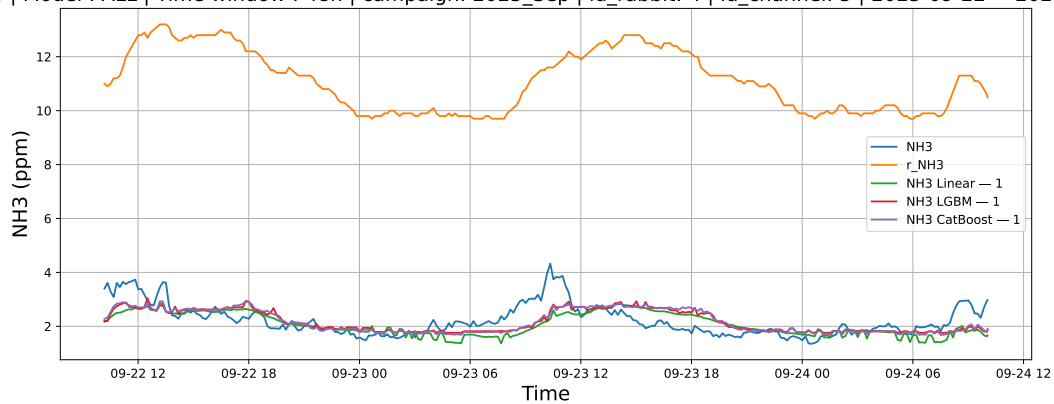
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



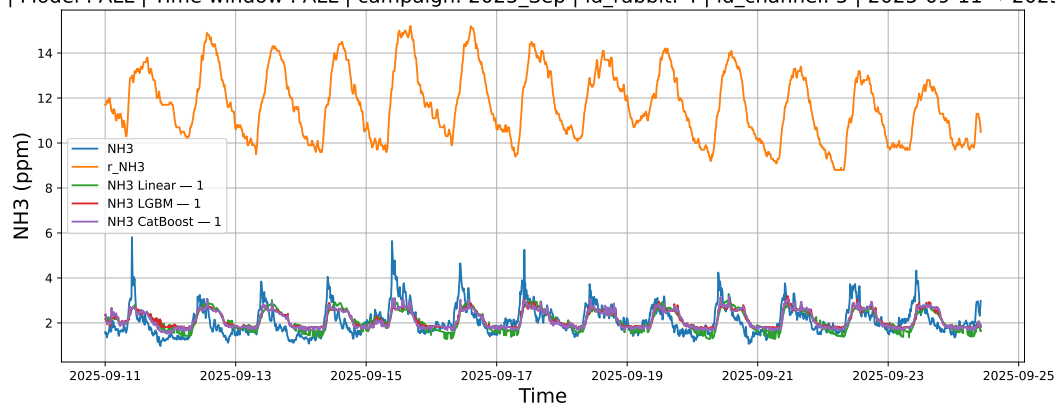
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

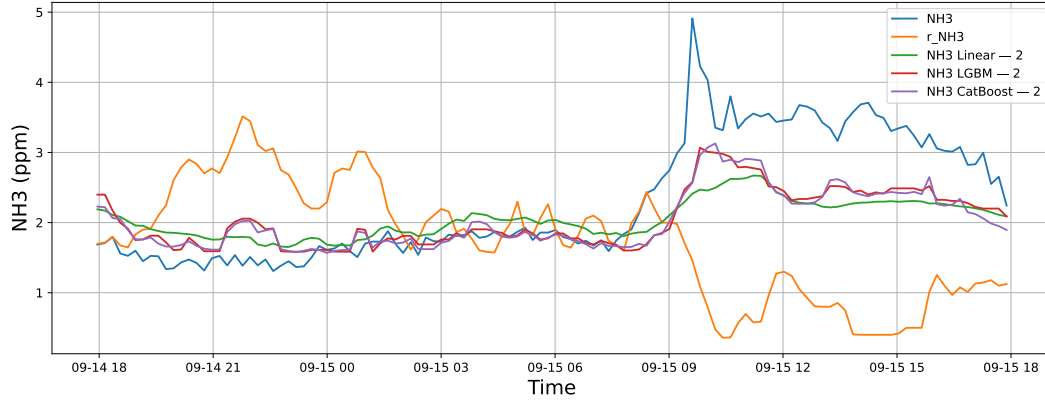
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

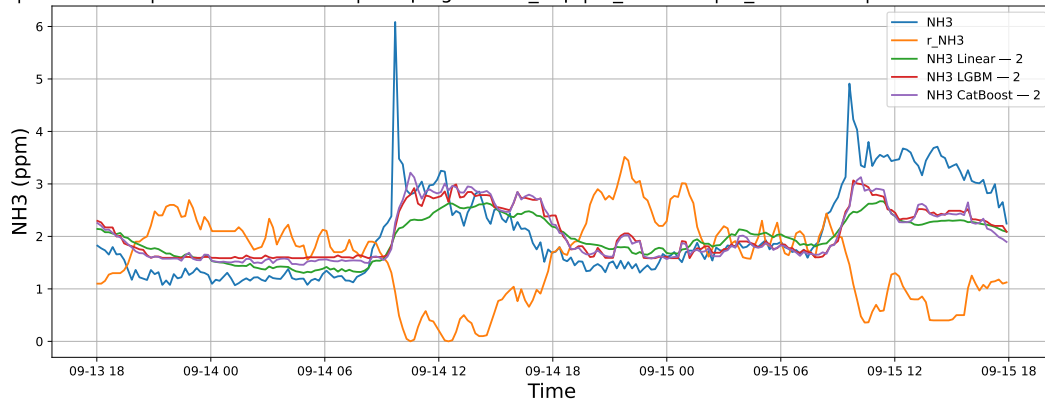
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

